Def: Fie $A \in \mathcal{M}_n(\mathbb{K})$ ($\mathbb{K} \in \{\mathbb{R}, \mathbb{C}\}$)

$\lambda \in \mathbb{K}$ s.n. **val proprie** pt $A \Leftrightarrow (\exists) X \in \mathcal{M}_{n,1}(\mathbb{K}), X \neq O_{n,1}$ (↑ vector propriu al lui A corresp. val. proprie λ.)

a.î. $A \cdot X = \lambda \cdot X$

$d(A)$ **set** mulț. val proprii ale lui A.

Obs: $(\exists) X \in \mathcal{M}_{n,1}(\mathbb{K}), X \neq O_{n,1}$ a.î. $AX = \lambda X$

$\Updownarrow$

$(A - \lambda I_n)X = O_{n,1}$

$\Updownarrow$

$\det(A - \lambda I_n) = 0$

Așadar: $\boxed{\lambda \in d(A) \Leftrightarrow \det(A - \lambda I_n) = 0}$ ① !!!

Def: Fie $A \in \mathcal{M}_n(\mathbb{K})$. Atunci $\boxed{P_A(\lambda) \overset{def}{=} \det(A - \lambda I_n)} =$ ↑

pol. caract ao. matr. A. $A = (a_{ij})_{1 \le i,j \le n}$

$= (-1)^n \lambda^n + (-1)^{n-1} \overline{I}_n(A) \lambda^{n-1}$

$+ \dots + \det A.$

↳ ① $\boxed{\lambda \in d(A) \Leftrightarrow P_A(\lambda) = 0}$ ② !!! ($\Leftrightarrow \lambda$ este răd. pol. caract)

Def: 2 matr. $A, B \in \mathcal{M}_n(\mathbb{K})$ s.n. asemenea ($\Leftrightarrow \exists$ C (matr. nesing.) a.î. $B = C^{-1} A C$

Prop 1: Dacă $A \sim B$, $\Rightarrow P_A(\lambda) = P_B(\lambda)$

Prop 2: Fie $f: V \to W$ ap lin (V, W $\mathbb{K}$-sp vect fin dim) și B_1, B_1' baze pt V; B_2, B_2' baze pt W. Atunci:

$$\boxed{\mathcal{M}_{B_1', B_2'}(f) = (M_{B_2 \to B_2'})^{-1} \cdot \mathcal{M}_{B_1, B_2}(f) \cdot M_{B_1 \to B_1'}} \quad ③$$

Prop 3: Fie V un $\mathbb{K}$-sp vect cu $\dim_{\mathbb{K}} V < +\infty$ și $f: V \to V$ endomorf.
Fie B_1, B_2 baze pt V. Atunci:

fol (3) $\Rightarrow$ $\boxed{M_{B_1}(f) = M^{-1}_{B_2 \to B_1} \cdot M_{B_2}(f) \cdot M_{B_2 \to B_1}}$

$\Downarrow$

$\boxed{M_{B_1}(f) \sim M_{B_2}(f)}$ $\left(\text{Top } P_{M_{B_1}(f)}(\lambda) = P_{M_{B_2}(f)}(\lambda)\right)$

Def: Fie V un $\mathbb{K}$-sp vect cu $\dim_{\mathbb{K}} V < +\infty$ și B o bază a lui V, $f: V \to V$ endomorfism. Atunci $P_f(\lambda) \overset{\text{def}}{=} P_{M_B(f)}(\lambda)$

P_f: pol. caract. ass. lui f. (endomorf.)

Teoremă: În ip. def. de mai sus, $\lambda \in d(f) \Leftrightarrow P_f(\lambda) = 0$

Dem: $\lambda \in d(f) \Leftrightarrow f - \lambda \cdot 1_V$ nu este izomorfism. $\Leftrightarrow$

$\Leftrightarrow$ $\underbrace{M_B(f - \lambda \cdot 1_V)}_{\text{|| (-pros } \dim_{\mathbb{K}} V = n.}$ nu este inv. $\Leftrightarrow \det(M_B(f) - \lambda I_n) = 0$

$M_B(f) - \lambda I_n$

$\det(M_B(f) - \lambda I_n) = 0$

$P_{M_B(f)}(\lambda) = 0$

$P_f(\lambda)$ ■

Ex 1: În sp. vect real $\mathcal{F}(\mathbb{R}) \overset{\text{not}}{=} \{f \mid f: \mathbb{R} \to \mathbb{R}\}$, com. subsp.
gen: $V = sp\{f_1, f_2\}$, unde $f_1, f_2: \mathbb{R} \to \mathbb{R}$, $f_1(x) = \cos x$, $f_2(x) = \sin x$

Fie $g: V \to V$, $g(f) = f'$. Să se arate că g endomorfism și să se determine $d(g)$.

Sol: $(\forall)\alpha, \beta \in \mathbb{R}$ și $f_1, f_2 \in V$.

$g(\alpha f_1 + \beta f_2) = (\alpha f_1 + \beta f_2)' = \alpha f_1' + \beta f_2' = \alpha g(f_1) + \beta g(f_2)$

$\{f_1, f_2\}$ sli?

Fie $\alpha, \beta \in \mathbb{R}$ a.î. $\alpha f_1 + \beta f_2 = 0$. Atunci $\alpha \cos x + \beta \sin x = 0$, $(\forall) x \in \mathbb{R}$

$x = 0$

$\Rightarrow \alpha = 0 \Rightarrow \beta \sin x = 0 (\forall) x \in \mathbb{R}$

$x = \pi/2 \Big/ \Rightarrow \beta = 0$.

Din $\alpha = \beta = 0 \Rightarrow \{f_1, f_2\}$ sli
, dar $\{f_1, f_2\} \subset G$ $\Big| \Rightarrow \underbrace{\{f_1, f_2\}}_{B}$ bază a lui V.

Val. proprii ale lui g sunt rădăcinile ecuației caracteristice
$P_g(\lambda) \overset{def}{=} P_{M_B(g)}(\lambda)$

$g(f_1) = g(\cos x) = -\sin x = -f_2 = 0 \cdot f_1 + (-1) \cdot f_2$ $\Big| \Rightarrow$
$g(f_2) = g(\sin x) = \cos x = f_1 = 1 \cdot f_1 + 0 \cdot f_2$ $\Big|$

$\Rightarrow M_B(g) = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$

$\Rightarrow P_g(\lambda) = \det(M_B(g) - \lambda I_2)$

$= \begin{vmatrix} -\lambda & 1 \\ -1 & -\lambda \end{vmatrix} = \lambda^2 + 1 \Rightarrow \Delta(g) = \emptyset$

$\uparrow$
$P_g(\lambda) = 0$ nu are răd reale.
$\Leftrightarrow P_g(\lambda) = 0$ are două răd. $\in \mathbb{C}$.

Def: Fie V, $\mathbb{K}$-spvect, $\dim_{\mathbb{K}} V < +\infty = n$ și $f: V \to V$ endomorfism
Fie $\lambda_0 \in \Delta(f)$. Atunci $P_f(\lambda) = (\lambda - \lambda_0)^{m_{\lambda_0}} \cdot Q(\lambda)$
, $Q \in \mathbb{K}_{n-m_{\lambda_0}}[X]$
cu $Q(\lambda_0) \neq 0$.

În acest caz, numim $\boxed{m_{\lambda_0}}$ MULTIPLICITATEA ALGEBRICĂ a valorii λ_0.
$\dim_{\mathbb{K}} V_{\lambda_0} \overset{not}{=} M_{\lambda_0}$ și s.n. MULTIPLICITATEA GEOMETRICĂ a valorii proprii λ_0.

obs: $\boxed{m_{\lambda_0}, M_{\lambda_0} \in \mathbb{N}}$

Teoremă: Fie V un $\mathbb{K}$-spvect cu $\dim_{\mathbb{K}} V = n$ și $f: V \to V$ endomorfism.
Atunci, dacă $\lambda \in \Delta(f) \Rightarrow 1 \leq n_\lambda \leq m_\lambda$

Dem: Fie $\lambda \in Sp(f)$ (sau $\in Sp(A)$) $v \in V, v \neq 0_V$ a.î. $f(v) = \lambda v$

$V_\lambda \overset{not}{=} \{ v \in V \mid f(v) = \lambda v \}$

$\Rightarrow V_\lambda \neq \{0\}$

$\Updownarrow$

$r_\lambda \overset{not}{=} \dim_{\mathbb{K}} V_\lambda \geq 1$

Fie $B_1 = \{v_1, v_2, \ldots, v_{r_\lambda}\} \subseteq V_\lambda$ o bază a lui V.

Completăm B_1 până la o bază a lui V, $B = B_1 \cup \{v_{r_\lambda+1}, v_{r_\lambda+2}, \ldots, v_n\}$.

Atunci $f(v_j) = \lambda v_j$, $j = \overline{1, r_\lambda}$

sau $\forall j = \overline{r_\lambda+1, n}$

$f(v_j) = a_{1j} v_1 + a_{2j} v_2 + \ldots + a_{nj} v_n$

$\Rightarrow M_B(f) = \begin{pmatrix} \lambda & 0 & \ldots & 0 & a_{1,r_\lambda+1} & \ldots & a_{1n} \\ 0 & \lambda & \ldots & 0 & a_{2,r_\lambda+1} & \ldots & a_{2n} \\ \vdots & & \ddots & & \vdots & & \vdots \\ 0 & 0 & \ldots & \lambda & a_{r_\lambda,r_\lambda+1} & \ldots & a_{r_\lambda,n} \\ & & & & \vdots & & \vdots \\ & & & & a_{n,r_\lambda+1} & \ldots & a_{nn} \end{pmatrix}$

r_λ linii

$f(v_1) \quad f(v_2) \quad \ldots \quad f(v_{r_\lambda}) \quad f(v_{r_\lambda+1}) \quad \ldots \quad f(v_n)$

$P_f(X) = P_{M_B(f)}(X) = \det(M_B(f) - X I_n)$

$= (\lambda - X)^{r_\lambda} \cdot Q(X)$

$\Downarrow$

i) $Q(\lambda) \neq 0 \Rightarrow r_\lambda = m_\lambda$

ii) $Q(\lambda) = 0 \Rightarrow r_\lambda < m_\lambda$

Consecință Dacă $\lambda \in Sp(f)$ a.î. $m_\lambda = 1$ atunci $\boxed{m_\lambda = r_\lambda = 1}$

(λ e valoare proprie simplă a lui f, $M_B(f)$)

Teoremă: HAMILTON-CAYLEY. Fie $A \in \mathcal{M}_n(\mathbb{K})$. Atunci $P_A(A) = O_{n \times n}$.

Dem: Fie $\lambda \in \mathbb{K}$. Atunci $P_A(\lambda) = a_n \lambda^n + a_{n-1} \lambda^{n-1} + \ldots + a_1 \lambda + a_0$ (1)

Adjuncta $(A - \lambda I_n)^*$ are fiecare element un polinom de grad $(n-1)$ în λ.

$(A-\lambda I_n)^* = \lambda^{n-1}B_{n-1} + \lambda^{n-2}B_{n-2} + \dots + \lambda B_1 + B_0$ (5)

$(A-\lambda I_n)(A-\lambda I_n)^* = \underbrace{[\det(A-\lambda I_n)]}_{P_A(\lambda)} \cdot I_n$ (6)

Înlocuim (4), (5) în (6) $\Rightarrow$

$(A-\lambda I_n)(\lambda^{n-1}B_{n-1} + \lambda^{n-2}B_{n-2} + \dots + \lambda B_1 + B_0) =$

$(a_n\lambda^n + a_{n-1}\lambda^{n-1} + \dots + a_1\lambda + a_0)I_n$

Identificând coef. ocelorași puteri ale lui $\lambda \Rightarrow$

A^n/λ^n: $-B_{n-1} = a_n I_n$

A^{n-1}/λ^{n-1}: $-B_{n-2} + A\cdot B_{n-1} = a_{n-1}I_n$

A^{n-2}/λ^{n-2}: $-B_{n-3} + A\cdot B_{n-2} = a_{n-2}I_n$

$\vdots \qquad \vdots$

A/λ: $A\cdot B_1 - B_0 = a_1 I_n$

λ^0: $A\cdot B_0 = a_0 I_n$

$\Rightarrow$ $-A^n B_{n-1} = a_n A^n$

$A^n B_{n-1} - A^{n-1}B_{n-2} = a_{n-1}A^{n-1}$

$A^{n-1}B_{n-2} - A^{n-2}B_{n-3} = a_{n-2}A^{n-2}$

$\vdots$

$A^2 B_1 - AB_0 = a_1 A$

$AB_0 = a_0 I_n$

"+"

$\Rightarrow O_{n\times n} = P_A(A)$

Diagonalizarea matricelor / endomorf. def. pe un sp. vect finit dimensional.

Fie $A \in \mathcal{M}_n(\mathbb{K})$. Zice că A este diagonalizabilă $\Leftrightarrow (\exists)\ T \in \mathcal{M}_n(\mathbb{K})$ nesingulară ai $T^{-1}\cdot A\cdot T = D$, unde $D \overset{not}{=} \begin{pmatrix} d_{11} & & 0 \\ & d_{22} & \\ 0 & & \ddots \\ & & & d_{nn} \end{pmatrix} \in \mathcal{M}_n(\mathbb{K})$

Obs: 1) $A \sim D \Rightarrow P_A(\lambda) = P_D(\lambda) = (d_{11}-\lambda)(d_{22}-\lambda)\dots(d_{nn}-\lambda)$

$\Rightarrow \sigma(A) = \{\underbrace{d_{11}, d_{22}, \dots, d_{nn}}_{\text{nu neapărat distincte.}}\}$

2) A diagonalizabilă $\Leftrightarrow P_A(\lambda)$ se poate descompune într-un produs de pol de grad 1 cu coef $\in \mathbb{K}$.

Def: Fie V un $\mathbb{K}$-sp vect fin. dim. și $f: V \to V$ endomorfism.
f s.n. diagonalizabil $\Leftrightarrow (\exists)$ o bază a lui V în care $M_B(f)$ să fie o matr. diagonală.

Criteriu de diagonalizare

Fie V un $\mathbb{K}$-sp vect cu $\dim_{\mathbb{K}} V = n$ și $f: V \to V$ endomorfism a.î.

$P_f(\lambda) = (-1)^n (\lambda - \lambda_1)^{m_1} (\lambda - \lambda_2)^{m_2} \dots (\lambda - \lambda_p)^{m_p}$

cu $m_1 + m_2 + \dots + m_p = n$

cu $\lambda_i \neq \lambda_j$, $(\forall)\ 1 \leq i, j \leq p$, $i \neq j$.

Afirmațiile următoare sunt $\Leftrightarrow$:

1) f este diagonalizabilă.
2) $(\exists)$ o bază a lui V formată din vectori proprii ai lui f.
3) $V = V_{\lambda_1} \oplus V_{\lambda_2} \oplus \dots \oplus V_{\lambda_p}$
4) $(\forall)\ 1 \leq i \leq p$, $m_{\lambda_i} = n_{\lambda_i}$

Diagonalizarea matricelor / endomorfismelor definite pe sp. vect. cu $\dim < +\infty$.

$\mathbb{K} \in \{\mathbb{R}, \mathbb{C}\}$

Def Fie $A \in \mathcal{M}_n(\mathbb{K})$. Atunci A este diagonalizabilă peste $\mathbb{K} \Leftrightarrow$ $A \sim D \in \mathcal{M}_n(\mathbb{K})$, D matr. diagonală. ($\Leftrightarrow (\exists)\ T$ nesingulară $\in \mathcal{M}_n(\mathbb{K})$ a.î. $T^{-1}AT = D$).

Obs: 1) $A \sim D \Rightarrow P_A(\lambda) = P_D(\lambda)$ (1)

D = matr. diag.,

Pres $D = \begin{pmatrix} d_{11} & & 0 \\ & d_{22} & \\ 0 & & d_{nn} \end{pmatrix}$

$\Rightarrow P_D(\lambda) \overset{\text{def}}{=} \det(D - \lambda I_n)$

$= (d_{11} - \lambda)(d_{22} - \lambda) \dots (d_{nn} - \lambda)$ (2)

$\Rightarrow \lambda \in \sigma(A) \overset{(1)}{\Leftrightarrow} P_A(\lambda) = 0$ (3)

(1), (2) $\Rightarrow \sigma(A) = \sigma(D) = \{d_{11}, d_{22}, \dots, d_{nn}\}$

$\Rightarrow$ [dacă $A \sim D$, D matr. diag. $\Rightarrow D$ are diag. principală valorile proprii ale lui A]

2) Dacă A diagonalizabilă peste $\mathbb{K}$, atunci $P_A(\lambda)$ se descompune în produs de polinoame de grad ≥ 1 cu coef. în $\mathbb{K}$.

(Cond. necesară de diagonalizare)

Ex: Dacă $A \in \mathcal{M}_3(\mathbb{R})$ și $P_A(\lambda) = (-1)^3(\lambda - 2)(\lambda^2 + 1)$

$\Rightarrow A$ nu diagonalizează peste $\mathbb{R}$ (A diagonalizează peste $\mathbb{C}$)

Def: Fie V un $\mathbb{K}$-sp. vect. cu $\dim_{\mathbb{K}} V < +\infty$ și $f: V \to V$ endomorfism. Atunci f este diagonalizabil peste $\mathbb{K} \Leftrightarrow (\exists)$ o bază a lui V relativ la care $M_B(f)$ este o matr. diagonală.

CRITERIU GENERAL DE DIAGONALIZARE

Fie V un $\mathbb{K}$-sp vect cu $\dim_{\mathbb{K}} V = n$ și $f: V \to V$ endomorfism.

Pres. $P_f(\lambda) = (-1)^n (\lambda - \lambda_1)^{m_1} (\lambda - \lambda_2)^{m_2} \dots (\lambda - \lambda_p)^{m_p}$

, unde $\lambda_1, \dots, \lambda_p \in \mathbb{K}$, $\lambda_i \neq \lambda_j$, $\forall i, j \in \{1, \dots, p\}$, $i \neq j$

$m_1 + m_2 + \dots + m_p = n$.

Atunci urm. afirm. sunt „$\Leftrightarrow$":

a) f diagonalizează peste $\mathbb{K}$.

b) ($\exists$) o bază a lui V formată din n vectori proprii ai lui f

c) $V_{\lambda_1} \oplus V_{\lambda_2} \oplus \dots \oplus V_{\lambda_p} = V$, pt că $V_{\lambda_i} \cap V_{\lambda_j} = \{0_V\}$

$\forall i, j \in \{1, \dots, p\}, i \neq j$.

d) $(\forall) k = \overline{1,p}$: $m_{\lambda_k} = n_{\lambda_k}$, unde $n_{\lambda_k} \overset{not}{=\!=} \dim_{\mathbb{K}} V_{\lambda_k}$

Dem $\boxed{a \Rightarrow b}$

Fie $B \subseteq V$, $B = \{v_1, \dots, v_n\}$ o bază a lui V a.î.

$$M_B(f) = \begin{pmatrix} d_{11} & & 0 \\ & \ddots & \\ 0 & & d_{nn} \end{pmatrix} \in \mathcal{M}_n(\mathbb{K})$$

$\Rightarrow f(v_1) = d_{11} v_1, \dots, f(v_n) = d_{nn} v_n$.

($\forall i \neq 0_V$, $\forall i = \overline{1,n}$ pt că aparțin unei baze)

$\Rightarrow d_{11}, \dots, d_{nn}$ sunt valori proprii pt $M_B(f)$, deci f

și $v_1, \dots, v_n$ sunt vectori proprii corresp. lui f.

Ex 1 Să se stabilească dacă endomorfismul $f: \mathbb{R}_3[X] \to \mathbb{R}_3[X]$,

$f(a + bX + cX^2 + dX^3) = a + d + bX + (c - 2d)X^2 + (a - 2c + 5d)X^3$

este diagonalizabil peste $\mathbb{R}$ și în caz afirmativ să se diagonalizeze

($\Leftrightarrow$ să se indice o bază pt $\mathbb{R}_3[X]$ relativ la care $M_{baza}(f)$ are

formă diagonală)

Sol: $\lambda \in d(f) \Leftrightarrow P_f(\lambda) = 0$

$P_f(\lambda) = P_{M_{B_c}(f)}(\lambda)$

$f(1) = 1 + X^3$; $f(X) = X$

$f(X^2) = X^2 - 2X^3$; $f(X^3) = 1 - 2X^2 + 5X^3$

$\Rightarrow$

$\Rightarrow M_{B_C}(f) = \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -2 \\ 1 & 0 & -2 & 5 \end{pmatrix}$ not

$\Rightarrow P_A(\lambda) \overset{\text{def}}{=} \det(A - \lambda I_4) = \begin{vmatrix} 1-\lambda & 0 & 0 & 1 \\ 0 & 1-\lambda & 0 & 0 \\ 0 & 0 & 1-\lambda & -2 \\ 1 & 0 & -2 & 5-\lambda \end{vmatrix}$

$= (1-\lambda) \cdot \begin{vmatrix} 1-\lambda & 0 & 1 \\ 0 & 1-\lambda & -2 \\ 1 & -2 & 5-\lambda \end{vmatrix}$

$= (1-\lambda)^2 \cdot \begin{vmatrix} 1-\lambda & 0 & 1 \\ 0 & 1-\lambda & -2 \\ 1 & -2 & 5-\lambda \end{vmatrix}$

$= (1-\lambda)[(\lambda-1)^2(5-\lambda) - (1-\lambda) - 4(1-\lambda)]$

$= (\lambda-1)^2[\lambda^2 - 6\lambda + 8 - 2]$

$= (\lambda-1)^2(\lambda^2 - 6\lambda) = \lambda(\lambda-1)^2(\lambda-6)$

$\Rightarrow \sigma(f) = \sigma(A) = \{0, 1, 6\}$ cu $m_{\lambda=0} = m_{\lambda=6} = 1$

și $m_{\lambda=1} = 2$

$m_{\lambda=0} = n_{\lambda=0} = 1$

$m_{\lambda=6} = n_{\lambda=6} = 1$

Ne interesează (ca să stabilim dacă f diagonalizează) dacă

$m_{\lambda=1} = n_{\lambda=1}$

$\| \dim_{\mathbb{R}} V_{\lambda=1}$

$V_{\lambda=1} = \{ P \in \mathbb{R}_3[X] \mid f(P) = \lambda P \}$

$\| 1$

Fie $P = a + bX + cX^2 + dX^3$; $a, b, c, d \in \mathbb{R}$

$\overset{f(P)=P}{\Longrightarrow} (a+d) + bX + (c-2d)X^2 + (a-2c+5d)X^3 = a + bX + cX^2 + dX^3$

$\Rightarrow \begin{cases} a + d = a \Rightarrow d = 0 \\ b = b \\ c - 2d = c \\ a - 2c + 5d = d \Rightarrow a = 2c \end{cases} \overset{(*)}{\Rightarrow} \begin{cases} d = 0 \\ a = 2c \end{cases}$

$\Rightarrow V_{\lambda=1} = \{2c + bX + cX^2 \mid b, c \in \mathbb{R}\} = \{bX + c(2 + X^2) \mid b, c \in \mathbb{R}\}$

$\Rightarrow$ o bază pt $V_{\lambda=1}$ poate fi $\{X, 2+X^2\} \Rightarrow n_{\lambda=1} = 2 = m_{\lambda=1}$

$= B_{V_{\lambda=1}} = B_1$

Dim $m_{\lambda=0} = n_{\lambda=0} = 1$

$m_{\lambda=1} = n_{\lambda=1} = 2$

$m_{\lambda=6} = n_{\lambda=6} = 1$

$\partial(f) = \{0,1,6\}$

Crit gen ⇒ f este diagonalizabilă pentru că ∂ diag (d)

Baza relativ la care matricea asociată lui f este o matr. diagonală este formată din reuniunea bazelor subspațiilor proprii

$V_{\lambda=0}, V_{\lambda=1}, V_{\lambda=6}$

Obs: f diagonalizabilă $\iff V_{\lambda=0} \oplus V_{\lambda=1} \oplus V_{\lambda=6} = (V = \mathbb{R}_3[X])$

$V_{\lambda=0} = \{P \in \mathbb{R}_3[X] \mid f(P) = 0\}$ $(= \ker f)$

$\Updownarrow \leftarrow P = a + bX + cX^2 + dX^3$

$$\iff \begin{cases} a+d=0 \\ b=0 \\ c-2d=0 \\ a-2c+5d=0 \end{cases} \iff \begin{cases} d=-a \\ b=0 \\ c=2d=-2a \end{cases}$$

$\Rightarrow V_{\lambda=0} = \{a - 2aX^2 - aX^3 \mid a \in \mathbb{R}\} = sp\{1-2X^2-X^3\}$

$\Rightarrow B_2 = \{1-2X^2-X^3\}$ bază pt $V_{\lambda=0}$

$V_{\lambda=6} = \{P \in \mathbb{R}_3[X] \mid f(P) = 6P\}$

$\Updownarrow \leftarrow P = a + bX + cX^2 + dX^3$

$a + d + bX + (c-2d)X^2 + (a-2c+5d)X^3 =$

$= 6a + 6bX + 6cX^2 + 6dX^3$

$$\iff \begin{cases} a+d=6a \Rightarrow \boxed{d=5a} \\ 6b=b \Rightarrow \boxed{b=0} \\ c-2d=6c \Rightarrow -2d=5c \\ a-2c+5d=6d \Rightarrow a-2c=d=5a \Rightarrow -2c=4a \end{cases}$$

$\Rightarrow \boxed{c=-2a}$

$\Rightarrow V_{\lambda=6} = \{a - 2aX^2 + 5aX^3 \mid a \in \mathbb{R}\} = sp\{1-2X^2+5X^3\}$

$\Rightarrow B_3 = \{1-2X^2+5X^3\}$ bază pt $V_{\lambda=6}$

$\Rightarrow B \overset{not}{=} B_1 \cup B_2 \cup B_3 = \{X, 1+X^2, 1-2X^2-X^3, 1-2X^2+5X^3\}$ bază a lui $\mathbb{R}_3[X]$ relativ la care f are matricea $D = \begin{pmatrix} 1 & & \\ & 0 & \\ & & 6 \end{pmatrix}$ ← val. proprii

Fie $\vec{a} \neq \vec{0}$, $\vec{a} \in V_3$, $\vec{a} = a_1\vec{i} + a_2\vec{j} + a_3\vec{k}$ și $f: V_3 \to V_3$, $f(\vec{x}) = \vec{a} \times \vec{x}$
Să se arate că f este endomorfism (temă), să se det $\delta(f)$ și să se stabilească dacă f este diagonalizabil (peste $\mathbb{R}$).

Sol: $\lambda \in \delta(f) \Leftrightarrow P_{M_{B_c}(f)}(\lambda) = 0$ unde $M_{B_c}(f) \overset{not}{=} A$

$(B_c)_{V_3} = \{\vec{i}, \vec{j}, \vec{k}\}$

$$f(\vec{i}) = \vec{a} \times \vec{i} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ a_1 & a_2 & a_3 \\ 1 & 0 & 0 \end{vmatrix} = a_3\vec{j} - a_2\vec{k}$$

$$f(\vec{j}) = \vec{a} \times \vec{j} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ a_1 & a_2 & a_3 \\ 0 & 1 & 0 \end{vmatrix} = -a_3\vec{i} + a_1\vec{k}$$

$$f(\vec{k}) = \vec{a} \times \vec{k} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ a_1 & a_2 & a_3 \\ 0 & 0 & 1 \end{vmatrix} = a_2\vec{i} - a_1\vec{j}$$

$$\Rightarrow M_{B_c}(f) = A = \begin{pmatrix} 0 & -a_3 & a_2 \\ a_3 & 0 & -a_1 \\ -a_2 & a_1 & 0 \end{pmatrix}$$

$$\Rightarrow P_A(\lambda) = \begin{vmatrix} -\lambda & -a_3 & a_2 \\ a_3 & -\lambda & -a_1 \\ -a_2 & a_1 & -\lambda \end{vmatrix} = -\lambda^3 - \lambda a_2^2 - \lambda a_1^2 - \lambda a_3^2$$

$$= -\lambda^3 - \lambda \|\vec{a}\|^2 = -\lambda(\lambda^2 + \|\vec{a}\|^2) \to P_A(\lambda)$$ nu are decât rădăcini reale $\Rightarrow f$ nu este diagonalizabil (peste $\mathbb{R}$).

$\delta(f) = \{0\}$

$V_{\lambda=0} = \{\vec{x} \in V_3 \mid f(\vec{x}) = \vec{0}\} = \{\vec{x} \in V_3 \mid \vec{a} \times \vec{x} = \vec{0}\}$ → Mulțimea vectorilor coliniari cu $\vec{a}$.

$= \{\alpha \cdot \vec{a} \mid \alpha \in \mathbb{R}\}$

(A) $\Leftrightarrow$ ~~T~~ $T^{-1}AT = J \overset{not}{=} \begin{pmatrix} 1 & & \\ & 1 & \\ & & 0_6 \end{pmatrix}$ (Ex 1)

$\uparrow$ not $M_{B_c}(f)$

unde în coloanele coord în $(B_c)_{\mathbb{R}^{3[X]}}$ ale vectorilor din $B_1 \cup B_2 \cup B_3$

Ex 3: Să se det termenul general al șirului $(x_n)_{n \geq 1}$ definit prin rel. de recurență:
$x_{n+2} = 15x_{n+1} - 56x_n$, $(\forall) n \geq 1$ și $x_1 = x_2 = 1$.

Sol: $X_n \overset{not}{=} \begin{pmatrix} x_n \\ x_{n-1} \end{pmatrix}$, $(\forall) n \geq 2$.

A4. $X_{n+2} = \begin{pmatrix} x_{n+2} \\ x_{n+1} \end{pmatrix} = \begin{pmatrix} 15x_{n+1} - 56x_n \\ x_{n+1} + 0 \cdot x_n \end{pmatrix} = A X_{n+1}$

, unde $A = \begin{pmatrix} 15 & -56 \\ 1 & 0 \end{pmatrix}$

$P_A(\lambda) = |15-\lambda)(-\lambda) + 56 = \lambda^2 - 15\lambda + 56$

$\lambda_{1,2} = \frac{15 \pm \sqrt{225 - 4 \cdot 56}}{2} = \frac{15 \pm 1}{2} \begin{cases} x_1 = 8 \\ x_2 = 7 \end{cases}$

$\Rightarrow P_A(\lambda) = (\lambda - 8)(\lambda - 7)$. A admite doar val proprii simple $\Rightarrow$

$\Rightarrow$ A diagonalizabilă $\Leftrightarrow (\exists) T \in \mathcal{M}_2(\mathbb{R})$ inversabilă a.î. $T^{-1}AT = D$

$\Leftrightarrow \boxed{A = TDT^{-1}}$

$V_{\lambda=7} = \left\{ \begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{R}^2 \mid A\begin{pmatrix} x \\ y \end{pmatrix} = 7\begin{pmatrix} x \\ y \end{pmatrix} \right\}$

$(A - 7I_2)\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$

$\begin{pmatrix} 15x - 56y \\ x \end{pmatrix} = \begin{pmatrix} 7x \\ 7y \end{pmatrix} \Rightarrow \begin{cases} 8x = 7y \\ 7x = x \end{cases}$

$\Rightarrow \boxed{x = 7y}$

$V_{\lambda=7} = Sp\left\{ \begin{pmatrix} 7 \\ 1 \end{pmatrix} \right\}$.

$V_{\lambda=8} = \left\{ \begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{R}^2 \mid A\begin{pmatrix} x \\ y \end{pmatrix} = 8\begin{pmatrix} x \\ y \end{pmatrix} \right\} \Leftarrow \begin{cases} 15x - 56y = 8x \\ x = 8y \end{cases} \Rightarrow x = 8y$

$\Rightarrow V_{\lambda=8} = Sp\left\{ \begin{pmatrix} 8 \\ 1 \end{pmatrix} \right\}$

Pt $T = \begin{pmatrix} 7 & 8 \\ 1 & 1 \end{pmatrix}$ are loc $A = TDT^{-1}$

$\boxed{X_n} = AX_{n-1} = A^2 X_{n-2} = \ldots = \boxed{A^{n-2} X_2}$

$A = TDT^{-1} \Rightarrow A^2 = (TDT^{-1})(TDT^{-1}) = TD(\underbrace{T^{-1}T}_{=I_2})DT^{-1} =$

$= TD^2T^{-1} \overset{\text{ind}}{\Rightarrow} A^n = TD^nT^{-1}$

Obs: Dacă $D = \begin{pmatrix} d_1 & 0 \\ 0 & d_2 \end{pmatrix} \in \mathcal{M}_2(\mathbb{R})$ și $d_1 \cdot d_2 \neq 0 \overset{\text{ind}}{\Rightarrow}$

$D^n = \begin{pmatrix} d_1^n & 0 \\ 0 & d_2^n \end{pmatrix}$

$T = \begin{pmatrix} 7 & 8 \\ 1 & 1 \end{pmatrix}$; $\det T = -1$; $A^* = \begin{pmatrix} 1 & -8 \\ -1 & 7 \end{pmatrix} \Rightarrow \text{inv} = \begin{pmatrix} -1 & 8 \\ 1 & -7 \end{pmatrix} = T^{-1}$

$\Rightarrow A^n = \begin{pmatrix} 7 & 8 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} 7^n & 0 \\ 0 & 8^n \end{pmatrix}\begin{pmatrix} -1 & 8 \\ 1 & -7 \end{pmatrix} = \begin{pmatrix} 7 & 8 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} -7^n & 8 \cdot 8^n \\ 7^n & -7 \cdot 8^n \end{pmatrix} =$

$= \begin{pmatrix} 8 \cdot 8^n - 7^{n+1} & 56 \cdot 8^n - 56 \cdot 7^n \\ 8^n - 7^n & 8 \cdot 8^n - 7^n \end{pmatrix}$

$$\begin{pmatrix} x_n \\ x_{n-1} \end{pmatrix} = \begin{pmatrix} 8^{n-1} - 7^{n-1} & 8\cdot 7^{n-1} - 7\cdot 8^{n-1} \\ 8^{n-2} - 7^{n-2} & 8\cdot 7^{n-2} - 7\cdot 8^{n-2} \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$\Leftrightarrow \begin{cases} x_n = 8^{n-1} - 7^{n-1} + 8\cdot 7^{n-1} - 7\cdot 8^{n-1} \\ x_{n-1} = \dots \end{cases}$$

$$\Leftrightarrow \boxed{x_n = -6\cdot 8^{n-1} + 7^n}$$

23.11.2021

EV1: Fie $\mathbb{K} \in \{\mathbb{R}, \mathbb{C}\}$, V un $\mathbb{K}$-sp vect cu $\dim_{\mathbb{K}} V < +\infty$, iar $f: V \to V$ endomorfism cu $f \neq 1_V$ (1_V), $f \neq 0$ (endomorfismul identic nul), $f^2 = f$ ($f \circ f$).

Să se determine $\sigma(f)$.

Obs $\lambda \in \sigma(f) \overset{\text{def}}{\iff} (\exists)\ v \in V \setminus \{0_V\}$ a.î. $f(v) = \lambda v \iff$

$\iff f(v) - \lambda v = 0_V \iff$

$\iff (f - \lambda 1_V)(v) = 0_V$

$\iff v \in \operatorname{Ker}[(f - \lambda 1_V)] \setminus \{0_V\}$

$\iff \lambda \in \sigma(f) \iff$ **$f - \lambda 1_V$ nu este izomorfism.**

Sol: Fie $\lambda \in \sigma(f) \iff \exists\, v \neq 0_V$, $v \in V$ a.î. $f(v) = \lambda v$

$\neq 0$ (!)

$\iff f(f(v)) = f(\lambda v) \iff$

$f^2 = f$

$\lambda v = f(v) = f^2(v) = f(f(v)) = \lambda^2 v$

$\iff \lambda^2 v = \lambda v \iff \lambda v(\lambda - 1) = 0_V$

$v \neq 0_V$

$\iff \lambda^2 - \lambda = 0 \iff \lambda \in \{0, 1\}$

Am arătat că $\sigma(f) \subseteq \{0, 1\}$

Să dem că $\{0, 1\} \subseteq \sigma(f)$

$0 \in \sigma(f) \overset{①}{\iff}$ f nu e izomorfism

Pres prin RA că f e izomorfism. Atunci $(\exists)\ f^{-1}$.

$f \circ f = f \mid \circ f^{-1}$

$f \circ_{-1} f = (f \circ f) \circ f^{-1}$

$1_V = f \circ f^{-1} = f \circ (f \circ f^{-1})$

$= f \circ 1_V = f$

$\Rightarrow$ Pres prin RA că f e izomorfism $\iff 0 \notin \sigma(f)$

$1 \in \sigma(f) \iff$ ① $f - 1_V$ nu este izo.

Pres prin RA că $f - 1_V$ izomorfism $\Rightarrow (\exists)(f - 1_V)^{-1}$.

$(f - 1_V)(v) = 0_V \iff (f - 1_V)^{-1}(0_V) = v$

$(f - 1_V)^{-1} \circ$

$(f - 1_V)(v) = (f - 1_V)^{-1}(0_V)$

$\Rightarrow 1_V(v) = 0_V$ ff ,,

$v \neq 0$

$1 \in \sigma(f) \iff$ ① $\underbrace{f - 1_V}_{\text{injj}}$ nu este izo.

g

Pres prin RA: g este izo $\Rightarrow (\exists) g^{-1}$.

$f(f(v)) = f((v)f)$

$(g)f = ((v)f)f$

$1-b \circ \lambda 0 = (v)($ $\underbrace{\lambda 1 - f}_{g}$ $) \circ g \iff \lambda 0 = (v) f - ((v) f) f$

$(v)(1_{-b} \circ \lambda 0) = (v)(1_{-b} \circ b \circ f) \Rightarrow$

$\forall v \in V$ (as)

ff $0 = f \iff \lambda 0 = (v) f$

Ex: Fie V un $\mathbb{K}$-sp vect cu $\dim_{\mathbb{K}} V = n$ și $f: V \to V$ endomorf. cu $f^2 = f$. Stabiliți dacă f diagonalizabil.

I Pres $f = 0$

$\lambda \in \sigma(f) \iff (\exists) v \in V$, $v \neq 0_V$ și $\underbrace{f(v)}_{0_V} = \lambda v$

$\lambda v = 0 \iff \boxed{\lambda = 0}$

$\sigma(f) = \{0\} \iff$

$V_{\lambda=0} \overset{\text{def}}{=} \{ v \in V \mid f(v) = 0_V \} = V \Rightarrow f$ diagonalizabil (evident dacă zic)

II Pres $f = 1_V$

$\lambda \in \sigma(f) \Leftrightarrow (\exists) v \in V \setminus \{0\}$ a.î. $\underbrace{f(v)}_{v} = \lambda v$

$\lambda v = v \iff \lambda = 1$

$\overline{1 = \lambda} \iff \begin{cases} v \neq 0 \\ 0 = v(1-\lambda) \end{cases}$

$\sigma(f) \subseteq \{1\}$

$0 = f - 1_V$ nu este izomorfism $\Rightarrow 1 \in \sigma(f)$ $\Big| \Rightarrow \sigma(f) = \{1\}$

$V_{\lambda=1} \overset{def}{=} \{v \in V \mid f(v) = v\} = V \Rightarrow f$ diagonalizabil

III $f \neq 1_V$, $f \neq 0$, $f^2 = f$. Am dem la curs că $\sigma(f) = \{0,1\}$

$V_{\lambda=0} = \{v \in V \mid f(v) = 0_V\} = \ker f$

$V_{\lambda=1} = \{v \in V \mid f(v) = v\} \subseteq \operatorname{Im} f$! ①

"$\supseteq$" ? Fie $w \in \operatorname{Im} f$ $\Rightarrow$ exista $v \in V$ a.i. $w = f(v)$

$(\exists)\ v \in V$ a.i. $f(v) = w \overset{f}{\Rightarrow}$

$\Rightarrow f(f(v)) = f(w)$

$f^2(v) = f(w)$

$\Big| \Rightarrow w = f(w)$

idem $f^2(v) = f(v) = w$ $\Big| \Rightarrow \operatorname{Im} f \subseteq V_{\lambda=1}$ ②

①② $\Rightarrow V_{\lambda=1} = \operatorname{Im} f$

Să dem că $\ker f + \operatorname{Im} f = \ker f \oplus \operatorname{Im} f$

Fie $v \in \ker f \cap \operatorname{Im} f \Rightarrow \begin{cases} v \in \ker f \Leftrightarrow f(v) = 0_V \\ v \in \operatorname{Im} f \Leftrightarrow (\exists) w \in V \text{ a.i. } f(w) = v \end{cases}$

$\Rightarrow f^2(w) = f(v)$

$f(w) = 0_V$

$\Rightarrow v = 0_V$

$\Rightarrow \ker f + \operatorname{Im} f = \ker f \oplus \operatorname{Im} f$ ③

Trb dem $\ker f \oplus \operatorname{Im} f = V$

$\dim_{\mathbb{K}} V = n < +\infty \Rightarrow \dim_{\mathbb{K}} V = \dim_{\mathbb{K}} \ker f + \dim_{\mathbb{K}} \operatorname{Im} f$

$= \dim_{\mathbb{K}} (\ker f + \operatorname{Im} f) + \underbrace{\dim_{\mathbb{K}} (\ker f \cap \operatorname{Im} f)}_{0}$

$\Rightarrow \dim_{\mathbb{K}} V = \dim_{\mathbb{K}} (\ker f + \operatorname{Im} f)$

$\Rightarrow V = \ker f + \operatorname{Im} f$ ④

③④ $\Rightarrow V = \ker f \oplus \operatorname{Im} f$

Ca f diagonalizabil este nec și suficient să existe o bază în V a.i. $V_{\lambda=0} \oplus V_{\lambda=1} = V$

$\overset{?}{\Rightarrow} \ker f \oplus \operatorname{Im} f = V$

$\Rightarrow$" Fie $v \in V$. Atunci $f(v) \in \operatorname{Im} f$ și $v = v - f(v) + \underbrace{f(v)}_{\in \operatorname{Im} f}$ (3)

$V \subseteq \operatorname{Ker} f \oplus \operatorname{Im} f$; $f(v - f(v)) = f(v) - f^2(v) = f(v) - f(v) = 0_V$,

$\Rightarrow v - f(v) \in \operatorname{Ker} f$. (4)

Din (2) și (4) $\Rightarrow V \subseteq \operatorname{Ker} f \oplus \operatorname{Im} f \subseteq V$

$\Rightarrow \operatorname{Ker} f \oplus \operatorname{Im} f = V$

$\Rightarrow V_{\lambda=0} \oplus V_{\lambda=1} = V \Leftrightarrow f$ diagonalizabil

07.12.2022

Spații vectoriale
reamintire cu produs scalar

Fie $\mathbb{K} \in \{\mathbb{R}, \mathbb{C}\}$.

Def Fie V un $\mathbb{K}$-sp vect. S.n. produs scalar pe V o funcție

$\langle \cdot, \cdot \rangle : V \times V \to \mathbb{K}$ a.î.:

i) $\langle v, v \rangle \geq 0, (\forall) v \in V$ și
$\langle v, v \rangle = 0_{\mathbb{K}} (=0) \Leftrightarrow v = 0_V$

ii) $\langle u, v \rangle = \overline{\langle v, u \rangle}$, $(\forall) u, v \in V$

iii) $\langle \alpha u + \beta v, w \rangle = \alpha \langle u, w \rangle + \beta \langle v, w \rangle$
$(\forall) \alpha, \beta \in \mathbb{K}$ și $(\forall) u, v, w \in V$.

Obs:

1°) $\langle u, \alpha v + \beta w \rangle \overset{ii)}{=} \overline{\langle \alpha v + \beta w, u \rangle} \overset{iii)}{=} \overline{\alpha \langle v, u \rangle + \beta \langle w, u \rangle}$
($\alpha, \beta \in \mathbb{K}$ și $u, v, w \in V$)
$= \bar{\alpha} \cdot \overline{\langle v, u \rangle} + \bar{\beta} \cdot \overline{\langle w, u \rangle} = \bar{\alpha} \langle u, v \rangle + \bar{\beta} \langle u, w \rangle$
($\Rightarrow$ pt $\mathbb{K} = \mathbb{R}$ este liniar și în al doilea argument)

2°) Dacă $v \in V$, at $\boxed{\langle v, 0_V \rangle = \langle 0_V, v \rangle = 0}$

Dem: $\langle 0_V, v \rangle = \langle v - v, v \rangle = \langle v, v \rangle - \langle v, v \rangle = 0$

$(V, \langle , \rangle)$ sn **SPAȚIU PREHILBERTIAN.**

Dacă $(V, \langle , \rangle)$ sp. prehilb și $\dim_{\mathbb{R}} V < +\infty$, at V o.n. **SPAȚIU EUCLIDIAN.**

Exemple:

1) $V = \mathbb{R}^n$, $\mathbb{K} = \mathbb{R}$

$\langle \underbrace{(x_1, \dots, x_n)}_{\in \mathbb{R}^n}, \underbrace{(y_1, \dots, y_n)}_{\in \mathbb{R}^n} \rangle \overset{def}{=} x_1 y_1 + x_2 y_2 + \dots + x_n y_n = \sum_{j=1}^{n} x_j y_j$

s.n. produs scalar **standard** (**canonic**) (**euclidian**)

2) $V = \mathbb{C}^n$, $\mathbb{K} = \mathbb{C}$; $\langle (x_1, \dots, x_n), (y_1, \dots, y_n) \rangle \overset{def}{=} \sum_{j=1}^{n} x_j \overline{y_j}$

s.n. produs scalar **STANDARD / CANONIC / EUCLIDIAN** pe $\mathbb{C}^n$.

3) $V = V_3$, $\mathbb{K} = \mathbb{R}$

$$\langle \vec{x}, \vec{y} \rangle \overset{\text{def}}{=} \begin{cases} 0, \text{ dacă } \vec{x} = \vec{0} \text{ sau } \vec{y} = \vec{0} \\ \|\vec{x}\| \cdot \|\vec{y}\| \cdot \cos(\vec{x}, \vec{y}), \text{ altfel} \end{cases}$$

4) $V = \mathcal{M}_n(\mathbb{R})$, $\mathbb{K} = \mathbb{R}$

$$\langle A, B \rangle \overset{\text{def}}{=} \operatorname{Tr}({}^tA \cdot B)$$

5) $V = \{ f : [a,b] \to \mathbb{R} \mid f \text{ cont} \}$, $\mathbb{K} = \mathbb{R}$ un mod. scalar este.

$$\langle f, g \rangle = \int_a^b f(x) g(x)\, dx.$$

6) $V = \{ f : [a,b] \to \mathbb{C} \mid f \text{ cont} \}$, $\mathbb{K} = \mathbb{C}$ un mod scalar este

$$\langle f, g \rangle = \int_a^b f(x) \overline{g(x)}\, dx$$

7) $V = \mathbb{R}_n[X]$, $\mathbb{K} = \mathbb{R}$, p.s. STANDARD:

$$\langle a_0 + a_1 X + \dots + a_n X^n, b_0 + b_1 X + \dots + b_n X^n \rangle \overset{\text{def}}{=} \sum_{j=0}^{n} a_j b_j$$

Def: Fie V un $\mathbb{K}$-sp vect. O ap $\| \cdot \| : V \to \mathbb{R}$ s.n. **NORMĂ** pe V dacă:

i) $\|v\| \geq 0$, $(\forall) v \in V$ și $\|v\| = 0 \Leftrightarrow v = 0_V$

ii) $\|\alpha v\| = |\alpha| \cdot \|v\|$

$(\forall) \alpha \in \mathbb{K}, (\forall) v \in V$

iii) $\|u + v\| \leq \|u\| + \|v\|$, $(\forall) u, v \in V$.

Ineg. **SCHWARZ**

$(V, \langle , \rangle)$ sp. prehilbertian.

Atunci $(\forall) x, y \in V$:

$$\boxed{|\langle x, y \rangle| \leq \sqrt{\langle x, x \rangle} \cdot \sqrt{\langle y, y \rangle}}$$

I Dacă $x = 0_V \Rightarrow |\langle x, y \rangle| = \sqrt{\langle x, x \rangle} \cdot \sqrt{\langle y, y \rangle}$

II Dacă $y = 0_V \Rightarrow |\langle x, y \rangle| = \sqrt{\langle x, x \rangle} \cdot \sqrt{\langle y, y \rangle}$

III Fie $x \in V$ și $y \in V \setminus \{0_V\}$.

Not $t = \dfrac{\langle x, y \rangle}{\langle y, y \rangle}$ $(\in \mathbb{K})$ $\qquad \overline{\langle x, y \rangle} \cdot \langle x, y \rangle = |\langle x, y \rangle|^2$

$$0 \leq \langle x - ty, x - ty \rangle = \langle x, x \rangle - \bar{t} \langle x, y \rangle - t \langle y, x \rangle + t \cdot \bar{t} \langle y, y \rangle =$$

$$= \langle x, x \rangle - \frac{|\langle x, y \rangle|^2}{\langle y, y \rangle} - \frac{|\langle x, y \rangle|^2}{\langle y, y \rangle} + \frac{|\langle x, y \rangle|^2}{\langle y, y \rangle} \quad (*)$$

$\langle x,x\rangle \geq \frac{|\langle x,y\rangle|^2}{\langle y,y\rangle}$ (2) $|\langle x,y\rangle| \leq \sqrt{\langle x,x\rangle}\cdot\sqrt{\langle y,y\rangle}$

Dacă în (*) avem $\langle x,x\rangle - \frac{|\langle x,y\rangle|^2}{\langle y,y\rangle} = 0 \Rightarrow$

$0 \leq \langle x-ty, x-ty\rangle \leq 0 \Rightarrow \langle x-ty, x-ty\rangle = 0$

$\Rightarrow \boxed{x = ty}$

Prop 1: Fie $(V, \langle,\rangle)$ un sp. prehilb. și

$\|\cdot\| : V \to \mathbb{R}$

$\|v\| \overset{def}{=} \sqrt{\langle v,v\rangle}$

Atunci $\|\cdot\|$ este o normă pe V.

Dem: i) $\|v\| \geq 0, \forall v \in V$

$\|v\| = 0 \Leftrightarrow \langle v,v\rangle = 0 \Leftrightarrow v = 0$

Fie $\alpha \in \mathbb{K}$ și $v \in V$.

ii) $\|\alpha v\| = \sqrt{\langle \alpha v, \alpha v\rangle} = \sqrt{\underbrace{\alpha\cdot\bar{\alpha}}_{|\alpha|^2}\langle v,v\rangle}$

$= |\alpha|\cdot\sqrt{\langle v,v\rangle} = |\alpha|\cdot\|v\|$

iii) $u, v \in V$. $\|u+v\| \leq \|u\| + \|v\|$

$\|u+v\|^2 = \langle u+v, u+v\rangle = \langle u,u\rangle + \langle u,v\rangle + \underbrace{\langle v,u\rangle}_{\overline{\langle u,v\rangle}} + \langle v,v\rangle$

$= \|u\|^2 + \|v\|^2 + 2\,\mathrm{Re}\langle u,v\rangle \overset{obs}{\leq}$ (*)

Obs: Dacă $z \in \mathbb{C}$ atunci $z + \bar{z} = 2(\mathrm{Re}\, z) \leq 2|z|$

$z = a + ib$ ↑

$a \leq \sqrt{a^2+b^2}$

(*) $\leq \|u\|^2 + \|v\|^2 + 2|\langle u,v\rangle| \overset{\text{Inegalitatea Schw.}}{\leq} \|u\|^2 + \|v\|^2 + 2\underbrace{\sqrt{\langle u,u\rangle}}_{\|u\|}\cdot\underbrace{\sqrt{\langle v,v\rangle}}_{\|v\|}$

$= (\|u\| + \|v\|)^2 \Rightarrow$

$\|u+v\| \leq \|u\| + \|v\|$

Def. Fie $(V, \langle,\rangle)$ un sp. prehilb.

a) Dacă $x, y \in V$ a.î. $\langle x,y\rangle = 0$, atunci x și y $\perp$.

b) Dacă $U, W \subseteq V$, spunem că U și W sunt $\perp$ $(U \perp W)$

$\Leftrightarrow (\forall) u \in U, (\forall) w \in W$ avem $\langle u, w\rangle = 0$

c) Dacă $\dim_{\mathbb{R}} V = n < +\infty$ și $B = \{v_1, \dots, v_n\} \subseteq V$ o bază. Atunci B

s.n.: bază ortogonală (OG) dacă $\langle v_i, v_j \rangle = 0$, $(\forall) i \neq j$, $i, j \in \overline{1,n}$

↳ bază ortonormată (ORTONORMALĂ) (ON) dacă este o bază formată din vectori ($\|v_i\| = 1$, $(\forall) i = \overline{1,n}$)

$\|v_i\| = 1$ def $\sqrt{\langle v_i, v_i \rangle}$

Prop 2: Fie $(V, \langle , \rangle)$ un sp. prehilbertian cu $\dim_{\mathbb{R}} V = n < +\infty$ și $\{v_1, \dots, v_n\}$ un sist de vectori nenuli și ortogonali, $(\forall) i, j \in \{1, \dots, n\}$, $i \neq j$.

At. $\{v_1, \dots, v_n\}$ este o bază pt V.

Dem: Fie $\alpha_1, \dots, \alpha_n \in \mathbb{K}$ a.î.

$\alpha_1 v_1 + \alpha_2 v_2 + \dots + \alpha_n v_n = 0_V \mid \cdot v_j$

Fie $j \in \overline{1,n}$

$\Rightarrow \alpha_j \langle v_j, v_j \rangle = 0$

$0 < \| v_j \| \neq 0_V$ $\Rightarrow \alpha_j = 0$

j arbitrar ales $\Rightarrow \alpha_j = 0$, $(\forall) j \in [1,n]$

$\Rightarrow \{v_1, \dots, v_n\}$ LI, dar $\dim_{\mathbb{R}} V = n$ $\Rightarrow \{v_1, v_2, \dots, v_n\}$ bază pt V.

Def: Fie $(V, \langle , \rangle)$ sp prehilb., $y \in V \setminus \{0_V\}$ și $x \in V$ s.n.

proiecția lui x pe y vectorul

$$pr_y x \overset{\text{def}}{=} \frac{\langle x, y \rangle}{\langle y, y \rangle} y.$$

Obs: $\boxed{x - pr_y x \perp pr_y x}$!!

$$\langle x - pr_y x, pr_y x \rangle = \left\langle x - \frac{\langle x, y \rangle}{\langle y, y \rangle} y, \frac{\langle x, y \rangle}{\langle y, y \rangle} y \right\rangle =$$

$$= \frac{\overline{\langle x, y \rangle}}{\langle y, y \rangle} \cdot \langle x, y \rangle - \frac{|\langle x, y \rangle|^2}{\langle y, y \rangle^2} \cdot \langle y, y \rangle = 0$$

B

ALGORITMUL GRAM-SCHMIDT.

$(V, \langle , \rangle)$ un sp. prehilb și $\{a_1, \dots, a_n\} \subseteq V$ un sli.

Definim $v_1 = a_1$

$v_2 = a_2 - pr_{v_1} a_2$

$v_3 = a_3 - pr_{v_1} a_3 - pr_{v_2} a_3$

...

$v_j = a_j - \sum_{k=1}^{j-1} pr_{v_k} a_j$ $\Big/\Big/$ $a_j - \sum_{k=1}^{j-1} \frac{\langle a_j, v_k \rangle}{\langle v_k, v_k \rangle} \cdot v_k$

$2 \le j \le n$. Not. $w_j = \frac{v_j}{\|v_j\|}$, $j = \overline{1,n}$

Atunci: a) $\{w_1, \dots, w_n\}$ este un sli ortogonal

b) $(\forall)\ 1 \le j \le n$, $sp\{a_1, \dots, a_j\} = sp\{w_1, \dots, w_j\}$ (ind)

c) $(\forall)\ 1 \le j \le n$, $\langle w_j, a_j \rangle = \|v_j\|$

Ex: Fie $f_1, f_2, f_3 : [-1,1] \to \mathbb{R}$, $f_1(x) = 1$, $f_2(x) = x$, $f_3(x) = x^2$ } jocă rolul lui a, iar g joacă rolul lui v

și $V = sp\{f_1, f_2, f_3\}$.

$V \underset{\text{subsp vect}}{\subset} \{f \mid f : [-1,1] \to \mathbb{R},\ f \text{ cont}\}$ cu produs scalar $\langle f, g \rangle = \int_{-1}^{1} f(x) g(x)\, dx$.

Să se det. o bază ON pt V. (Ex)

Obs: într-un sp. prehilb. finit dim, ($\exists$) cel puțin o bază ON.

Teoremă (complement ORTOGONAL)

Teoremă: Fie $(V, \langle , \rangle)$ un sp. prehilb cu $\dim_{\mathbb{R}} V = n$ și $U \underset{\text{subsp vect}}{\subseteq} V$ a.î. $\dim_{\mathbb{R}} U = m < n$. Atunci ($\exists !$) un subsp $U^{\perp} \subseteq V$ a.î. $U \perp U^{\perp}$ și $U \oplus U^{\perp} = V$

cu $\dim_{\mathbb{R}} U^{\perp} = n - m$ (GRASSMANN)

și $U^{\perp} = \{v \in V \mid v \perp u, (\forall) u \in U\}$

Dem: Fie $B_1 = \{a_1, a_2, \dots, a_m\} \subseteq U$ o bază a lui U. Compl. B_1 până la o bază $B = \{a_1, a_2, \dots, a_m, a_{m+1}, \dots, a_n\}$ a lui V.

Ap. bazei B proc. de ortogon. GRAM-SCHMIDT, obținem $B' = \{v_1, v_2, \dots, v_n\}$ bază ortogonală pt V.

Fie $U^\perp = sp\{v_{m+1}, \dots, v_n\}$. Vom arăta că acest spațiu are prop. cerute.

- $\underbrace{U + U^\perp}_{(U = sp\{a_1, \dots a_m\} \overset{IGS}{=} sp\{v_1, \dots, v_m\})} \overset{\supseteq}{=} sp\{v_1, \dots, v_m, v_{m+1}, \dots, v_n\} = V$
- Să arătăm că $U + U^\perp = U \oplus U^\perp$

Fie $v \in U \cap U^\perp$ a.î. $v \in U$
$v \in U^\perp$ (U ortogonal) $\Rightarrow$

$\Rightarrow \exists!\ \alpha_1, \alpha_2, \dots, \alpha_m, \dots, \alpha_n \in \mathbb{K}$ a.î.

$(\star)\ v = \alpha_1 v_1 + \dots + \alpha_m v_m = \alpha_{m+1} v_{m+1} + \dots + \alpha_n v_n$

$\Rightarrow \alpha_1 v_1 + \dots + \alpha_m v_m - \alpha_{m+1} v_{m+1} - \dots - \alpha_n v_n = 0$ /$\Leftarrow$
$\{v_1, v_2, \dots, v_m, \dots, v_n\}$ LI

$\Rightarrow \boxed{\alpha_j = 0, (\forall)\ j \in \overline{1,n}}$

$\Rightarrow v = 0_V \Rightarrow U + U^\perp = U \oplus U^\perp$.

$\star$ I: Ar că $\boxed{U \perp U^\perp}$ (adică $\langle u, v \rangle = 0$, $(\forall)\ u \in U$ și $v \in U^\perp$)

ALGA=D-S10

07.12.2022.

1. Pe $\mathbb{R}^3$, ne cons p.s standard.

Fie $B = \{f_1 = (\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}), f_2 = (\frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}, -\frac{2}{\sqrt{6}}),$
$f_3 = (\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}, 0)\} \subseteq \mathbb{R}^3$. Si $x = (2,3,4) \in \mathbb{R}^3$.

Să se arate că B este o bază ON pt $\mathbb{R}^3$ relativ la p.s. standard și apoi să se det coord lui x rel. la baza B.

Obs Fie $(V, \langle , \rangle)$ sp. prehilbertian și $\{v_1, \dots, v_n\} \subseteq V \setminus \{0\}$, iar $\alpha_1, \alpha_2, \dots, \alpha_n \in \mathbb{K}^*$. Atunci $\{v_1, \dots, v_n\}$ sist. ort. $\Leftrightarrow$ $\{\alpha_1 v_1, \alpha_2 v_2, \dots, \alpha_n v_n\}$ sist. ort.

$f_1 = \frac{1}{\sqrt{3}}(\overbrace{1,1,1}^{v_1})$; $f_2 = \frac{1}{\sqrt{6}}(\overbrace{1,1,-2}^{v_2})$; $f_3 = \frac{1}{\sqrt{2}}(\overbrace{1,-1,0}^{v_3})$

$\langle v_1, v_2 \rangle = 1 \cdot 1 + 1 \cdot 1 + 1 \cdot (-2) = 0$
$\langle v_1, v_3 \rangle = 1 \cdot 1 + 1 \cdot (-1) + 1 \cdot 0 = 0$
$\langle v_2, v_3 \rangle = 1 \cdot 1 + 1 \cdot (-1) + (-2) \cdot 0 = 0$
$\Big\} \xrightarrow{\text{obs}} \langle f_i, f_j \rangle = 0$ dacă $i \neq j$, $i, j \in \{1,2,3\}$

$\|f_1\| = \sqrt{\langle f_1, f_1 \rangle}$
$\|\frac{1}{\sqrt{3}} v_1\| = \frac{1}{\sqrt{3}} \|v_1\| = \frac{1}{\sqrt{3}} \sqrt{\langle v_1, v_1 \rangle} = \frac{1}{\sqrt{3}} \sqrt{1^2+1^2+1^2} = \frac{\sqrt{3}}{\sqrt{3}} = 1$

$\|f_2\| = \frac{1}{\sqrt{6}} \|v_2\| = \frac{1}{\sqrt{6}} \sqrt{1^2+1^2+(-2)^2} = 1$

$\|f_3\| = \frac{1}{\sqrt{2}} \|v_3\| = \frac{1}{\sqrt{2}} \sqrt{1^2+(-1)^2+0} = 1$

$\Rightarrow$ B este un sistem ortonormat de vectori.

B OG $\Rightarrow$ B SLI
Card $B = 3 = \dim_{\mathbb{R}} \mathbb{R}^3 = 3$ $\Big| \Rightarrow$ B este o bază pt $\mathbb{R}^3$ $\Big| \Rightarrow$
B ON

B este o bază ortonormată pt $\mathbb{R}^3$.

$\Downarrow$

$\exists!\ a, b, c \in \mathbb{R}$: $x = a f_1 + b f_2 + c f_3$ / $\cdot f_1$

$\Rightarrow \langle x, f_1 \rangle = a \langle f_1, f_1 \rangle + b \langle f_2, f_1 \rangle + c \langle f_3, f_1 \rangle$
$\langle f_1, f_1 \rangle = \|f_1\|^2 = 1$; $\langle f_2, f_1 \rangle = 0$; $\langle f_3, f_1 \rangle = 0$

$\Rightarrow a = \langle x, f_1 \rangle = 3\sqrt{3}$

La fel $\Rightarrow b = \langle x, f_2 \rangle = \frac{1}{\sqrt{6}}(-3) = -\frac{3}{\sqrt{6}}$

$\Rightarrow c = \langle x, f_3 \rangle = \frac{1}{\sqrt{2}}(-1) = -\frac{1}{\sqrt{2}}$

$\Rightarrow [x]_{\mathcal{B}} = \begin{pmatrix} \frac{3\sqrt{2}}{2} \\ -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} \end{pmatrix}$

Ex2: Pe sp. vect real $\mathcal{C}([-1,1]) = \{f \mid f: [-1,1] \to \mathbb{R}, f \text{ cont}\}$ considerăm pe ps $\langle f|g \rangle \overset{\text{def}}{=} \int_{-1}^{1} f(x)g(x)\,dx$.

Fie $V = sp\{f_1, f_2, f_3\} \subseteq \mathcal{C}([-1,1])$

unde $f_1(x) = 1$, $f_2(x) = x$, $f_3(x) = x^2$ $(\forall)\, x \in [-1,1]$.

Să se det o bază ortonormată a lui V relativ la ps considerat.

$? \{f_1, f_2, f_3\}$ SLi.

Fie $\alpha, \beta, \gamma \in \mathbb{R}$ a.î. $\alpha f_1 + \beta f_2 + \gamma f_3 = 0$. Atunci

$\alpha + \beta x + \gamma x^2 = 0$, $(\forall)\, x \in [-1,1]$

$x = 0 \Rightarrow \alpha = 0$

$x = 1 \Rightarrow \beta + \gamma = 0$

$x = -1 \Rightarrow -\beta + \gamma = 0$ $\Big| \Rightarrow \beta = \gamma = 0$ $\Big] \Rightarrow$

$\Rightarrow \alpha = \beta = \gamma = 0 = \lambda \{f_1, f_2, f_3\}$ SLi

Cum $\{f_1, f_2, f_3\}$ SG pt V $\Big] \Rightarrow$

$\Rightarrow \{f_1, f_2, f_3\}$ o bază pt V.

Fol. alg. GRAM-SCHMIDT, vom constr. o bază ON pornind de la $\{f_1, f_2, f_3\}$, $\{g_1, g_2, g_3\}$ pt V.

$g_1 \overset{\text{def}}{=} f_1 \Rightarrow g_1(x) = 1$, $(\forall) x \in [-1,1]$

$g_2 = f_2 - \frac{\langle f_2, g_1 \rangle}{\langle g_1, g_1 \rangle} g_1 \Rightarrow g_2(x) = x - \frac{\int_{-1}^{1} x\,dx}{\int_{-1}^{1} dx} = x$

$g_3 = f_3 - \frac{\langle f_3, g_1 \rangle}{\langle g_1, g_1 \rangle} g_1 - \frac{\langle f_3, g_2 \rangle}{\langle g_2, g_2 \rangle} g_2$

$= x^2 - \frac{\int_{-1}^{1} x^2\,dx}{\int_{-1}^{1} dx} \cdot 1 - \frac{\int_{-1}^{1} x^3\,dx = 0}{\int_{-1}^{1} x^2\,dx} \cdot x$

$= x^2 - \frac{\frac{x^3}{3}\Big|_{-1}^{1}}{x\Big|_{-1}^{1}} = x^2 - \frac{\frac{1}{3}(1+1)}{(1+1)} = x^2 - \frac{1}{3}$

$\Rightarrow$ o bază OG pt $V = \{g_1, g_2, g_3\}$

o bază ON pt V va fi $\left\{ \frac{g_1}{\|g_1\|}, \frac{g_2}{\|g_2\|}, \frac{g_3}{\|g_3\|} \right\}$

$\|g_1\|^2 = \langle g_1, g_1\rangle = \int_{-1}^{1} dx = 2$

$\|g_2\|^2 = \langle g_2, g_2\rangle = \int_{-1}^{1} x^2\, dx = \frac{2}{3}$.

$\|g_3\|^2 = \langle g_3, g_3\rangle = \int_{-1}^{1} \left(x^2 - \frac{1}{3}\right)^2 dx = \int_{-1}^{1} \left(x^4 - \frac{2}{3}x^2 + \frac{1}{9}\right) dx$

$= \frac{1}{5}(1+1) - \frac{2}{3}\cdot\frac{1}{3}(1+1) + \frac{1}{9}(1+1)$

$= 2\left(\frac{1}{5} - \frac{5}{9} + \frac{1}{9}\right) = 2\cdot\frac{9-5}{45} = 2\cdot\frac{4}{45} = \frac{8}{45}$.

! $\|g_1\| = \sqrt{\langle g_1, g_1\rangle}$

O bază ON pt V: $\left\{\frac{1}{\sqrt{2}} g_1, \sqrt{\frac{3}{2}}\cdot\sqrt{\frac{45}{8}} g_3\right\}$

$= \left\{\frac{1}{\sqrt{2}}, \sqrt{\frac{3}{2}}x, \sqrt{\frac{45}{8}}\left(x^2 - \frac{1}{3}\right)\right\}$

Ex 3: Pe $\mathbb{R}^4$ se cons. ps. standard și vect.

$a_1 = (1,0,-1,0)$, $a_2 = (0,5,-5,1)$, $a_3 = (2,0,-2,2)$ și $a_4 = (1,-2,1,3)$

Să se det. o bază ON pt $V = \mathrm{sp}\{a_1, a_2, a_3, a_4\}$ față de ps considerat.

? $\{a_1, a_2, a_3, a_4\}$ sci.

$$\begin{pmatrix} a_1 & a_2 & a_3 & a_4 \\ 1 & 0 & 2 & 1 \\ 0 & 5 & 0 & -2 \\ -1 & -5 & -2 & 1 \\ 0 & 1 & 2 & 3 \end{pmatrix} \xrightarrow{L_3 \to L_3 + L_1} \begin{pmatrix} 1 & 0 & 2 & 1 \\ 0 & 5 & 0 & -2 \\ 0 & -5 & 0 & 2 \\ 0 & 1 & 2 & 3 \end{pmatrix} \begin{matrix} L_3 \to L_3 + L_2 \\ \to \\ L_4 \to L_4 + \frac{1}{5}L_2 \end{matrix}$$

$$\to \begin{pmatrix} 1 & 0 & 2 & 1 \\ 0 & 5 & 0 & -2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \to \begin{pmatrix} 1 & 0 & 2 & 1 \\ 0 & 1 & 2 & 3 \\ 0 & -5 & 0 & 2 \\ 0 & 5 & 0 & -2 \end{pmatrix} \xrightarrow{L_4 \to L_4 + L_3} \begin{pmatrix} 1 & 0 & 2 & 1 \\ 0 & 1 & 2 & 3 \\ 0 & -5 & 0 & 2 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$L_3 \to L_3 + 5L_2$

$$\to \begin{pmatrix} \textcircled{1} & 0 & 2 & 1 \\ 0 & \textcircled{1} & 2 & 3 \\ 0 & 0 & \textcircled{10} & 17 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$\Rightarrow$ vect coresp $\{a_1, a_2, a_3\}$ coresp. col. cu pivot form o bază a lui V.

Ap. GRAM-SCHMIDT. Vom constr. o bază o.G. pornind de la $\{a_1, a_2, a_3\}$, $\{b_1, b_2, b_3\}$ pt V.

$b_1 \overset{\text{def}}{=} a_1 \Rightarrow b_1 = (1,0,-1,0)$

$$b_2 = a_2 - \frac{\langle a_2, b_1\rangle}{\langle b_1, b_1\rangle}\cdot b_1 = (0,5,-5,1) - \frac{0\cdot 1 + 0\cdot 5 + (-1)\cdot(-5) + 0\cdot 1}{1\cdot 1 + 0\cdot 0 + (-1)\cdot(-1) + 0\cdot 0}(1,0,-1,0)$$

$$= (0,5,-5,1) - \frac{5}{2}(1,0,-1,0)$$

$$= \left(-\frac{5}{2}, 5, -5 + \frac{5}{2}, 1\right)$$

$$= \left(-\frac{5}{2}, 5, -\frac{5}{2}, 1\right) = \frac{1}{2}(-5, 10, -5, 2)$$

$b_3 = a_3 - \frac{\langle a_3, b_1\rangle}{\langle b_1, b_1\rangle} \cdot b_1 - \frac{\langle a_3, b_2\rangle}{\langle b_2, b_2\rangle} \cdot b_2$

$= (2,0,-2,2) - \frac{4}{2}(1,0,-1,0) - \frac{2}{\frac{154}{4}}\left(-\frac{5}{2}, 5, -\frac{5}{2}, 1\right)$

$\langle a_3, b_1\rangle = 2\cdot 1 + 0\cdot 0 + (-2)\cdot(-1) + 0\cdot 2 = 4$

$\langle a_3, b_2\rangle = 2\cdot\left(-\frac{5}{2}\right) + 0 + (-2)\cdot\left(+\frac{5}{2}\right) + 2\cdot 1 = 2$

$\langle b_2, b_2\rangle = \left(+\frac{5}{2}\right)\cdot\left(+\frac{5}{2}\right) + 5\cdot 5 + \left(+\frac{5}{2}\right)\cdot\left(+\frac{5}{2}\right) + 1\cdot 1$

$= \frac{25}{4} + 25 + \frac{25}{4} + 1 = \frac{154}{4} = \frac{77}{2}$

$= (2,0,-2,2) - (2,0,-2,0) - \frac{4^2}{77}\cdot\frac{1}{2}(-5,10,-5,2)$

$= (0,0,0,2) - \frac{2}{77}(-5,10,-5,2)$

$= \frac{2}{77}\left((0,0,0,77) - (-5,10,-5,2)\right)$

$= \frac{2}{77}(5,-10,5,75) = \frac{10}{77}(1,-2,1,15)$

2) o bază OG pt V = ~~$\{b_1, b_2, b_3\}$~~ $\{(1,0,-1,0), (-5,10,-5,2), (1,-2,1,15)\}$

o bază ON pt V va fi $\left\{\frac{b_1}{\|b_1\|}, \frac{b_2}{\|b_2\|}, \frac{b_3}{\|b_3\|}\right\}$.

~~$\|b_1\|^2 = \langle b_1, b_1\rangle = 2$~~

$\left\{\frac{w_1}{\|w_1\|}, \frac{w_2}{\|w_2\|}, \frac{w_3}{\|w_3\|}\right\}$

$\|w_1\| = \sqrt{2}$

$\|w_2\| = \sqrt{25+100+25+4} = \sqrt{154}$

$\|w_3\| = \sqrt{1+4+1+225} = \sqrt{231}$

=) -

- 2) $\left\{\frac{1}{\sqrt{2}}(1,0,-1,0), \frac{1}{\sqrt{154}}(-5,10,-5,2), \frac{1}{\sqrt{231}}(1,-2,1,15)\right\}$

Exu: Pe $\mathbb{R}_3[X]$ se consideră p.s. standard

$\langle a_0+a_1X+a_2X^2+a_3X^3, b_0+b_1X+b_2X^2+b_3X^3\rangle \overset{\text{def}}{=\!=} a_0b_0+a_1b_1+a_2b_2+a_3b_3.$

Fie $V = \{P \in \mathbb{R}_3[X] \mid P(2) = 0\} \underset{\text{subsp vect}}{\subseteq} \mathbb{R}_3[X]$

Să se det o bază pt $V^\perp$ și să se găsească $pr_V(x^3-20x^2-5x-3)$

Sol: Să det. o bază pt V.

$\forall P \in V$. Din $P(2)=0 \Rightarrow (X-2) \mid P \Leftrightarrow$

$P=(X-2)Q$, $Q \in \mathbb{R}_2[X]$

$\Leftrightarrow P=(X-2)(a_0+a_1X+a_2X^2) = a_0\underbrace{(X-2)}_{P_1} + a_1\underbrace{(X^2-2X)}_{P_2} + a_2\underbrace{(X^3-2X^2)}_{P_3}$

$\Rightarrow \{P_1,P_2,P_3\}$ SG pt V.

$\{P_1,P_2,P_3\}$ SLi (coef. pol. de grade 2 câte 2 dif) $\Rightarrow$

$\Rightarrow \{P_1,P_2,P_3\}$ o bază pt V. $\Rightarrow \dim_{\mathbb{R}} V=3 \Rightarrow \boxed{\dim_{\mathbb{R}} V^\perp = 1.}$

$V^\perp = \{P \in \mathbb{R}_3[X] \mid P \perp Q, \forall Q \in V\} = \{P \in \mathbb{R}_3[X] \mid P \perp P_j, j \in \{1,2,3\}\}$

$P = a_0+a_1X+a_2X^2+a_3X^3$

$= \{a_0+a_1X+a_2X^2+a_3X^3 \in \mathbb{R}_3[X] \mid -2a_0+a_1=0, -2a_1+a_2=0, -2a_2+a_3=0\} =$

$= \{a_0+a_1X+a_2X^2+a_3X^3 \in \mathbb{R}_3[X] \mid a_1=2a_0, a_2=4a_0, a_3=8a_0\}$

~~$= \{a_0+2a_0$~~

$= \{a_0(\underbrace{1+2X+4X^2+8X^3}_{P_4 \neq 0}) \mid a_0 \in \mathbb{R}\} \Rightarrow \{P_4\}$ o bază pt $V^\perp$.

$pr_V S$, unde $S \overset{not}{=} X^3-20X^2-5X-3$.

$V \oplus V^\perp = \mathbb{R}_3[X] \Rightarrow \exists!\ S_1 \in V$ și $S_2 \in V^\perp$ a.î.

$S = S_1 + S_2$

$\quad //not \quad //not$

$pr_V S \quad pr_{V^\perp} S$

$S_1 \in V, S_2 \in V^\perp \Rightarrow \exists!\ \alpha_1,\alpha_2,\alpha_3,\alpha_4 \in \mathbb{R}$ a.î.

$S = \underbrace{\alpha_1P_1+\alpha_2P_2+\alpha_3P_3}_{S_1} + \underbrace{\alpha_4P_4}_{S_2} \mid \cdot P_4 >$

$\Rightarrow \langle S,P_4\rangle = \alpha_4\langle P_4,P_4\rangle$

$\langle P_4,P_4\rangle = 1^2+2^2+4^2+8^2 = 21+64 = 85$

$\langle S,P_4\rangle = 1\cdot1-20\cdot2-5\cdot4-3\cdot8 = -85$

$\Rightarrow -85 = \alpha_4 85 \Rightarrow \alpha_4 = -1$

$\Rightarrow S_1 \overset{not}{=} pr_V S = S+P_4$

$= 9X^3-16X^2-3X-2$

Ex5: Fie $V = \{(x_1,\dots,x_5) \in \mathbb{R}^5 \mid \begin{matrix} x_1+x_2+x_3-x_4+x_5=0 \\ x_1+2x_2-x_4+x_5=0 \\ 3x_1+x_2+5x_3-3x_4+3x_5=0 \end{matrix}\} \subseteq \mathbb{R}^5$ subsp vect

Să se det $V^\perp$ și $pr_V u$ rel la prod. scal. standard pe $\mathbb{R}^5$.

$$\begin{pmatrix} 1 & 1 & 1 & -1 & 1 \\ 1 & 2 & 0 & -1 & 1 \\ 3 & 1 & 5 & -3 & 3 \end{pmatrix} \xrightarrow[L_3 \to L_3 - 3L_1]{L_2 \to L_2 - L_1} \begin{pmatrix} 1 & 1 & 1 & -1 & 1 \\ 0 & 1 & -1 & 0 & 0 \\ 0 & -2 & 2 & 0 & 0 \end{pmatrix}$$

$$\to \begin{pmatrix} 1 & 1 & 1 & -1 & 1 \\ 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow{L_1 \to L_1 - L_2} \left(\begin{array}{cc|ccc} 1 & 0 & 2 & -1 & 1 \\ 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right)$$
$$x_1\ x_2\ x_3\ x_4\ x_5$$

$$\Rightarrow \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{pmatrix} = \begin{pmatrix} -2x_3 + x_4 - x_5 \\ x_3 \\ x_3 \\ x_4 \\ x_5 \end{pmatrix}$$

$$= x_3 \begin{pmatrix} -2 \\ 1 \\ 1 \\ 0 \\ 0 \end{pmatrix} + x_4 \begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} + x_5 \begin{pmatrix} -1 \\ 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

$$\Rightarrow V = \mathrm{Sp}\left\{ \begin{pmatrix} -2 \\ 1 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} \right\}.$$

$\Rightarrow$ O bază pt V este $\{(-2,1,1,0,0), (1,0,0,1,0), (-1,0,0,0,1)\}$ (v_1, v_2, v_3) cu $\dim_{\mathbb{R}} V = 3$

cu $\dim_{\mathbb{R}} V^\perp = 2$

$$V^\perp = \{ x \in \mathbb{R}^5 \mid x \perp v_j, j \in \{1,2,3\} \}.$$
$$(x_1, x_2, x_3, x_4, x_5)$$

$x \perp v_1 \Rightarrow \langle (x_1,\dots,x_5), (-2,1,1,0,0) \rangle = 0$

$\Leftrightarrow -2x_1 + x_2 + x_3 = 0$

$x \perp v_2 \Rightarrow x_1 + x_4 = 0$

$x \perp v_3 \Rightarrow -x_1 + x_5 = 0$

$$\begin{matrix} x_1 & x_2 & x_3 & x_4 & x_5 \end{matrix}$$
$$\begin{pmatrix} -2 & 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ -1 & 0 & 0 & 0 & 1 \end{pmatrix} \to \begin{pmatrix} 1 & 0 & 0 & 1 & 0 \\ -2 & 1 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 & 1 \end{pmatrix} \xrightarrow[L_3 \to L_3 + L_1]{L_2 \to L_2 + 2L_1} \begin{pmatrix} 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 2 & 0 \\ 0 & 0 & 0 & 1 & 1 \end{pmatrix}$$

$$\to \begin{pmatrix} \boxed{1} & 0 & 0 & 1 & 0 \\ 0 & \boxed{1} & 1 & 2 & 0 \\ 0 & 0 & 0 & \boxed{1} & 1 \end{pmatrix} \to \begin{pmatrix} 1 & 0 & 0 & 0 & -1 \\ 0 & 1 & 1 & 0 & -2 \\ 0 & 0 & 0 & 1 & 1 \end{pmatrix} \xrightarrow[L_1 \to L_1 - L_3]{L_2 \to L_2 - 2L_3} \begin{pmatrix} \boxed{1} & 0 & 0 & 0 & -1 \\ 0 & \boxed{1} & 1 & 0 & -2 \\ 0 & 0 & 0 & \boxed{1} & 1 \end{pmatrix}$$
$$x_1\ x_2\ x_3\ x_4\ x_5$$

$$\Rightarrow \begin{matrix} x_1 = x_5 \\ x_2 = -x_3 + 2x_5 \\ x_4 = -x_5 \end{matrix} \Rightarrow V^\perp = \{(x_1,\dots,x_5) \mid x_1 = x_5,\ x_2 = -x_3 + 2x_5,\ x_4 = -x_5\}$$

$$V^\perp = \{ x_3(0,1,1,0,0) + x_5(1,2,0,-1,1) \mid x_3, x_5 \in \mathbb{R} \}$$
$$\Rightarrow V^\perp = \mathrm{Sp}\{(0,1,1,0,0), (1,2,0,-1,1)\} \Rightarrow \text{baza}$$

ALGEBRA - SIO - TEMA

? pr_V^U ; $U = (1,1,3,-1,4) \in \mathbb{R}^5$

Stim că $V \oplus V^\perp = \mathbb{R}^5$ $\Rightarrow \exists!\ U_1 \in V,\ U_2 \in V^\perp$ a.î. $U = U_1 + U_2$.

pr_V^U

dar $\begin{cases} U_1 = \alpha_1 v_1 + \alpha_2 v_2 + \alpha_3 v_3 \\ U_2 = \alpha_4 v_4 + \alpha_5 v_5 \end{cases}$; $\alpha_1, \alpha_2, \alpha_3, \alpha_4, \alpha_5 \in \mathbb{R}$ și $v_1, v_2, v_3 \in V$ și $v_4, v_5 \in V^\perp$.

$pr_{V^\perp}^U$ $\Rightarrow \boxed{U_1 = U - U_2}$

$U = U_1 + U_2 \mid \cdot \langle \cdot, v_4 \rangle \Rightarrow \langle U, v_4 \rangle = \alpha_4 \langle v_4, v_4 \rangle + \alpha_5 \langle v_5, v_4 \rangle$

$U = U_1 + U_2 \mid \cdot v_5 \Rightarrow \langle U, v_5 \rangle = \alpha_4 \langle v_4, v_5 \rangle + \alpha_5 \langle v_5, v_5 \rangle$

$\langle U, v_4 \rangle = 1 \cdot \overset{0}{0} + 1 \cdot (-1) + 3 \cdot 1 + (-1) \cdot \overset{0}{0} + 4 \cdot \overset{0}{0} = 2$

$\langle U, v_5 \rangle = 1 \cdot 1 + 1 \cdot 2 + 3 \cdot \overset{0}{0} + (+1) \cdot (+1) + 4 \cdot 1 = 8.$

$\langle v_4, v_4 \rangle = 0 \cdot 0 + (-1)^2 + 1^2 + 0^2 + 0^2 = 2$

$\langle v_4, v_5 \rangle = 0 \cdot 1 + (-1) \cdot 2 + 1 \cdot 0 + 0 + 0 = -2$

$\langle v_5, v_5 \rangle = 1 + 4 + 0 + 1 + 1 = 7.$

$\Rightarrow \begin{cases} 2 = \alpha_4 \cdot 2 + \alpha_5 \cdot (-2) \\ 8 = \alpha_4 \cdot (-2) + \alpha_5 \cdot 7 \end{cases}$ ④

$5\alpha_5 = 10 \Rightarrow \boxed{\alpha_5 = 2} \Rightarrow 2 = 2\alpha_4 - 4 \Rightarrow 2\alpha_4 = 6 \Rightarrow \boxed{\alpha_4 = 3}$

$\Rightarrow U_1 = U - U_2$

$pr_V^U = U - pr_{V^\perp}^U$

$pr_V^U = (1,1,3,-1,4) - \alpha_4(0,-1,1,0,0) - \alpha_5(1,2,0,-1,1)$

$= (1,1,3,-1,4) - (0,-3,3,0,0) - (2,4,0,-2,2)$

$\Rightarrow \boxed{pr_V^U = (-1,0,0,1,2)}$

Ex 1 cont 1.

Pe $\mathbb{R}^5$ se cons. p.s. standard și $V = \{(x_1,\dots,x_5) \in \mathbb{R}^5 \mid \begin{matrix} x_1+x_2+x_3-x_4+x_5=0 \\ x_1+2x_2+(-x_4)+x_5=0 \\ 3x_1+x_2+5x_3-3x_4+3x_5=0 \end{matrix}\}$

Să se det $V^\perp$ și $pr_V u$, unde $u = (1,1,3,-1,4)$

Seminarul trecut:

$$B_V = \{(-2,1,1,0,0), (1,0,0,1,0), (-1,0,0,0,1)\}.$$
$$\quad v_1 \qquad\quad v_2 \qquad\quad v_3$$

$$V^\perp = \{x \in \mathbb{R}^5 \mid x \perp v_j,\ 1 \le j \le 3\}.$$
$$\quad x = (x_1,\dots,x_5)$$

$$= \{(x_1,x_2,x_3,x_4,x_5) \in \mathbb{R}^5 \mid \begin{matrix} -2x_1+x_2+x_3=0 \\ x_1+x_4=0 \Rightarrow x_4=-x_1 \\ -x_1+x_5=0 \end{matrix}\} \quad \rightarrow x_3 = 2x_1 - x_2.$$
$$\Downarrow$$
$$x_5 = x_1$$

$$\Rightarrow V^\perp = \{(x_1, x_2, 2x_1-x_2, -x_1, x_1) \mid x_1, x_2 \in \mathbb{R}\}$$

$$= \{x_1(1,0,2,-1,1) + x_2(0,1,-1,0,0) \mid x_1,x_2 \in \mathbb{R}\}$$

$$\Rightarrow V^\perp = \mathrm{Sp}\{\underbrace{(1,0,2,-1,1)}_{w_1}, \underbrace{(0,1,-1,0,0)}_{w_2}\}$$

$\{w_1, w_2\}$ SLi

$\Rightarrow \{w_1, w_2\}$ o bază pt $V^\perp$

$\Rightarrow \dim_{\mathbb{R}} V^\perp = 2$

$u \in \mathbb{R}^5 = V \oplus V^\perp \Rightarrow \exists!\ u_1, u_2$ ($u_1 \in V$, $u_2 \in V^\perp$) și $a,b,c,d,e \in \mathbb{R}$

$$\text{și } u = \underbrace{(a v_1 + b v_2 + c v_3)}_{pr_V u} + \underbrace{d w_1 + e w_2}_{pr_{V^\perp} u} \quad / \cdot w_1 \Rightarrow \text{(1)}, \quad \cdot w_2 \Rightarrow \text{(2)}$$

$$\text{(1)}: \langle u, w_1\rangle = d\langle w_1, w_1\rangle + e\langle w_2, w_1\rangle$$
$$\langle u, w_2\rangle = d\langle w_1, w_2\rangle + e\langle w_2, w_2\rangle$$

$\langle u, w_1\rangle = (1,0,6,1,4) = 12$

$\langle u, w_2\rangle = (0,1,-3,0,0) = -2$

$\langle w_1, w_1\rangle = (1,0,4,1,1) = 7$

$\langle w_2, w_2\rangle = (0,1,1,0,0) = 2$

$\langle w_1, w_2\rangle = (0,0,-2,0,0) = -2$

scalari!!!

$$\Rightarrow \begin{cases} 12 = 7d - 2e \\ -2 = -2d + 2e \end{cases} \Rightarrow \begin{cases} 10 = 5d \\ d - e = 1 \end{cases} \Rightarrow \begin{cases} d = 2 \\ e = 1 \end{cases}$$

$\Rightarrow \begin{cases} 12 = 7d - 2e \\ -2 = -2d + 2e \end{cases}$ (1)

$5d = 10 \Rightarrow d = 2 \Rightarrow 14 - 2e = 12$

$\Rightarrow 2e = 2 \Rightarrow e = 1$

$\Rightarrow v = u_1 + u_2$

$\Rightarrow \mathrm{pr}_V v = v - \mathrm{pr}_{V^\perp} v$

$\mathrm{pr}_V v \quad \mathrm{pr}_{V^\perp} v$

$= (1,1,3,-1,4) - (2(1,0,2,-1,1) + (0,1,-1,0,0))$

$= (1,1,3,-1,4) - (2,1,3,-2,2)$

$\Rightarrow \mathrm{pr}_V v = (-1,0,0,+1,2).$

Fie $A \in \mathcal{M}_{m,n}(\mathbb{R})$

$\| \quad (a_1 | a_2 | \dots | a_n)$ unde $a_j \in \mathbb{R}^m = \mathcal{M}_{m,1}(\mathbb{R})$, $1 \le j \le n$.

Not. $\mathbb{R}^m$-subspace $\mathcal{J}(A) \overset{\text{not}}{=} \mathcal{J}\{a_1, \dots, a_n\} = \{x_1 a_1 + x_2 a_2 + \dots + x_n a_n \mid x_1, \dots, x_n \in \mathbb{R}\}$.

$= \{A \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \mid x_1, \dots, x_n \in \mathbb{R}\} = \{Ax \mid x \in \mathbb{R}^n = \mathcal{M}_{n,1}(\mathbb{R})\}$.

$(\Rightarrow \dim_{\mathbb{R}} \mathcal{J}(A) = \mathrm{rang}(A))$

Not. $\mathcal{W}(A) \overset{\text{not}}{=} \{ x \in \underbrace{\mathbb{R}^n}_{= \mathcal{M}_{n,1}(\mathbb{R})} \mid Ax = O_{m,n} \}$. $(\Rightarrow \dim_{\mathbb{R}} \mathcal{W}(A) = n - \mathrm{rang}(A))$.

2) Are loc $\Leftrightarrow r(A) = n$ ($\Leftrightarrow$ A are col. liniar independente)

$\Leftrightarrow \mathrm{rang}(\underbrace{{}^tA \cdot A}_{\in \mathcal{M}_n(\mathbb{R})}) = n$. $(\Leftrightarrow \mathcal{J}({}^tA \cdot A)^{-1})$.

$\mathcal{M}_{n,1}(\mathbb{R}) \ni \mathbb{R}^n : \langle x, y \rangle = {}^t x \cdot y$

Sol: $\dim_{\mathbb{R}} \mathcal{W}(A) = n - \mathrm{rang}(A)$

$\dim_{\mathbb{R}} \mathcal{W}({}^tA \cdot A) = n - \mathrm{rang}({}^tA \cdot A)$ $\Big|$ $\Rightarrow$ ca nr dem em. an R-sup.

nr dem $\mathcal{W}(A) = \mathcal{W}({}^tA \cdot A)$

"$\subseteq$" Fie $x \in \mathcal{W}(A)$. At. $Ax = O_{m,1}$ $\Big| \cdot {}^tA$ st

$\Rightarrow ({}^tA A)x = \underbrace{{}^tA \cdot O_{m,1}}_{O_{n,1}} \Rightarrow x \in \mathcal{W}({}^tA \cdot A)$.

"$\supseteq$" Fie $x \in \mathcal{W}({}^tA \cdot A)$. At. $({}^tA \cdot A)x = O_{n,1}$

$\| \quad {}^tA \cdot (Ax)$

$\cdot {}^t y \quad \Big|$

$\Rightarrow \underbrace{{}^t x \cdot {}^t A}_{{}^t(Ax)} \cdot (Ax) = \underbrace{{}^t x \cdot O_{n,1}}_{0}$

$x \in \mathcal{W}(A)$

$\| \quad \uparrow$

$({}^t x {}^t A) \cdot (Ax) = 0$

$(Ax)^t (Ax) = 0 \Leftrightarrow \langle Ax, Ax \rangle = 0 \Leftrightarrow \|Ax\|^2 = 0 \Rightarrow Ax = O_{m,1}$

arătat că $\mathcal{N}(A) = \mathcal{N}({}^tA \cdot A)$, așadar $\dim_{\mathbb{R}} \mathcal{N}(A) = \dim_{\mathbb{R}} \mathcal{N}({}^tA \cdot A)$.
Deducem $\operatorname{rang}(A) = \operatorname{rang}({}^tA \cdot A)$.

Obs: Dacă $A \in \mathcal{M}_{m,n}(\mathbb{R})$ are coloanele liniar independente, atunci $(\exists)\,({}^tA \cdot A)^{-1}$

Metoda celor mai mici pătrate

$\mathbb{R}^n$ $\alpha x = \beta$ ← legea fizicii
↳ de dim mărimi.

$\alpha_1 x = \beta_1$, $\alpha_2 x = \beta_2$, ..., $\alpha_n x = \beta_n$

$$\underbrace{\begin{pmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{pmatrix}}_{a} x = \underbrace{\begin{pmatrix} \beta_1 \\ \vdots \\ \beta_n \end{pmatrix}}_{b} \quad ; \quad (\not\exists)\ \boxed{Ax = b} \ (*) \text{ sist. incomp.}$$

Sn. soluție în sensul celor mai mici pătrate sau pseudosoluție pentru (*) acel x pentru care suma pătratelor erorilor este minimă

$$(\alpha_1 x - \beta_1)^2 + (\alpha_2 x - \beta_2)^2 + \dots + (\alpha_n x - \beta_n)^2 = \|ax - b\|^2 = \bar{\varepsilon}(x)$$

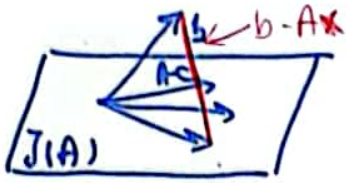
$\bar{x}$ = sol în sensul celor mai mici pătrate pentru (*)
dacă $\bar{\varepsilon}(\bar{x}) = \min_{x \in \mathbb{R}} \bar{\varepsilon}(x)$.

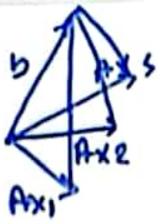
$$\bar{\varepsilon}(x) = \langle ax - b, ax - b \rangle = \langle ax, ax \rangle - \langle ax, b \rangle - \langle b, ax \rangle + \langle b, b \rangle$$
$$= \underbrace{({}^ta \cdot a)}_{\alpha} x^2 - 2({}^ta \cdot b)x + {}^tb \cdot b$$

$$\min_{x \in \mathbb{R}} \bar{\varepsilon}(x) = \bar{\varepsilon}(\bar{x}), \quad \boxed{\bar{x} = \frac{\langle a, b \rangle}{\langle a, a \rangle}}$$

↓ calc $\bar{x}$ dacă în (*) în loc de b avem $\operatorname{pr}_a b = \frac{\langle b, a \rangle}{\langle a, a \rangle} a$

Fie $A \in \mathcal{M}_{m,n}(\mathbb{R})$ și $b \in \mathcal{J}(A) \Big/ \Rightarrow Ax = b$ sist. incompatibil. S.n.
Fie $Ax = b$
sol în sensul celor mai mici pătrate pt sistemul $Ax = b$, orice vector $\bar{x} \in \mathbb{R}^n = \mathcal{M}_{n,1}(\mathbb{R})$ pentru care $\|Ax - b\|$ min.





$c \in \mathbb{R}^n = \mathcal{M}_{n,1}(\mathbb{R})$. $\|Ax-b\| = \|A\bar{x}-b\|$ dacă $\boxed{A\bar{x}-b \perp J(A)}$

$\Rightarrow (\forall)\, c \in \mathbb{R}^n = \mathcal{M}_{n,1}(\mathbb{R}) : \langle Ac, A\bar{x}\rangle = 0$

$\Rightarrow (\text{scriu } \forall)\, c \in \mathbb{R}^n = \mathcal{M}_{n,1}(\mathbb{R}) : \quad {}^t(Ac)(A\bar{x}-b) = 0$

$${}^tc\,({}^tAA\bar{x} - {}^tA\cdot b) = 0$$

$$\text{fie } c = ({}^tA\cdot A)\bar{x} - {}^tA\cdot b \;\Big|\Rightarrow$$

$${}^tc\cdot c = 0 \;(*)\; \langle c, c\rangle = 0 \Rightarrow c = 0_{n,1} \Rightarrow$$

$$\Rightarrow ({}^tA\cdot A)\bar{x} = {}^tA\cdot b$$

Pp. $\operatorname{rang} A = n$. Atunci $\exists\, ({}^tA\cdot A)^{-1} \;\Big|\Rightarrow$

$$\boxed{\bar{x} = ({}^tA\cdot A)^{-1}\,{}^tA\cdot b}$$

și $\boxed{A\bar{x} = pr_{J(A)}\, b}$

Ex 3: Dacă $\operatorname{rang} A = n$ și A are coloanele OG, at.

$pr_{J(A)}\, b \in \mathbb{R}^m$ este suma proiecțiilor lui b pe coloanele lui A.

Mai exact dacă $A = (a_1 | \dots | a_n)$, $\{a_1, \dots, a_n\}$ sistem OG atunci

$$\Rightarrow pr_{J(A)}\, b = \sum_{k=1}^{n} \frac{\langle b, a_k\rangle}{\|a_k, a_k\rangle}\, a_k$$

Sol.

$$pr_{J(A)}\, b = A\bar{x} = A({}^tA\cdot A)^{-1}\,{}^tA\cdot b \overset{t}{=}$$

$$({}^tA\cdot A) = \begin{pmatrix} {}^ta_1\cdot a_1 & {}^ta_1\cdot a_2 \dots \\ {}^ta_2\cdot a_1 & {}^ta_2\cdot a_2 \dots \\ \vdots & \vdots \\ {}^ta_n\cdot a_1 & {}^ta_n\cdot a_2 \dots \end{pmatrix} = \begin{pmatrix} \langle a_1, a_1\rangle & \langle a_1, a_2\rangle \dots \\ \langle a_2, a_1\rangle & \langle a_2, a_2\rangle \dots \\ \vdots & \vdots \\ \langle a_n, a_1\rangle & \langle a_n, a_2\rangle \dots \langle a_n, a_n\rangle \end{pmatrix}$$

$$= \begin{pmatrix} \|a_1\|^2 \neq 0 & & 0 \\ & \ddots & \\ 0 & & \|a_n\|^2 \neq 0 \end{pmatrix}$$

$a_i \perp a_j$ pt $i \neq j$, $i, j \in \{1, \dots, n\}$

$$\overset{t}{=} A \cdot \begin{pmatrix} \frac{1}{\|a_1\|^2} & & 0 \\ & \ddots & \\ 0 & & \frac{1}{\|a_n\|^2} \end{pmatrix} {}^tA\cdot b$$

$${}^tA\cdot b = \begin{pmatrix} {}^ta_1\cdot b \\ {}^ta_2\cdot b \\ \vdots \\ {}^ta_n\cdot b \end{pmatrix} = \begin{pmatrix} \langle a_1, b\rangle \\ \langle a_2, b\rangle \\ \vdots \\ \langle a_n, b\rangle \end{pmatrix}$$

$$\pi_{J(A)} b = A \begin{pmatrix} \frac{1}{\|a_1\|^2} & & 0 \\ & \ddots & \\ 0 & & \frac{1}{\|a_n\|^2} \end{pmatrix} \begin{pmatrix} \langle a_1, b\rangle \\ \vdots \\ \langle a_n, b\rangle \end{pmatrix}$$

$$= A \begin{pmatrix} \frac{\langle a_1, b\rangle}{\|a_1\|^2} \\ \vdots \\ \frac{\langle a_n, b\rangle}{\|a_n\|^2} \end{pmatrix} = \frac{\langle a_1, b\rangle}{\langle a_1, a_1\rangle} \cdot a_1 + \frac{\langle a_2, b\rangle}{\langle a_2, a_2\rangle} a_2 + \ldots + \frac{\langle a_n, b\rangle}{\langle a_n, a_n\rangle} \cdot a_n$$

$$= \pi_{a_1} b + \ldots + \pi_{a_n} b.$$

Exu: Fie $b = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$ și $A = \begin{pmatrix} 1 & 2 \\ -1 & 1 \\ 0 & 1 \end{pmatrix}$ (a_1 a_2). Arăt că $\det(A|b) = \begin{vmatrix} 1 & 2 & -1 \\ -1 & 1 & 0 \\ 0 & 1 & 1 \end{vmatrix}$

$= 1 + 1 + 2 = 4 \neq 0 \Rightarrow b \notin J(A))$

Să se det. $\pi_{J(A)} b$ în 3 moduri.

I) $\pi_{J(A)} b = A (^tA \cdot A)^{-1} \cdot {}^tA \cdot b$, se calc.

II) O bază pt $J(A)$ este $\{a_1, a_2\}$. Atunci $[J(A)]^\perp = \left\{ \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \middle| \begin{matrix} x_1 - x_2 = 0 \\ 2x_1 + x_2 + x_3 = 0 \end{matrix} \right\}$

$x_2 = x_1$
$\Rightarrow x_3 = -3x_1$

$$= \left\{ x_1 \underbrace{\begin{pmatrix} 1 \\ 1 \\ -3 \end{pmatrix}}_{a_3} \middle| x_1 \in \mathbb{R} \right\}$$

$b \in \mathbb{R}^3 = J(A) \oplus [J(A)]^\perp \Rightarrow (\exists)! \alpha, \beta, \gamma \in \mathbb{R}$ a.î.

$b = \underbrace{\alpha a_1 + \beta a_2}_{\pi_{J(A)} b} + \underbrace{\gamma (a_3)}_{\pi_{(J(A))^\perp} b} \quad | \cdot a_3$

$\Rightarrow \langle b, a_3\rangle = \gamma \langle a_3, a_3\rangle$

$-4 = \gamma \cdot 11 \Rightarrow \gamma = \frac{-4}{11}$

$\Rightarrow \underbrace{\pi_{J(A)} b} = b + \frac{4}{11} a_3 = \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} + \frac{4}{11} \begin{pmatrix} 1 \\ 1 \\ -3 \end{pmatrix}$

$= \underbrace{\frac{1}{11} \begin{pmatrix} -7 \\ 4 \\ -1 \end{pmatrix}}$ ✓

III) Aplic. proc. de OG G-S șir de vectori $\{a_1, a_2\}$.

$v_1 = a_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$

$v_2 = a_2 - \frac{\langle a_2, v_1\rangle}{\langle v_1, v_1\rangle} \cdot v_1 = \begin{pmatrix} 2 \\ 1 \end{pmatrix} - \frac{1}{2} \begin{pmatrix} -1 \\ 0 \end{pmatrix}$

$= \frac{1}{2} \begin{pmatrix} 3 \\ 3 \\ 2 \end{pmatrix}$

$\Rightarrow J(A) = \mathrm{Sp}\{a_1, a_2\} = \mathrm{Sp}\{v_1, v_2\} = J \begin{pmatrix} b_1 & b_2 \\ -1 & 3 \\ 0 & 3 \\ 1 & 2 \end{pmatrix}$

$\Rightarrow \pi_{J(A)} b = \pi_{J(b_1, b_2)} b \overset{ex3}{=} \pi_{b_1} b + \pi_{b_2} b = \frac{\langle b, b_1\rangle}{\langle b_1, b_1\rangle} b_1 + \frac{\langle b, b_2\rangle}{\langle b_2, b_2\rangle} b_2$

$= \frac{-1}{2} \begin{pmatrix} -1 \\ 0 \end{pmatrix} - \frac{1}{22} \begin{pmatrix} 3 \\ 3 \\ 2 \end{pmatrix} = -\frac{1}{22} \left[\begin{pmatrix} 11 \\ 0 \end{pmatrix} + \begin{pmatrix} 3 \\ 3 \\ 2 \end{pmatrix} \right] = -\frac{1}{22} \begin{pmatrix} 14 \\ 3 \\ 2 \end{pmatrix} = \frac{1}{11} \begin{pmatrix} -7 \\ -4 \\ -1 \end{pmatrix}$ ✓

Teorema de existență și unicitate a complementului ortogonal

Fie $(V, \langle\ \rangle)$ sp. metrice ai $\dim_{\mathbb{K}} V = n < +\infty$ și $U \underset{\text{subsp vect}}{\subseteq} V$ cu $\dim_{\mathbb{K}} U < \dim_{\mathbb{K}} V \Rightarrow m < n \Rightarrow (\exists!)\ U^{\perp} \underset{\text{subsp vect}}{\subseteq} V$ ai $(= m)$

U și $U^{\perp}$ nu fie subsp ort și $U \oplus U^{\perp} = V$. În acest caz, $U^{\perp}$ ($U \perp U^{\perp}$)

s.n. comp ortogonal al lui U.

În plus, $\dim_{\mathbb{K}} U^{\perp} = \dim_{\mathbb{K}} V - \dim_{\mathbb{K}} U$ și

$U^{\perp} = \{x \in V \mid \underbrace{x \perp U}_{!!}\}$

$(x \perp u, (\forall) u \in U) \Leftrightarrow \langle x, u\rangle = 0$, $(\forall) x \in V$ și $u \in U$.

Dem:

Se cons. $B_1 = \{a_1, \dots, a_m\} \subseteq U$ o bază pt U.

Se comp B_1 până la o bază a lui V, $B = \{a_1, \dots, a_m, a_{m+1}, \dots, a_n\} \subseteq V$.

Apl. proc. GRAM-SCHMIDT lui $B \Rightarrow$

$B' = \{v_1, \dots, v_m, v_{m+1}, \dots, v_n\}$ o bază ortogonală pt V, pentru care are loc $U = sp.\{a_1, \dots, a_m\} = \{v_1, \dots, v_m\}$.

Cursul trecut: $\cdot\ U + U^{\perp} = U \oplus U^{\perp}$

$\cdot\ U + U^{\perp} = V$

Mai avem de arătat: 1) $U \perp U^{\perp}$

2) Unicitatea lui $U^{\perp}$.

1) Fie $x \in U$ și $y \in U^{\perp}$. Vom arăta că $\langle x, y\rangle = 0$.

$\exists!\ \beta_1, \dots \beta_m \in \mathbb{K}$ ai $x = \sum_{i=1}^{m} \beta_i v_i$

$\exists!\ \beta_{m+1}, \dots, \beta_n \in \mathbb{K}$ ai $y = \sum_{j=m+1}^{n} \beta_j v_j$

At $\langle x,y\rangle = \langle \sum_{i=1}^{m} \beta_i v_i , \sum_{j=m+1}^{n} \beta_j v_j \rangle$ $\quad \langle \alpha x, \beta y\rangle = \alpha \langle x, \beta y\rangle = \alpha \bar{\beta} \langle x,y\rangle$

$= \sum_{i=1}^{m} \beta_i \sum_{j=m+1}^{n} \bar{\beta}_j \underbrace{\langle v_i, v_j\rangle}_{=0 \text{ pt că } i\neq j \text{ și } B' \text{ bază OG pt } V.}$

$\underbrace{\in U}\ \underbrace{\in U^\perp}$

$= 0$ ■

a) Pp că mai există $U_1 \subseteq V$ a.î. $U_1 \perp U$ și $U_1 \oplus U = V$

Vom arăta că $U_1 = U^\perp$. Arătăm, mai întâi că $U_1 \subseteq U^\perp$ și apoi $U^\perp \subseteq U_1$, de unde fol.

Fie $z \in U_1 \subseteq V = U \oplus U^\perp$ $\Rightarrow \exists!\ z_1 \in U$ și $z_2 \in U^\perp$ a.î. $z = z_1 + z_2$.

Atunci $\langle z_1, z_2\rangle = \langle z_1, (z - z_2), z_2\rangle = \langle z_1, z_2\rangle + \langle z_2, z_2\rangle$
$\quad\quad\quad

...: a) Fie $u \in U$. At

$\|v-u\|^2 = \langle v-u, v-u\rangle = \langle v,v\rangle - \langle v,u\rangle - \langle u,v\rangle + \langle u,u\rangle$; $v = v_1 + v_2$, $u = u$

$\Rightarrow \langle v, u\rangle = \langle v_1+v_2, u\rangle = \langle v_1, u\rangle$

$\Rightarrow \|v-u\|^2 = \langle v,v\rangle - \langle v_1,u\rangle - \langle u, v_1\rangle + \langle u,u\rangle$

$\Rightarrow \langle \underbrace{v_1-u}_{\in U} + \underbrace{v_2}_{\in U^\perp}, \underbrace{v_1-u}_{\in U} + \underbrace{v_2}_{\in U^\perp}\rangle = \langle v_1-u, v_1-u+v_2\rangle + \langle v_2, v_1-u+v_2\rangle$

$= \langle v_1-u, v_1-u\rangle + \|v_2\|^2 = \|v_1-u\|^2 + \underbrace{\|v_2\|^2}_{\geq 0} \geq \|v_1-u\|^2$

b) $\|v-u\|^2 = \|v_1-u\|^2 + \|v_2\|^2$ / $\Rightarrow \|v_2\|^2 = 0 \Rightarrow v_2 = 0_V \Rightarrow$ $\boxed{v = v_1}$

$\|v-u\| = \|v-v_1\|$, dar $\|v-u\| = \|v-v_1\| \rightarrow \|v-u\| = 0 \Rightarrow v-u = 0_V \Rightarrow \boxed{v = u}$ $\Rightarrow$

$\Rightarrow \boxed{u = v_1}$.

Obs* Dacă $A, B, C \in \mathcal{M}_n(\mathbb{R})$ a.î. $AB = CA = I_n$. Să arătăm $B = C$.

$AB = I_n \mid \cdot C_{st} \Rightarrow CAB = C$

$CA = I_n \mid \cdot B_{dr} \Rightarrow CAB = B$ $\Rightarrow$ $B = C$. ∎

Matrici ortogonale

Dacă priv. $\mathbb{R}^n$ ca sp. de coloane,

$\langle x, y\rangle = \sum_{k=1}^{n} x_k y_k = {}^t x \cdot y$

$\begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix}$

Teorema 2: Fie Q o matr $\in \mathcal{M}_n(\mathbb{R})$. Atunci sp. sunt „$\Leftrightarrow$":

1) Q are coloanele ortonormate (${}^t Q \cdot Q = I_n$)
2) $Q^{-1} = {}^t Q$
3) Q are liniile ortonormate ($Q \cdot {}^t Q = I_n$)
4) Q păstrează lungimea vectorilor ; $\|Qx\| = \|x\|$, $(\forall) x \in \mathbb{R}^n$
5) Q păstrează prod. scalar ; $\langle Qx, Qy\rangle = \langle x, y\rangle$, $(\forall) x, y \in \mathbb{R}^n$ // $= \mathcal{M}_{n,1}(\mathbb{R})$

1) $\Rightarrow$ 2) $\Rightarrow$ 3) $\Rightarrow$ 4) $\Rightarrow$ 5) $\Rightarrow$ 1).

Obs: O matr Q cu una din prop 1-5 s.n. **matrice OG.**

1) $\Rightarrow$ 2) $ {}^tQ \cdot Q = \bar{I}_n \underset{\text{obs}}{\Rightarrow} {}^tQ \cdot Q = Q \cdot {}^tQ = \bar{I}_n$

$\Rightarrow Q^{-1} = {}^tQ$.

2) $\Rightarrow$ 3) evident

3) $\Rightarrow$ 4) Fie $x \in \mathbb{R}^n = \mathcal{M}_{n,1}(\mathbb{R})$. Atunci $\|Qx\|^2 = \langle Qx, Qx \rangle$

$\overset{t}{=} (Qx)Qx$

$= x\,{}^tQ\,Qx$

$= {}^tx\,({}^tQQ)\,x \underset{\text{obs}}{=}$

$= {}^tx\,(Q \cdot {}^tQ)x = \langle x, x \rangle$

$= \|x\|^2 \Rightarrow \|Qx\| = \|x\|$.

4) $\Rightarrow$ 5) Fie $x, y \in \mathbb{R}^n = \mathcal{M}_{n,1}(\mathbb{R})$. Atunci $\underbrace{\|Q(x+y)\|^2}_{\|\|} \overset{4}{=} \|x+y\|^2$

$\langle Qx+Qy, Qx+Qy \rangle$

$\|Qx\|^2 + 2\langle Qx, Qy \rangle + \|Qy\|^2$

$\|x\|^2 \quad \|y\|^2$

$\|x+y\|^2 = \langle x+y, x+y \rangle$

$= \|x\|^2 + 2\langle x, y \rangle + \|y\|^2$

$\Rightarrow \langle Qx, Qy \rangle = \langle x, y \rangle$

5) $\Rightarrow$ 1) Fie $e_j \overset{\text{not}}{=} \begin{pmatrix} 0 \\ \vdots \\ 1_j \\ \vdots \\ 0 \end{pmatrix}$ j, $1 \le j \le n$. Atunci: $\underset{5)}{\langle Qe_j, Qe_i \rangle} = \langle e_j, e_i \rangle$

$\| \qquad \| \qquad \| \qquad \

descompunerea QR a unei matrice $A \in \mathcal{M}_n(\mathbb{R})$ nesingulare.

$A = Q \cdot R$, unde Q este o matr. ortogonală ; $Q, R \in \mathcal{M}_n(\mathbb{R})$

$\downarrow {}^tQ \cdot$

$\Rightarrow {}^tQ \cdot A = R$. Fie $A = (a_1 | a_2 | \dots | a_n)$ cu $a_j \in \mathbb{R}^n = \mathcal{M}_{n,1}(\mathbb{R})$, $\forall\, 1 \le j \le n$.

$\det A \neq 0 \Rightarrow \{a_1, \dots, a_n\}$ sli $\Rightarrow$ putem aplic. proc. GRAM-SCHMIDT sistemului $\{a_1, \dots, a_n\}$. $\Rightarrow$

$\exists$ $\{v_1, \dots, v_n\}$ sli cu vectorii 2 câte 2 ortogonali.

Alegem Q matr. cu coloanele $= (w_1 | w_2 | \dots | w_n)$

, unde $w_j = \frac{v_j}{\|v_j\|}$, $1 \le j \le n$. ($\Rightarrow Q$ matr. OG)

Din ${}^tQ \cdot A = R = (r_{ij})_{1 \le i,j \le n}$ și $Q = (w_1 | w_2 | \dots | w_n)$ $\Rightarrow$

pt $1 \le j \le n \Rightarrow r_{jj} = {}^tw_j \cdot a_j = \langle w_j, a_j \rangle = \|v_j\|$

Din proc G-S: $v_j = a_j - \sum_{k=1}^{j-1} \frac{\langle a_j, v_k \rangle}{\langle v_k, v_k \rangle} \cdot v_k \;\Big|\cdot v_j \Rightarrow$

$\Rightarrow \underbrace{\langle v_j, v_j \rangle}_{\|v_j\|^2} = \langle a_j, v_j \rangle \Leftrightarrow \|v_j\| = \langle a_j, \frac{v_j}{\|v_j\|} \rangle$

$= \langle a_j, w_j \rangle$

$= \langle w_j, a_j \rangle$

Ex: Fie $A = \begin{pmatrix} \frac{1}{2} & -1 & 5 \\ 2 & 0 & -3 \\ -1 & -1 & -1 \end{pmatrix}$

a) Să se arate că A are coloanele liniar independente.

b) Să se det. descomp $A = QR$, Q = matrice OG, iar $R \in \mathcal{M}_3(\mathbb{R})$ superior triunghiulară.

$\det A = 16 - 3 + 14 = 27 \neq 0 \Rightarrow$ A are coloanele l. indep. !!!

Prin $A = (a_1 | a_2 | a_3)$. Aplic. G-S $\exists$ sli $\{a_1, a_2, a_3\}$.

$v_1 \overset{\text{def}}{=} a_1$

$v_2 = a_2 - \frac{\langle a_2, v_1 \rangle}{\langle v_1, v_1 \rangle} v_1 = \begin{pmatrix} -1 \\ 0 \\ -1 \end{pmatrix} - \frac{-3}{9} \begin{pmatrix} \frac{1}{2} \\ -2 \\ 2 \end{pmatrix}$

$= \frac{1}{3} \left[\begin{pmatrix} -3 \\ 0 \\ -3 \end{pmatrix} + \begin{pmatrix} \frac{1}{2} \\ -2 \\ 2 \end{pmatrix} \right]$

$= \frac{1}{3} \begin{pmatrix} -2 \\ -2 \\ -1 \end{pmatrix} = -\frac{1}{3} \begin{pmatrix} 2 \\ 2 \\ 1 \end{pmatrix}$

$$v_3 = a_3 - \frac{\langle a_3, v_1\rangle}{\langle v_1, v_1\rangle} v_1 - \frac{\langle a_3, v_2\rangle}{\langle v_2, v_2\rangle} v_2$$

$$= \begin{pmatrix} 5 \\ -3 \\ -7 \end{pmatrix} - \frac{+7}{9} \begin{pmatrix} 1 \\ -2 \\ 2 \end{pmatrix} + \frac{+3 \cdot (-1)}{1 \cdot 9} \begin{pmatrix} 2 \\ 2 \\ 1 \end{pmatrix} \cdot \left(-\frac{1}{3}\right)$$

$$= \begin{pmatrix} 5 \\ -3 \\ -7 \end{pmatrix} + \frac{1}{3} \begin{pmatrix} 1 \\ -2 \\ 2 \end{pmatrix} + \frac{1}{3} \begin{pmatrix} 2 \\ 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 5 \\ -3 \\ -7 \end{pmatrix} + \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 6 \\ -3 \\ -6 \end{pmatrix} = 3 \begin{pmatrix} 2 \\ -1 \\ -2 \end{pmatrix}$$

$\|v_1\|^2 = 9 \Rightarrow \|v_1\| = 3$

$\|v_2\| = 1$

$\|v_3\| = 3$

$$\Rightarrow Q = \left(\frac{v_1}{\|v_1\|} \middle| \frac{v_2}{\|v_2\|} \middle| \frac{v_3}{\|v_3\|} \right)$$

$$Q = \begin{pmatrix} 1/3 & -2/3 & 2/3 \\ -2/3 & -2/3 & -1/3 \\ 2/3 & -1/3 & -2/3 \end{pmatrix}$$

$$\Rightarrow R = {}^tQ \cdot A = \frac{1}{3} \begin{pmatrix} 1 & -2 & 2 \\ -2 & -2 & -1 \\ 2 & -1 & -2 \end{pmatrix} \begin{pmatrix} 1 & -1 & 5 \\ -2 & 0 & -3 \\ 2 & -1 & -7 \end{pmatrix} = \dots$$

$$= \frac{1}{3} \begin{pmatrix} 9 & -3 & -3 \\ 0 & 3 & 3 \\ 0 & 0 & 27 \end{pmatrix} = \begin{pmatrix} 3 & -1 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 3 \end{pmatrix}$$

$\mathbb{K} \in \{\mathbb{R}, \mathbb{C}\}$.

Pe $\mathbb{R}^n = \mathcal{M}_{n,1}(\mathbb{R})$ cu p.s. standard:

$\langle x, y \rangle \overset{def}{=} {}^t x \cdot y$

Pe $\mathbb{C}^n = \mathcal{M}_{n,1}(\mathbb{C})$ cu p.s. standard:

$\langle x, y \rangle \overset{def}{=} {}^t x \cdot \bar{y}$

Fie $A \in \mathcal{M}_n(\mathbb{K})$, not. $A = (a_1 | \dots | a_n)$ $(a_j \in \mathbb{K}^n = \mathcal{M}_{n,1}(\mathbb{K}))_{1 \le j \le n}$

Def: 1) $A \in \mathcal{M}_n(\mathbb{R})$ se numește OG $\Leftrightarrow (\exists) A^{-1}$ a.i. $A^{-1} = {}^t A$ ($\Leftrightarrow$ $\{a_1, \dots, a_n\}$ este o bază ON în $\mathbb{R}^n$)

2) $A \in \mathcal{M}_n(\mathbb{C})$ se numește UNITARĂ $\Leftrightarrow (\exists) A^{-1}$ a.i. $A^{-1} = {}^t\bar{A}$ ($\Leftrightarrow$ $\{a_1, \dots, a_n\}$ este o bază ON în $\mathbb{C}^n$)

3) $A \in \mathcal{M}_n(\mathbb{R})$ se numește SIMETRICĂ $\Leftrightarrow A = {}^t A$.

4) $A \in \mathcal{M}_n(\mathbb{C})$ se numește HERMITICĂ $\Leftrightarrow A = {}^t\bar{A}$.

Propoziție

1) Dacă $A \in \mathcal{M}_n(\mathbb{C})$ este mat. HERM. $\Rightarrow (\forall) v \in \mathbb{C}^n = \mathcal{M}_{n,1}(\mathbb{C})$ are loc $\boxed{{}^t\bar{v} \cdot A \cdot v \in \mathbb{R}}$

2) Dacă $A \in \mathcal{M}_n(\mathbb{C})$ este mat. HERM $\Rightarrow \sigma(A) \subseteq \mathbb{R}$

3) Dacă $A \in \mathcal{M}_n(\mathbb{R})$ este sim $\Rightarrow \sigma(A) \subseteq \mathbb{R}$.

Dem:

1) Fie $v \in \mathbb{C}^n = \mathcal{M}_{n,1}(\mathbb{C})$ și $z \overset{not}{=} {}^t\bar{v} \cdot A \cdot v \Rightarrow \bar{z} = {}^t v \cdot \bar{A} \cdot \bar{v} = {}^t({}^t v \cdot \bar{A} \cdot \bar{v})$ ($\bar{z} = {}^t z$)

$= {}^t\bar{v} \cdot {}^t({}^t v \cdot \bar{A})$

$= {}^t\bar{v} \cdot \boxed{{}^t\bar{A}} \cdot v = {}^t\bar{v} \cdot A \cdot v = z \Rightarrow z \in \mathbb{R}$.

2) Fie $\lambda \in \sigma(A)$. Atunci $(\exists) v \in \mathbb{C}^n \setminus \{0_{n,1}\}$ a.i. $Av = \lambda v$ | ${}^t\bar{v}$

$\mathbb{R} \ni {}^t\bar{v} \cdot A \cdot v \in \mathbb{R} = \lambda({}^t\bar{v} \cdot v) = \lambda({}^t\bar{v} \cdot \bar{v}) = \lambda \underbrace{\langle \bar{v}, \bar{v} \rangle}_{>0} \Rightarrow \lambda \in \mathbb{R}$.

Teorema 1

Fie A o matr. simetrică. Atunci $\exists\, Q \in \mathcal{M}_n(\mathbb{R})$ o matr. OG ($\Leftrightarrow Q^{-1} = {}^tQ$) a.î. $Q^{-1}AQ = \Lambda = \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix}$ unde $\{\lambda_1, \dots \lambda_n\}$ sunt val. proprii ale lui A, nu neapărat distincte.

!!! **Consecință:** Orice matrice simetrică, diagonalizează este diagonalizabilă peste $\mathbb{R}$.

Teorema 2

Fie A o matr. hermitică. Atunci $\exists\, U \in \mathcal{U}_n(\mathbb{C})$ o matr. unitară ($\Leftrightarrow U^{-1} = {}^t\bar{U}$) a.î. $U^{-1}AU = \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix}$, unde $\{\lambda_1, \dots \lambda_n\}$ sunt val. proprii ale lui A, nu neapărat distincte.

Consecință: Orice matrice hermitică, diagonalizează (este diag. peste $\mathbb{C}$)

Prop

Fie A o matr. simetrică și $\lambda, \mu \in \delta(A)$ a.î. $\lambda \neq \mu$. Atunci $V_\lambda \perp V_\mu$.

Dem: Fie $x \in V_\lambda$ și $y \in V_\mu$. $\quad \langle x, y \rangle = {}^t x \cdot y$

$\Downarrow \qquad \Downarrow$

$Ax = \lambda x \quad t_x \cdot / \; Ay = \mu y \;\Rightarrow$

$t(Ax) = \lambda\, {}^t x \Leftrightarrow {}^t x \cdot {}^t A = \lambda\, {}^t x \Rightarrow {}^t x \cdot A = \lambda\, {}^t x \;|\cdot y_{dr}$

$\Rightarrow {}^t x \cdot A \cdot y = \lambda\, {}^t x \cdot y = \lambda \langle x, y \rangle$ ①

$\Rightarrow {}^t x \cdot A \cdot y = \mu \langle x, y \rangle$ ②

① ② $\Rightarrow (\lambda - \mu)\langle x, y \rangle = 0$

$\lambda - \mu \neq 0$

$\Rightarrow \langle x, y \rangle = 0.$

Lema 3

Fie $A \in \mathcal{M}_n(\mathbb{R}) = {}^tA$ și $P_A(\lambda) = (-1)^n (\lambda-\lambda_1)^{m_1}(\lambda-\lambda_2)^{m_2}\dots(\lambda-\lambda_p)^{m_p}$

(cu $m_1+m_2+\dots+m_p = n$ și $\lambda_i \neq \lambda_j, \forall i,j \in \{1,\dots,p\}$, $i \neq j$)

Obs $A = {}^tA \Rightarrow A$ diag ($\Leftrightarrow \forall 1 \le j \le p : m_j = \dim_{\mathbb{R}} V_{\lambda_j}$)

Pt fiecare $1 \le j \le p$, fie $\{v_1^{(j)}, v_2^{(j)}, \dots, v_{m_j}^{(j)}\}$ o bază ON pt V_{λ_j}.

Atunci:

$Q \overset{not}{=} (v_1^{(1)} | \dots | v_{m_1}^{(1)} | v_1^{(2)} | \dots | v_{m_2}^{(2)} | \dots | v_1^{(p)} | \dots | v_{m_p}^{(p)})$ este o matr OG pt care $Q^{-1} \cdot A \cdot Q = \begin{pmatrix} \lambda_1 \overbrace{\dots}^{m_1\cdot\lambda_1} \lambda_1 & & & \\ & \lambda_2 \overbrace{\dots}^{m_2\cdot\lambda_2} \lambda_2 & & \\ & & \ddots & \\ & & & \lambda_p \overbrace{\dots}^{m_p} \lambda_p \end{pmatrix} \overset{not}{=} \Delta$

Dem:

$AQ = (Av_1^{(1)} | Av_{m_1}^{(1)} | Av_1^{(2)} | \dots | Av_{m_2}^{(2)} | \dots | Av_1^{(p)} | \dots | Av_{m_p}^{(p)})$

$Q^{-1} = {}^tQ$

$\Rightarrow {}^tQAQ = \begin{pmatrix} {}^tv_1^{(1)} \cdot Av_1^{(1)} & {}^tv_1^{(1)} \cdot A \cdot v_2^{(1)} & \dots & {}^tv_1^{(1)} A v_{m_1}^{(1)} & \dots & {}^tv_1^{(1)} A v_{m_p}^{(p)} \\ \underbrace{\lambda_1 v_1^{(1)}}_{\lambda_1 \langle v_1^{(1)}, v_1^{(1)} \rangle = \lambda_1} & \lambda_1 \langle v_1^{(1)}, v_2^{(1)} \rangle = 0 & & \lambda_1 \langle v_1^{(1)}, v_{m_1}^{(1)} \rangle = 0 & & \lambda_p \langle v_1^{(1)}, v_{m_p}^{(p)} \rangle = 0 \\ \dots & \dots & & \dots & & \dots \end{pmatrix}$

$= \Delta$

${}^tQ = \begin{pmatrix} {}^tv_1^{(1)} \\ \vdots \\ {}^tv_{m_p}^{(p)} \end{pmatrix}$

Forme pătratice

Def: Fie $A = {}^tA \in \mathcal{M}_n(\mathbb{R})$. Atunci $f: \mathbb{R}^n (= \mathcal{M}_{n,1}(\mathbb{R})) \to \mathbb{R}$, def prin

$f(\underset{\mathbb{R}^n}{x}) = \langle Ax, x \rangle = {}^t(Ax) \cdot x = {}^t x \cdot A \cdot x$

$A = (a_{ij})_{1 \le i,j \le n}$

$= a_{11}x_1^2 + a_{22}x_2^2 + \dots + a_{nn}x_n^2 + 2\sum_{1 \le i < j \le n} a_{ij} x_i x_j$

. At f sn. FORMĂ PĂTRATICĂ asociată lui A.

Ex Să se scrie mat A asociată formei pătratice:

$$f(x) = 3x_3^2 - 8x_1x_2 - 4x_1x_3 - 4x_2x_3, \quad f: \mathbb{R}^3 (= \mathcal{M}_{3,1}(\mathbb{R})) \to \mathbb{R}$$

$$\Downarrow$$

$$A = \begin{pmatrix} 0 & -4 & -2 \\ -4 & 0 & -2 \\ -2 & -2 & 3 \end{pmatrix}$$

Def: Fie $f: \mathbb{R}^n (= \mathcal{M}_{n,1}(\mathbb{R})) \to \mathbb{R}$ o formă pătratică și

$\mathcal{B} = \{v_1, \dots, v_n\} \subseteq \mathbb{R}^n (= \mathcal{M}_{n,1}(\mathbb{R}))$ o bază în $\mathbb{R}^n$.

Spunem că f are forma canonică relativ la baza $\mathcal{B}$ $\Leftrightarrow$

$(\exists)\ \alpha_1, \dots, \alpha_n \in \mathbb{R}$ a.î. $(\forall) y_1, \dots, y_n \in \mathbb{R}$ să aibă loc:

$$f(y_1v_1 + y_2v_2 + \dots + y_nv_n) = \alpha_1 y_1^2 + \alpha_2 y_2^2 + \dots + \alpha_n y_n^2.$$

Teorema 4

Fie $A = {}^tA \in \mathcal{M}_n(\mathbb{R})$ și $f: \mathbb{R}^n \to \mathbb{R}$, formă pătr. asociată: $f(x) = {}^tx \cdot A \cdot x$.

Fie $Q \in \mathcal{M}_n(\mathbb{R})$ o mat. ort. a.î. $Q^{-1} \cdot A \cdot Q = \Delta \Leftrightarrow$

(am dem $(\exists)$ lui Q în Teorema 1)

$(Q_1 | Q_2 | \dots | Q_n)$ $(=)$ $\{Q_1, \dots, Q_n\} \subseteq \mathbb{R}^n$ o bază ort în $(\mathbb{R}^n)$.

Atunci: $(\forall) y_1, \dots, y_n \in \mathbb{R}$ fie $y = (y_1, \dots, y_n)$

$$f(y_1Q_1 + \dots + y_nQ_n) =$$

fie $x = Qy \Rightarrow f(x) = {}^tx \cdot A \cdot x = {}^t(Qy) \cdot A \cdot Qy$

$$= {}^ty \cdot {}^tQ \cdot A \cdot Qy$$

$$= {}^ty \cdot \Delta y$$

$$\Delta = \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix}$$

$$= (y_1 \dots y_n) \underbrace{\begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix} \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix}}$$

$$\begin{pmatrix} \lambda_1 y_1 \\ \vdots \\ \lambda_n y_n \end{pmatrix}$$ (cu $\lambda_1, \dots, \lambda_n$ nu neap dist)

$$= \sum_{j=1}^{n} \lambda_j y_j^2$$

1 dar $f(y_1Q_1 + \dots + y_nQ_n) = \lambda_1 y_1^2 + \dots + \lambda_n y_n^2$

$\Rightarrow$ f are forma canonică în rap cu baza $\{Q_1, \dots, Q_n\}$ a lui $\mathbb{R}^n$.

mat. transf. ortogonală (rezultă și metoda valorilor și vectorilor proprii), să se reducă la forma canonică forma pătratică

$f: \mathbb{R}^3 (= \mathcal{M}_{3,1}(\mathbb{R})) \to \mathbb{R}$; $f(x) = 3x_1^2 + 8x_1x_2 - 4x_1x_3 - 4x_2x_3$.

Pentru a reduce și a trata lui $\mathbb{R}^3$ în rap cu care f are rasp. formă canonică

$$\Rightarrow A = \begin{pmatrix} 0 & -4 & -2 \\ -4 & 0 & -2 \\ -2 & -2 & 3 \end{pmatrix}$$

$$\det(A - \lambda I_3) = \begin{vmatrix} -\lambda & -4 & -2 \\ -4 & -\lambda & -2 \\ -2 & -2 & 3-\lambda \end{vmatrix} = \cancel{\lambda^2(3-\lambda) + 6}$$

$$= -\begin{vmatrix} \lambda & 4 & 2 \\ 4 & \lambda & 2 \\ 2 & 2 & \lambda - 3 \end{vmatrix} = -\big(\lambda^2(\lambda-3) + 16 + 16 - 4\lambda - 4\lambda - 16(\lambda-3) \big)$$

$$= \begin{vmatrix} 4-\lambda & -4 & -2 \\ \lambda - 4 & -\lambda & -2 \\ 0 & -2 & 3-\lambda \end{vmatrix} = (4-\lambda) \begin{vmatrix} 1 & -4 & -2 \\ -1 & -\lambda & -2 \\ 0 & -2 & 3-\lambda \end{vmatrix}$$

$\uparrow C_1 \to C_1 + C_2$

$$= (4-\lambda) \begin{vmatrix} 0 & -4 & -2 \\ 0 & -\lambda - 4 & -4 \\ 0 & -2 & 3-\lambda \end{vmatrix}$$

$$= (4-\lambda)[(\lambda+4)(\lambda-3) - 8] = (4-\lambda)(\lambda^2 + \lambda - 20)$$

$\lambda_1 = 4 \quad \lambda_2 = 4 \quad \lambda_3 = -5$

$\Rightarrow d(A) \in \{-5, 4\}$ cu $m_{\lambda=-5} = 1$ și $m_{\lambda=4} = 2$.

În cont, det bază ON pt $V_{\lambda=-5}$ și $V_{\lambda=4}$.

$\lambda = -5$: $V_{\lambda=-5}$: $A - \lambda I_3 = A + 5I_3 = \begin{pmatrix} 5 & -4 & -2 \\ -4 & 5 & -2 \\ -2 & -2 & 8 \end{pmatrix} \xrightarrow{L_1 \to L_1 + L_2} \begin{pmatrix} 1 & 1 & -4 \\ -4 & 5 & -2 \\ -2 & -2 & 8 \end{pmatrix}$

$\xrightarrow[L_3 \to L_3 + 2L_1]{L_2 \to L_2 + 4L_1} \begin{pmatrix} 1 & 1 & -4 \\ 0 & 9 & -18 \\ 0 & 0 & 0 \end{pmatrix} \xrightarrow{L_2 \to L_2 \cdot \frac{1}{9}} \begin{pmatrix} 1 & 1 & -4 \\ 0 & 1 & -2 \\ 0 & 0 & 0 \end{pmatrix} \xrightarrow{L_1 \to L_1 - L_2} \begin{pmatrix} 1 & 0 & -2 \\ 0 & 1 & -2 \\ 0 & 0 & 0 \end{pmatrix}$

Atunci $(A + 5I_3)\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \Leftrightarrow \begin{cases} x - 2z = 0 \\ y - 2z = 0 \end{cases} \Leftrightarrow \begin{cases} x = 2z \\ y = 2z \end{cases}$

$\Rightarrow \begin{pmatrix} x \\ y \\ z \end{pmatrix} = z\begin{pmatrix} 2 \\ 2 \\ 1 \end{pmatrix} \Rightarrow \{v_1\}$ o bază pt $V_{\lambda=-5}$, iar $\left\{ \frac{v_1}{\|v_1\|} \right\}$ pt că ai doar un vect pt proc G-S

$= \left\{ \frac{1}{3}\begin{pmatrix} 2 \\ 2 \\ 1 \end{pmatrix} \right\}$ este o bază ON pt $V_{\lambda=-5}$

Forme pătratice (cont.)

Def.: Fie $f:\mathbb{R}^n \to \mathbb{R}$ o formă pătratică. (poz. si neg. def...)

L1: Fie A o matr. sim. $\in \mathcal{M}_n(\mathbb{R})$. Atunci următoarele sunt echivalente:

a) $(\forall)\ x \in \mathbb{R}^n \setminus \{0_{n,1}\}$, ${}^tX \cdot A \cdot X > 0$. (${}^tX \cdot A \cdot X < 0$)

b) $\sigma(A) \subseteq (0, +\infty)$ ($\sigma(A) \subseteq (-\infty, 0)$)

c) Dacă $\Delta_k \overset{not}{=} \det((a_{ij})_{1 \le i,j \le k})$, $(\forall) k = \overline{1,n}$, atunci $\Delta_k > 0$, $(\forall) k = \overline{1,n}$,

, unde $A \overset{not}{=} (a_{ij})_{1 \le i,j \le n}$ ($(-1)^k \Delta_k > 0$, $(\forall) k = \overline{1,n}$)

T2: TEOREMA INERȚIEI

Fie f o formă pătr., $f:\mathbb{R}^n \to \mathbb{R}$ și o bază $B_1 = \{u_1, u_2, \dots, u_n\} \subseteq \mathbb{R}^n$;

$B_2 = \{v_1, v_2, \dots, v_n\} \subseteq \mathbb{R}^n = \mathcal{M}_{n,1}(\mathbb{R})$, două baze ale lui $\mathbb{R}^n$ și

$\alpha_1, \dots, \alpha_n$ și $\beta_1, \dots, \beta_n \in \mathbb{R}$ a.î.

$f(y_1u_1 + y_2u_2 + \dots + y_nu_n) = \alpha_1 y_1^2 + \alpha_2 y_2^2 + \dots + \alpha_n y_n^2$.

$f(y_1v_1 + y_2v_2 + \dots + y_nv_n) = \beta_1 y_1^2 + \beta_2 y_2^2 + \dots + \beta_n y_n^2$

, $(\forall) y_1, \dots, y_n \in \mathbb{R}$. Atunci:

$\text{card}\{i \mid i \in \{1,\dots,n\}, \alpha_i > 0\} = \text{card}\{i \mid i \in \{1,\dots,n\}, \beta_i > 0\} \overset{not}{=} p$.

$\text{card}\{j \mid j \in \{1,\dots,n\}, \alpha_j < 0\} = \text{card}\{j \mid j \in \{1,\dots,n\}, \beta_j < 0\} \overset{not}{=} q$.

Obs: 1) Perechea (p,q) s.n. SIGNATURA formei pătr. f.

2) forma pătr. f este poz. def. ($\Leftrightarrow$ f are signătura $(n,0)$)

3) forma pătr. f este neg. def. ($\Leftrightarrow$ f are sig. $(0,n)$)

Ex1: Să se determine $\lambda \in \mathbb{R}$ pt care forma pătratică $f(x) = 3x_1^2 + 4x_2^2 + 5x_3^2 + 4x_1x_2 - \lambda x_1x_3$, $\left(\begin{smallmatrix} x_1 \\ x_2 \\ x_3 \end{smallmatrix}\right)$ ($f:\mathbb{R}^3 \to \mathbb{R}$)

este poz. def.

$\Rightarrow A = \begin{pmatrix} 3 & 2 & -\frac{\lambda}{2} \\ 2 & 4 & 0 \\ -\frac{\lambda}{2} & 0 & 5 \end{pmatrix}$!!!

Sol: $\Delta_1 = 3 > 0$; $\Delta_2 = 8$; $\Delta_3 = 60 + 0 + 0 - \frac{4\lambda^2}{4} - 20$

$= 40 - \lambda^2 > 0$

$\lambda^2 < 40 \Rightarrow \boxed{\lambda \in (-2\sqrt{10}, 2\sqrt{10})}$

Pt $\lambda=4$: $A-\lambda I_3 = A-4I_3 = \begin{pmatrix} -4 & -4 & -2 \\ -4 & -4 & -2 \\ -2 & -2 & -1 \end{pmatrix} \to \begin{pmatrix} 2 & 2 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$

$\Rightarrow (A-4I_3)\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \Leftrightarrow \{ 2x+2y+z=0$

$\Leftrightarrow z = -2x-2y$

$\Rightarrow \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x \\ y \\ -2x-2y \end{pmatrix} = x\underbrace{\begin{pmatrix} 1 \\ 0 \\ -2 \end{pmatrix}}_{a_2} + y\underbrace{\begin{pmatrix} 0 \\ 1 \\ -2 \end{pmatrix}}_{a_3}$

~~$\Rightarrow$ o bază ON pt $V_{\lambda=4} = \{ e_2 = \frac{v_2}{\|v_2\|} =$~~

Pt a obț. o bază ON pt $V_{\lambda=4}$, aplic. G-S luăm $\{a_2, a_3\}$

$v_2 \overset{\text{not}}{=} a_2 = \begin{pmatrix} 1 \\ 0 \\ -2 \end{pmatrix}$; $v_3 = a_3 - \frac{\langle a_3, v_2 \rangle}{\langle v_2, v_2 \rangle} v_2$

$= \begin{pmatrix} 0 \\ 1 \\ -2 \end{pmatrix} - \frac{4}{5}\begin{pmatrix} 1 \\ 0 \\ -2 \end{pmatrix}$

$= \frac{1}{5}\begin{pmatrix} 0 \\ 5 \\ -10 \end{pmatrix} - \frac{1}{5}\begin{pmatrix} 4 \\ 0 \\ -8 \end{pmatrix}$

$= \frac{1}{5}\begin{pmatrix} -4 \\ 5 \\ -2 \end{pmatrix}$

2) $\{v_2, v_3\}$ bază OG pt $V_{\lambda=4}$.

3) $\{\frac{v_2}{\|v_2\|}, \frac{v_3}{\|v_3\|}\}$ bază ON pt $V_{\lambda=4}$.

4) $\{ e_2 \overset{\text{not}}{=} \frac{v_2}{\|v_2\|} = \frac{1}{\sqrt{5}}\begin{pmatrix} 1 \\ 0 \\ -2 \end{pmatrix}; e_3 \overset{\text{not}}{=} \frac{v_3}{\|v_3\|} = \frac{1}{3\sqrt{5}}\begin{pmatrix} -4 \\ 5 \\ -2 \end{pmatrix} \}$

5) f are forma canonică $f(y_1e_1 + y_2e_2 + y_3e_3) = -5y_1^2 + 4y_2^2 + 4y_3^2$, (! contează ordinea)

relativ la baza ON: $\{e_1 = \frac{1}{3}\begin{pmatrix} 2 \\ 2 \\ 1 \end{pmatrix}, e_2 = \frac{1}{\sqrt{5}}\begin{pmatrix} 1 \\ 0 \\ -2 \end{pmatrix}, e_3 = \frac{1}{3\sqrt{5}}\begin{pmatrix} -4 \\ 5 \\ -2 \end{pmatrix}\}$

a lui $\mathbb{R}^3$.

Def: Fie $f: \mathbb{R}^n \to \mathbb{R}$ o formă pătratică. Atunci

a) f s.n. pozitiv definită $\Leftrightarrow (\forall)\ x \in \mathbb{R}^n (= \mathcal{M}_{n,1}(\mathbb{R})) \setminus \{0_{n,1}\}$ are loc $f(x) > 0$.

b) f s.n. negativ definită $\Leftrightarrow (\forall)\ x \in \mathbb{R}^n (= \mathcal{M}_{n,1}(\mathbb{R})) \setminus \{0_{n,1}\}$ are loc $f(x) < 0$.

Teorema Sylvester: Fie $A = {}^tA = (a_{ij})_{1 \le i,j \le n} \in \mathcal{M}_n(\mathbb{R})$.

Atunci, următ. sunt $\Leftrightarrow$

a) $(\forall)\ x \in \mathbb{R}^n \setminus \{0_{n,1}\}$: ${}^tx \cdot Ax > 0$ $({}^tx \cdot A \cdot x < 0)$

b) $\sigma(A) \subseteq (0, +\infty)$ $(\sigma(A) \subseteq (-\infty, 0))$

c) Dacă pt $1 \le k \le n$, not $\Delta_k \overset{\text{not}}{=} \det((a_{ij})_{1 \le i,j \le k})$, atunci $\Delta_k > 0$ $(\forall) k = \overline{1,n}$
$((-1)^k \Delta_k > 0, (\forall) k = \overline{1,n})$

Ecuații diferențiale liniare de ordin superior cu coeficienți constanți

Forma generală:

$$a_n y^{(n)} + a_{n-1} y^{(n-1)} + \dots + a_1 y' + a_0 y = f(x) \quad (1)$$

$a_n \neq 0 \Rightarrow$ ec. dif de ordin n

$a_0, \dots, a_n \in \mathbb{R}$; f fct. cont.; $f: I \subseteq \mathbb{R} \to \mathbb{R}$ (interval)

Teorema de existență și unicitate a soluției pentru problema Cauchy asociată

Dacă $f: I \to \mathbb{R}$ cont, $x_0 \in I$ și $y_0, y_1, \dots, y_{n-1}$ fix, at $\exists!$ $y: I \to \mathbb{R}$ de n ori derivabilă a.î. să fie sol pt ec 1 și $y(x_0) = y_0$, $y'(x_0) = y_1$, $\dots$, $y^{(n-1)}(x_0) = y_{n-1}$

Alg. de rezolvare a ec. dif (1)

Fie

(2) $a_n y^{(n)} + a_{n-1} y^{(n-1)} + \dots + a_1 y' + a_0 y = 0$ ec. dif. omogenă asoc. ec (1).

Sol. gen a ec. (1) este: $y(x) = y_h(x) + y_p(x)$

$y_h(x)$ → sol part (1)

$y_p(x)$ → sol gen (2)

Det lui y_h:

Fie $S = \{\varphi \mid \varphi: I \to \mathbb{R}, \varphi$ sol pt ec dif (2)$\}$. At se dem S este sp. vect $\subseteq \mathbb{R}$ de dimensiune n.

Ec. caracteristică asociată lui (2):

$$a_n \lambda^n + a_{n-1} \lambda^{n-1} + \dots + a_1 \lambda + a_0 = 0 \quad (3)$$

i) Dacă $\lambda \in \mathbb{R}$ este o rădăcină multiplă de ordin $m \geq 1$ a ec. caract (3), ei îi corespund sol. liniar independente:

$f_1(x) = e^{\lambda x}$, $f_2(x) = x e^{\lambda x}$, $f_3(x) = x^2 e^{\lambda x}$, $\dots$, $f_m(x) = x^{m-1} e^{\lambda x}$ ale ecuației dif (2).

ii) Dacă $\lambda_{1,2} = \alpha \pm i\beta$ ($\alpha, \beta \in \mathbb{R}$) sunt rădăcini multiple de ordin $s \geq 1$ pt ec. caract (3), lor le corespund sol liniar indep ale ec dif (2) unm:

$\varphi_1 = e^{\alpha x}\cos\beta x$, $\varphi_2(x) = e^{\alpha x}\sin\beta x$, $\varphi_3(x) = xe^{\alpha x}\cos\beta x$, $\varphi_4(x) = xe^{\alpha x}\sin\beta x$, ... , $\varphi_{2s-1}(x) = x^{s-1}e^{\alpha x}\cos\beta x$, $\varphi_{2s}(x) = x^{s-1}e^{\alpha x}\sin\beta x$.

Reuniunea tuturor solutiilor liniar indep. obț. pornind de la sol. ec. caract coresp. o rădăc pt S.

$y_{sg}(x) = C_1\varphi_1(x) + C_2\varphi_2(x) + \dots + C_n\varphi_n(x)$; $C_1, \dots, C_n \in \mathbb{R}$

Ex 2: Să se det. sol. gen. pt urm. ec liniare omogene:

a) $y^{(5)} - y^{(4)} + y^{(3)} = 0$

b) $y^{(5)} - 2y^{(4)} + 2y^{(3)} - 4y'' + y' - 2y = 0$

c) $y^{(4)} - 4y^{(3)} + 10y'' - 20y' + 25y = 0$.

---

a) Ec. caract. as:

~~$y^5 - y^4 + y^3 = 0$~~ $\lambda^5 - \lambda^4 + \lambda^3 = 0$

$\lambda^3(\lambda^2 - \lambda + 1) = 0$; sol sunt $\lambda_1 = 0$, $m_{\lambda_1} = 3$

~~$\lambda^3(\lambda-2)(\lambda+1) = 0$~~ $\lambda_{2,3} = \frac{1 \pm i\sqrt{3}}{2}$, $m_{\lambda_{2,3}} = 1$

S

$\Rightarrow$ lui $\lambda_1 = 0$ îi corespund sol. liniar indep: $1, x, x^2$.

$\lambda_{2,3} = \frac{1 \pm i\sqrt{3}}{2}$ îi corresp: $e^{\frac{x}{2}}\cos\left(\frac{\sqrt{3}}{2}x\right)$, $e^{\frac{x}{2}}\sin\left(\frac{\sqrt{3}}{2}x\right)$

($\alpha = 1/2$, $\beta = \frac{\sqrt{3}}{2}$ ... scalează cel exponențial)

$\Rightarrow y_{sg}(x) = C_1 + C_2x + C_3x^2 + C_4e^{\frac{x}{2}}\cos\left(\frac{\sqrt{3}}{2}x\right) + C_5e^{\frac{x}{2}}\sin\left(\frac{\sqrt{3}}{2}x\right)$.

; $C_1, \dots, C_5 \in \mathbb{R}$.

(pt $\lambda_1 = 0$: $\rho_1 = e^{0\cdot x} = 1$; $\varphi_2 = xe^{0\cdot x} = x$; $\rho_3 = x^2e^{0\cdot x} = x^2$)

b) Ec. caract as:

$\lambda^5 - 2\lambda^4 + 2\lambda^3 - 4\lambda^2 + \lambda - 2 = 0$

$\lambda^4(\lambda-2) + 2\lambda^2(\lambda-2) + (\lambda-2) = 0$

$(\lambda-2)(\lambda^4 + 2\lambda^2 + 1) = 0$

$(\lambda-2)(\lambda^2+1)^2 \rightarrow \lambda_1 = 2$, $m_{\lambda_1} = 1$

$\rightarrow \lambda_{2,3} = \pm i$, $m_{\lambda_{2,3}} = 2$.

1) $\varphi_1(x) = e^{2x}$

($\alpha = 0$, $\beta = 1$) $\varphi_2(x) = \cos x$; $\varphi_3(x) = \sin x$; $\varphi_4(x) = x\cos x$; $\varphi_5(x) = x\sin x$

$\alpha, \beta \in \mathbb{R}$

2) $y_{sg}(x) (= y(x)) = C_1\rho_1(x) + C_2\varphi_2(x) + \dots + C_5\rho_5(x) =$

$= C_1e^{2x} + (C_2 + C_4x)\cos x + (C_3 + C_5x)\sin x$; $C_1, \dots, C_5 \in \mathbb{R}$.

c) $\lambda^4 - 4\lambda^3 + 14\lambda^2 - 20\lambda + 25 = 0 \Leftrightarrow (\lambda^2 - 2\lambda + 5)^2 = 0$

$(\lambda^2)^2 \quad 2\cdot 2\cdot\lambda^2\cdot\lambda \quad 5^2$

$\lambda^4 \; 4\lambda^2 \; 25$

$-4\lambda^3 + 10\lambda^2 - 20\lambda$

$\Rightarrow \lambda_{1,2} = \frac{2 \pm 4i}{2} = 1 \pm 2i$, $m_{\lambda_1} = m_{\lambda_2} = 2$.

$\alpha = 1$

$\beta = 2$

$\Rightarrow$ e) $e^x\cos 2x, e^x \sin 2x, xe^x\cos 2x, xe^x \sin 2x$.

$\Rightarrow$ Sol. generală este:

$y(x) = e^x [(c_1 + c_2 x)\cos 2x + (c_3 + c_4 x)\sin 2x]$; $c_1, \dots, c_4 \in \mathbb{R}$

Det. lui y_p

I Met. coef. nedet (COEFICIENȚILOR NEDETERMINAȚI)

$\rightarrow$ se aplică doar când f este **cvasipolinom** sau sumă de cvasipol.

$\Downarrow$

$f(x) = e^{\alpha x}(\underbrace{P_n(x)}_{\text{pol de gr } n}\cos\beta x + \underbrace{Q_m(x)}_{\text{pol de gr } m}\sin\beta x)$

, $\alpha, \beta \in \mathbb{R}$ cu $P_n(x), Q_m(x) \in \mathbb{R}[X]$.

Fie s ord. de multiplicitate al lui $\alpha + i\beta$ ca rădăcină a ec. caract. (convenție: $s = 0$ dacă $\alpha + i\beta$ nu este răd. a ec. caract)

Fie $N \overset{\text{not}}{=} \max\{m, n\}$. Căutăm y_p de forma:

$$y_p(x) = x^s e^{\alpha x}[\tilde{P}_N(x)\cos\beta x + \tilde{Q}_N(x)\sin\beta x],$$

unde $\tilde{P}_n(x), \tilde{Q}_n(x) \in \mathbb{R}_N[X]$ sunt pol. cu coef. nedeterm. care se obț. înlocuind y_p în ec. dif (1) și identificând coeficienții

Form. lui Leibniz:

Fie $f, g \in \mathcal{C}^n(I)$. Atunci

$$\boxed{(f\cdot g)^{(n)}(x) = \sum_{k=0}^{n} C_n^k\, g^{(k)}(x)\cdot f^{(n-k)}(x)}$$

Ex: Să se rez. ec. dif neomogenă:

$$y'' - 9y = e^{3x}\cos x$$

$\boxed{y_h}$: $\lambda^2 - 9 = 0 \Leftrightarrow \lambda_{1,2} = \pm 3$ cu $m_{\lambda=1, \lambda=2} = 1$.

$\Rightarrow \{f_1(x) = e^{3x}, f_2(x) = e^{-3x}\}$ bază în sp. sol. ec. dif omogene asociată $\Rightarrow y_h(x) = c_1 f_1(x) + c_2 f_2(x)$

$= c_1 e^{3x} + c_2 e^{-3x}$; $c_1, c_2 \in \mathbb{R}$

$= e^{3x}\cos x = e^{\alpha x}[P_n(x)\cos\beta x + Q_m(x)\sin\beta x]$

$\Downarrow$

$a=3 \; ; \beta=1 \Rightarrow \alpha+i\beta = 3+i$ nu e rădăcină a ec. caract $\Rightarrow s=0$

$P_n(x)=1$; $Q_m(x)=0$

$n=0$ $\quad$ $m=-\infty$

$N \overset{not}{=} \max\{m,n\} = 0$.

$\swarrow$ 0 nu este sol. ec. dif. neomo.

$\Rightarrow y_p$ are forma

$$\boxed{y_p(x) = e^{3x}(a\cos x + b\sin x)}$$

$y_p'' - 9y_p = e^{3x}\cos x$ (1)

$y_p(x) = e^{3x}(a\cos x + b\sin x)$ (3)

$(f\cdot g)^{(2)}(x) \overset{Leib}{=} f''(x)\cdot g(x) + 2f'(x)\cdot g'(x) + f(x)\cdot g''(x)$] $\Rightarrow$

$\Rightarrow y_p''(x) = 9e^{3x}(a\cos x + b\sin x) + 6e^{3x}(-a\sin x + b\cos x)$

$+ e^{3x}(-a\cos x - b\sin x)$ (2) (se simplif e^{3x})

(2), (3) $\to$ (1) $\Rightarrow 6(-a\sin x + b\cos x) + (-a\cos x - b\sin x) = \cos x$ $\Big/ \Rightarrow$

$\{\sin x, \cos x\}$ SLI

$\begin{cases} -6a - b = 0 \\ 6b - a = 1 \;/\cdot(-6) \Rightarrow -36b + 6a = -6 \end{cases}$ (+)

$+37b = +6 \Rightarrow b = \frac{6}{37}$

$\Rightarrow a = -\frac{1}{37}$

$$\Rightarrow \boxed{y_p = -\frac{1}{37}e^{3x}(-\cos x + 6\sin x)}$$ ← sol gen a ec. dif de rezolvat

$y(x) = y_{\hat{o}}(x) + y_p(x)$

$= c_1 e^{3x} + c_2 e^{-3x} + \frac{e^{3x}}{37}(-\cos x + 6\sin x)$; $c_1, c_2 \in \mathbb{R}$

Temă:

Să se rez. ec. dif lin. neomo:

$y'' - 5y' = 3x^2 + \sin 5x$

$y_p = (y_p)_1 + (y_p)_2$; unde $(y_p)_1$ este o sol part a ec. dif $y'' - 5y' = 3x^2$

$(y_p)_2$ ← ——— ” $y'' - 5y = \sin 5x$

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$y'' - 5y' = 3x^2 + \sin 5x$

Suma de cvasi polinoame

$\Rightarrow y(x) = y_h(x) + y_{p_1}(x) + y_{p_2}(x)$.

y_h: ec. caract. omog.: $\lambda^2 - 5\lambda = 0$

$\lambda(\lambda - 5) = 0 \Rightarrow \begin{cases} \lambda_1 = 0, m_{\lambda=1} = 1 \\ \lambda_2 = 5, m_{\lambda=5} = 1 \end{cases}$; $\lambda_1, \lambda_2 \in \mathbb{R}$.

$\Rightarrow f_1(x) = 1$; $f_2(x) = e^{5x}$

$\Rightarrow \boxed{y_h = C_1 + C_2 e^{5x}}$; $C_1, C_2 \in \mathbb{R}$.

y_{p1}: $y'' - 5y' = 3x^2 \Rightarrow f(x) = 3x^2 = e^{\alpha x}[P_n(x)\cos\beta x + Q_m(x)\sin\beta x]$

$\Downarrow$

$\alpha = 0, \beta = 0 \Rightarrow \alpha + \beta i = 0$, nod. a ec caract $\Rightarrow S = 1$.

$\Downarrow$

$P_n(x) = 3x^2$; $Q_m(x) = 0$

$\Downarrow$; $\Downarrow$

$n = 2$; $m = -\infty$

$\Rightarrow \boxed{N = 2}$

$\Rightarrow y_{p_1}(x) = x^1 \cdot [(a + bx + cx^2)] = ax + bx^2 + cx^3$.

$y_{p_1}'' - 5y_{p_1}' = 3x^2$ (1)

$y_{p_1}'(x) = a + 2bx + 3cx^2$ (2)

$y_{p_1}''(x) = 2b + 6cx$ (3)

(2), (3) $\rightarrow$ (1) $\Rightarrow 2b + 6cx - 5a - 10bx - 15cx^2 = 3x^2$

$\Rightarrow \begin{cases} -15c = 3 \Rightarrow \boxed{c = -\frac{1}{5}} \\ 2b - 5a = 0 \Rightarrow -5a = 6 \Rightarrow \boxed{a = -\frac{6}{5}} \\ 6c - 10b = 0 \Rightarrow -10b = 30 \Rightarrow \boxed{b = -3} \end{cases}$

$\Rightarrow \boxed{y_{p1}(x) = -\frac{6}{5}x - 3x^2 - 5x^3}$

y_{p_2}: $y'' - 5y' = \sin 5x \Rightarrow f(x) = \sin 5x = e^{\alpha x}[P_n(x)\cos\beta x + Q_m(x)\sin\beta x]$

$\Downarrow$

$\alpha = 0, \beta = 5 \Rightarrow \alpha + \beta i = 5i$, nu este nod a ec. caract $\Rightarrow S = 0$

$\Downarrow$

$P_n(x) = 0$; $Q_m(x) = 1$

$\Downarrow$; $\Downarrow$

$n = -\infty$; $n = 0$

$\Rightarrow N = 0$

$\Rightarrow y_{p_2}(x) = a\cos 5x + b\sin 5x$

$y'' - 5y' = \sin 5x$

$y'(x) = -5a\sin 5x + 5b\cos 5x$

$y''(x) = -25a\cos 5x - 25b\sin 5x$

$\Rightarrow (5b - 25a)\cos 5x + (-5a - 25b)\sin 5x = \sin 5x$

$\Rightarrow \begin{cases} 5b - 25a = 0 \Rightarrow b - 5a = 0 \Rightarrow b = 5a \Rightarrow a = \frac{b}{5} \\ 5a + 25b = 1 \end{cases}$

$\Rightarrow 26b = 1 \Rightarrow b = \frac{1}{26} \Rightarrow a = \frac{1}{130}$

$$\Rightarrow \boxed{y_{p_2}(x) = \frac{1}{130}\left(\cos 5x + 5\sin 5x\right)}$$

$\Rightarrow y(x) = y_h(x) + y_{p_1}(x) + y_{p_2}(x)$

$\Rightarrow y(x) = c_1 + c_2 e^{5x} - \frac{6}{5}x - 3x^2 - 5x^3 + \frac{1}{130}(\cos 5x + 5\sin 5x)$; $c_1, c_2 \in \mathbb{R}$.

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Ex: Fol. transf. ortog., să se reducă la forma cano, forma pătr:

$f: \mathbb{R}^3 \to \mathbb{R}$

$f(x) = -5x_1^2 + 5x_2^2 + 5x_3^2 - 4(-2x_1x_2 - 2x_1x_3 - 2x_2x_3)$

, unde $x \in \mathbb{R}^3$ rel. la care f are resp. forma can.

$$A = \begin{pmatrix} 5 & -1 & -1 \\ -1 & 5 & -1 \\ -1 & -1 & 5 \end{pmatrix}$$

$$P_A(\lambda) = \det(A - \lambda I_3) = \begin{vmatrix} 5-\lambda & -1 & -1 \\ -1 & 5-\lambda & -1 \\ -1 & -1 & 5-\lambda \end{vmatrix} = (5-\lambda)^3 - 1 - 1 - 3(5-\lambda) =$$

$$\xrightarrow{C_1 \to C_1 + C_2 + C_3} \begin{vmatrix} 3-\lambda & -1 & -1 \\ 3-\lambda & 5-\lambda & -1 \\ 3-\lambda & -1 & 5-\lambda \end{vmatrix} = (3-\lambda) \begin{vmatrix} 1 & -1 & -1 \\ 1 & 5-\lambda & -1 \\ 1 & -1 & 5-\lambda \end{vmatrix} =$$

$L_2 \to L_2 - L_1$, $L_3 \to L_3 - L_1$

$$= \begin{vmatrix} 1 & -1 & -1 \\ 0 & 6-\lambda & 0 \\ 0 & 0 & 6-\lambda \end{vmatrix} = (3-\lambda)(\lambda-6)^2$$

$\Downarrow$

$\sigma(A) \in \{3, 6\}$

cu $m_{\lambda=3} = 1$; $m_{\lambda=6} = 2$.

$$\lambda = 3 \Rightarrow V_{\lambda=3} = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 \;\middle|\; A\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \lambda \begin{pmatrix} x \\ y \\ z \end{pmatrix} \right\}$$

$$= \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 \;\middle|\; (A - 3I_3)\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \right\}.$$

$$A - 3I_3 = \begin{pmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{pmatrix} \xrightarrow{L_1 \leftrightarrow L_2} \begin{pmatrix} -1 & 2 & -1 \\ 2 & -1 & -1 \\ -1 & -1 & 2 \end{pmatrix} \xrightarrow[L_3 \to L_3 - L_1]{L_2 \to L_2 + 2L_1} \begin{pmatrix} -1 & 2 & -1 \\ 0 & 3 & -3 \\ 0 & -3 & 3 \end{pmatrix}$$

$$\to \begin{pmatrix} -1 & 2 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix} \to \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix}$$

Acum: $(A - 3I_3)\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \Leftrightarrow \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \Leftrightarrow \begin{cases} x - z = 0 \\ y - z = 0 \end{cases} \Leftrightarrow$

$\Leftrightarrow \begin{pmatrix} x \\ y \\ z \end{pmatrix} = z \underbrace{\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}}_{v}$ $\Rightarrow \{v_1\}$ este o bază pt $V_{\lambda=3}$ și

$\{q_1 \overset{\text{not}}{=} \frac{v_1}{\|v_1\|} = \frac{1}{\sqrt{3}}\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}\}$ este o bază ON pt $V_{\lambda=3}$

$V_{\lambda=6} = \{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 \mid (A - 6\tilde{I}_3) \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \}$.

$A - 6\tilde{I}_3 = \begin{pmatrix} -1 & -1 & -1 \\ -1 & -1 & -1 \\ -1 & -1 & -1 \end{pmatrix} \to \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix} \to \begin{pmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$

$\Rightarrow (A - 6\tilde{I}_3) \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$ (=) $x + y + z = 0$

$\Rightarrow x = -y - z$ $\quad y = -x - z$

$\Rightarrow \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x \\ -x - z \\ z \end{pmatrix} = x \underbrace{\begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}}_{v_2} + z \underbrace{\begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix}}_{v_3}$

$\Rightarrow \{v_2, v_3\}$ obtinut pt $V_{\lambda=6}$.

$\{v_2, v_3\}$ li $\Rightarrow$ Aplic. proc. G-S lui $V_{\lambda=6}$.

$w_2 = v_2$

$w_3 = v_3 - \frac{\langle v_3, w_2 \rangle}{\langle w_2, w_2 \rangle} w_2 = v_3 - \frac{1}{2} \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}$

$= \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix} - \frac{1}{2} \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} = \frac{1}{2} \left(\begin{pmatrix} 0 \\ -2 \\ 2 \end{pmatrix} - \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} \right)$

$= \frac{1}{2} \begin{pmatrix} -1 \\ -1 \\ 2 \end{pmatrix}$ fără la normă

$\Rightarrow \left\{ \frac{w_2}{\|w_2\|} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}, \frac{w_3}{\|w_3\|} = \frac{1}{\sqrt{6}} \begin{pmatrix} -1 \\ -1 \\ 2 \end{pmatrix} \right\}$ bază ON pt $V_{\lambda=6}$.

$\|\cdot\|$ dat de q_2 $\quad \|\cdot\|$ dat de q_3

$\Rightarrow f(y_1 e_1 + y_2 e_2 + y_3 e_3) = 3y_1^2 + 6y_2^2 + 6y_3^2$, $\forall y_1, y_2, y_3 \in \mathbb{R}$.

Ec. Diferențiale de ordinul I

Forma gen: $y' = f(x, y)$, unde $f : D (\subseteq \mathbb{R}^2) \to \mathbb{R}$ md

$\Leftrightarrow (y'(x) = f(x, y(x)))$ $\quad$ $y \in C^1$

$dy(x) = y'(x) dx$ $\quad \frac{dy}{dx} = f(x, y)$ (=) $P(x, y) dx + Q(x, y) dy = 0$

$df(1,2) = \frac{\partial f}{\partial x}(1,2) dx + \frac{\partial f}{\partial y}(1,2) dy$

$(x_0,y_0)\in\Delta$, pereche fix. a $\begin{cases} y'=f(x,y) \\ y(x_0)=y_0 \end{cases}$ n.n. problema Cauchy.

Teorema de existență și unicitate a soluției unei probleme Cauchy.

$f:\Delta\ (\subseteq \mathbb{R}^2)\neq\emptyset \to \mathbb{R}$ a.î. $(\forall)\ (x,y)\in\Delta$, $(\exists)\ \frac{\partial f}{\partial y}(x,y)$ și $\frac{\partial f}{\partial y}:\Delta\to\mathbb{R}$ să fie continuă (Δ mod, f cont). Atunci: $(\forall)\ (x_0,y_0)\in\Delta$, $(\exists)\ \overline{I}\subseteq\mathbb{R}$ $\overline{I}$ interval deschis

și $\exists!\ y:\overline{I}\to\mathbb{R}$ derivabilă: $(\forall)x\in\overline{I}\Rightarrow(x,y(x))\overset{x_0}{\in}\Delta$

$(\forall)x\in\overline{I}\Rightarrow y'(x)=f(x,y(x))$ și $y(x_0)=y_0$.

Ecuații diferențiale cu variabile separabile

Forma gen: $y'=a(x)\cdot b(y)$, unde $a:\underset{\text{int}}{I}(\subseteq\mathbb{R})\to\mathbb{R}$ cont.

și $b:\underset{\text{int}}{J}(\subseteq\mathbb{R})\to\mathbb{R}$, $b\in C^{(1)}(J)$.

$$\frac{dy}{dx}=a(x)\cdot b(y)$$

Pas 1 Rez. ec. $b(y)=0\Rightarrow y_0,y_1,\dots$

$\Downarrow$

$y=y_0, y=y_1,\dots$ sol constante ec. de rezolvat

Pas 2 Pt $y\in J$ a.î. $b(y)\neq 0$ avem:

$$\frac{y'(x)}{b(y(x))}=a(x)\ (=)\ \int$$

$$\Rightarrow \int\frac{y'(x)}{b(y(x))}dx=\int a(x)dx$$

$y=y(x)\rightsquigarrow dy=y'(x)dx$

$$\Rightarrow \boxed{\int\frac{1}{b(y)}dy=\int a(x)dx}$$

$$\boxed{B(y)=A(x,\underset{\mathbb{R}}{c})}$$

sol. gen. a ec. cu v.s.

Ex 2: Să se det. sol. pt fiecare din urm. ec. dif.

a) $\frac{x}{y}\sqrt{1+y^2}+\sqrt{1+x^2}\cdot y'=0 \quad (=) \sqrt{1+x^2}\cdot y' = -\frac{x}{y}\sqrt{1+y^2}$

$y\neq 0$

$$\frac{y}{\sqrt{1+y^2}}dy = -\frac{x}{\sqrt{1+x^2}}dx \ (\Leftrightarrow \int$$

$$\Leftrightarrow \int \frac{y}{\sqrt{1+y^2}}dy = -\int \frac{x}{\sqrt{1+x^2}}dx$$

$$\Leftrightarrow \sqrt{1+y^2} = -\sqrt{1+x^2} + C, \ C\in \bar{I}\ (\subseteq \mathbb{R})$$

$$\sqrt{1+y^2}+\sqrt{1+x^2} = C, \ \underline{C\in(0,+\infty)} \ !!!$$

$y(\sqrt{3})=\sqrt{3} \Rightarrow C=4. \quad \Leftrightarrow \sqrt{1+y^2} = 4-\sqrt{1+x^2} \Rightarrow (1+y^2) = (4-\sqrt{1+x^2})^2.$

$\Leftrightarrow y=\sqrt{3}$

$\Rightarrow$ Ec. diferențiale omogene $y' = f\left(\frac{y}{x}\right)$, unde $f:\bar{I}\ (\subseteq\mathbb{R}) \to \mathbb{R}$ ind. funcție de clasă C^1

Se rezolvă fol. sch de funcție $\boxed{z=\frac{y}{x}}$ $(y=y(x) \longmapsto z=z(x))$

b) $y' = \frac{4x-3y}{3x-2y}, \ x\neq 0, \ y\neq \frac{3}{2}x$

$$y' = \frac{4-3\left(\frac{y}{x}\right)}{3-2\frac{y}{x}} \ !!!$$

$\Leftarrow$

. Facem sch de funcție $z=\frac{y}{x} \Rightarrow y=zx$

$\Rightarrow y' = z'\cdot x + z$!! z efct !!

$\Rightarrow z'\cdot x + z = \frac{4-3z}{3-2z} \ (\Rightarrow z'\cdot x = \frac{2z-3}{-z+} + \frac{3z-4}{2z-3}$

$$\Rightarrow z'\cdot x = \frac{-2z^2+6z-4}{2z-3}$$

$$\Rightarrow z'\cdot x = \frac{-2(z^2-3z+2)}{2z-3}$$. ec. dif. cu var. sep. în x și z

$z^2-3z+2=0 \ (\Rightarrow (z-1)(z-2)=0 \Rightarrow \begin{cases} z_1=1 \Rightarrow y=x \\ z_2=2 \Rightarrow y=2x \end{cases}$ sol. singulare pt ecuație de rezolvat.

Pt $z\neq\{1,2\}$: $\frac{dz\cdot(2z-3)}{-2(z^2-3z+2)} = \frac{dx}{x} \ /\int$

$$\Leftrightarrow -\frac{1}{2}\int \frac{2z-3}{z^2-3z+2}dz = \int \frac{1}{x}dx$$

$$\Leftrightarrow -\frac{1}{2}\ln|z^2-3z+2| = \ln x + C$$

$$\Leftrightarrow \ln|z^2-3z+2| = -\ln^2 x + C \ (\Leftrightarrow (z^2-3z+2)x^2 = K, \ K\in\mathbb{R}^*$$

$y^2 - 3xy + 2x^2 = k$, $k \in \mathbb{R}^*$ $\Big| \Rightarrow y^2 - 3xy + 2x^2 = k$, $k \in \mathbb{R}$.

$y = x$, $y = 2x$

c) $y' = \frac{y}{x} + \cos^2\left(\frac{y}{x}\right)$; $z = \frac{y}{x} \Leftrightarrow y = zx$

$\Rightarrow z'x + z = z + \cos^2 z$

$\Rightarrow z'x = \cancel{z} + \cancel{z} + \cos^2 z$; $\cos^2 z = 0 \Rightarrow z = \frac{k\pi}{2} \Rightarrow \frac{y}{x} = \frac{k\pi}{2}$, $k \in \mathbb{Z}$, soluții singulare.

$z = \frac{\pi}{2} + k\pi \Rightarrow \frac{y}{x} = \frac{\pi}{2} + k\pi$

$\Rightarrow y = x\left(\frac{\pi}{2} + k\pi\right)$

$y = \frac{xk\pi}{2}$

$\frac{zx}{\cos^2 z}$ $\quad \frac{z'}{\cos^2 z} = \frac{1}{x}$

Pentru $z \neq \frac{\pi}{2} + k\pi$

$\frac{dz}{\cos^2 z} = \frac{dx}{x} \Big| \int \Rightarrow$ $\operatorname{tg} z = \ln|x| + c$

$\operatorname{tg} z - \ln|x| = c$

$\operatorname{tg} \frac{y}{x} - \ln|x| = c$

$\Rightarrow y = x \operatorname{arctg}(\ln|x| + c)$, $c \in \mathbb{R}$

III $y' = f(ax + by + c)$, $f: I \left(\subseteq \mathbb{R}\right) \to \mathbb{R}$, $f \in C^{(1)}(I)$.

interval

$a, b, c \in \mathbb{R}$; $b \neq 0$

Se rez. pol. sch de fcț. $z = ax + by + c$

d) $y' = \frac{8x - 4y + 1}{4x - 2y + 1}$, $y \neq \frac{4x+1}{2}$

$= \frac{\cancel{8}x - 4y + 1}{\cancel{4}x} = \frac{2(4x - 2y + 1) - 1}{4x - 2y + 1}$

$z = 4x - 2y + 1 \Leftrightarrow 2y = 4x - z + 1$

$y = \frac{4x - z + 1}{2} \Rightarrow y' = 2 - \frac{z'}{2}$

Ec de var. derivate:

$\Rightarrow 2 - \frac{z'}{2} = \frac{2z - 1}{z} \Leftrightarrow -\frac{z'}{2} = \frac{2}{z} + \frac{2z - 1}{z} = -\frac{1}{z}$

$-1z' = +\frac{2}{z}$

$z' = \frac{2}{z} \Rightarrow \frac{dz}{dx} = \frac{2}{z}$

$\Leftrightarrow z\,dz = 2\,dx \Big| \int$

$\Rightarrow \frac{z^2}{2} = 2x + c \Leftrightarrow$

$z^2 = 4x + c$, $c \in \mathbb{R}$

(z) $(4x - 2y + 1)^2 = 4x + c$

IV $y' = f\left(\frac{a_1x+b_1y+c_1}{a_2x+b_2y+c_2}\right)$; $f: I \subseteq \mathbb{R} \to \mathbb{R}$, $f \in C^{(1)}(I)$ int. I

unde $a_1, b_1, c_1, a_2, b_2, c_2 \in \mathbb{R}$ a.i. $\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix} \neq 0$.

$\Leftrightarrow$ Fie sist $\begin{cases} a_1x+b_1y+c_1=0 \\ a_2+b_2y+c_2=0 \end{cases}$ comp det.

Fie (x_0, y_0) sol ac. sist. liniar.

Vom rel. sch. de fct. $z = \underset{(z(x))}{y} = \frac{y-y_0}{x-x_0}$

e) $y' = \frac{x-y}{x+y+2}$, $\boxed{y \neq -x-2}$

$\begin{vmatrix} 1 & -1 \\ 1 & 1 \end{vmatrix} = 1+1 = 2 \neq 0$ $\quad \begin{pmatrix} 1 & -1 & 0 \\ 1 & 1 & 2 \end{pmatrix} \overset{L_2 \to L_2 - L_1}{\to} \begin{pmatrix} 1 & -1 & 0 \\ 0 & 2 & 2 \end{pmatrix}$

$\to \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & 1 \end{pmatrix} \to \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}$

$\Rightarrow (x_0, y_0) = (-1, 1)$

$z = \frac{y-1}{x+1}$

$\Rightarrow y = z(x+1)+1$

$\Rightarrow y' = z'(x+1) + z$

$\Rightarrow z'(x+1)+z = \frac{x-z(x+1)+1}{x+z(x+1)+1}$ $\Rightarrow z'(x+1)+z = \frac{(x+1)(1-z)}{(x+1)(1+z)}$

$z'(x+1) = \frac{1-z}{1+z} - z$

$z'(x+1) = \frac{1-z-z-z^2}{1+z} = \frac{1-2z-z^2}{1+z}$

; $1-2z-z^2 = 0$

$z^2+2z-1=0$

$\Delta = 4+4 = 8$

$z_{1,2} = \frac{-2 \pm 2\sqrt{2}}{2}$

$z_{1,2} = -1 \pm \sqrt{2}$

$\Rightarrow$ sol singulare.

$\Rightarrow \begin{cases} y_1 = (-1-\sqrt{2})(x+1)+1 \\ y_2 = (-1+\sqrt{2})(x+1)+1 \end{cases}$

, sol. singulare pt ec.

$\frac{(z^2+2z-1)'}{!} \frac{(z+1)dz}{1-2z-z^2} = \frac{dx}{x+1} \Big| \int$

$-\frac{1}{2}\int \frac{2(z+1)}{z^2+2z-1} dz = \int \frac{1}{x+1} dx$

$z \leftarrow \frac{y-1}{x+1}$; $-\frac{1}{2}\ln|z^2+2z-1| = \ln|x+1| + \beta$

$-\frac{1}{2}\ln\left|\left(\frac{y-1}{x+1}\right)^2 + 2\frac{y-2}{x+1} - 1\right| = \ln|x+1| + \beta$

z) $(z^2+2z-1)(x+1)^2 = k \in \mathbb{R}^*$. ; $z \leftarrow \frac{y-1}{x+1}$. de rezolvat.

$\left(\ln\left[(x+1)\sqrt{\left|\left(\frac{y-1}{x+1}\right)^2 + 2\frac{y-2}{x+1} - 1\right|}\right] = t\right)$

" k (fără radical $\wedge 2$)

Ecuații diferențiale liniare de ordin superior cu coef. constanți

Forma generală:

(1): $a_n y^{(n)} + a_{n-1} y^{(n-1)} + \dots + a_1 y' + a_0 y = f(x), (\forall) x \in \bar{I}$.

, unde $a_0, a_1, \dots, a_n \in \mathbb{R}$, $a_n \neq 0$ ($\Leftrightarrow$ ec. (1) este de ordin n)

și $f: \bar{I} (\subseteq \mathbb{R}) \to \mathbb{R}$ continuă

Ec. dif. omogenă asociată ec (1):

(2): ~~$a_n \lambda^n + a_{n-1} \lambda^{n-1}$~~

$a_n y^{(n)} + a_{n-1} y^{(n-1)} + \dots + a_1 y' + a_0 y = 0$

Fie S mulț. $= \{f: \bar{I} \to \mathbb{R} \mid f$ este sol. pt ec. dif (2)$\}$.

Teoremă:

a) S este un sp. vect $\mathbb{R}$ de dim n.

b) Sol generală a ec. dif (1) este:

$$y(x) = \underbrace{y_h(x)}_{\text{notat } y_0(x)} + y_p(x), (\forall) x \in \bar{I}.$$

, $y_h(x)$ ($y_0(x)$) = sol. gen a ec dif (2)

, $y_p(x)$ = sol. part. a ec. dif (1).

c) Metoda variației constantelor pentru det. lui y_p.

Fie $\{f_1, \dots, f_n\}$ o bază pt S. At ($\exists$) n funcții $C_1(\cdot), C_2(\cdot), \dots, C_n(\cdot)$ deriv pe $\bar{I}$ ($C_k(\cdot): \bar{I} \to \mathbb{R}; k = \overline{1,n}$) astfel încât o sol. particulară a ec. dif (1) să fie:

$y_p(x) = C_1(x) f_1(x) + \dots + C_n(x) f_n(x)$, unde

$$\begin{cases} C_1' f_1 + C_2' f_2 + \dots + C_n' f_n = 0 \\ C_1' f_1' + C_2' f_2' + \dots + C_n' f_n' = 0 \\ \vdots \\ C_1' f_1^{(n-2)} + C_2' f_2^{(n-2)} + \dots + C_n' f_n^{(n-2)} = 0 \\ C_1' f_1^{(n-1)} + C_2' f_2^{(n-1)} + \dots + C_n' f_n^{(n-1)} = \underbrace{f(x)}_{a_n} \end{cases}$$

!!!

Ex 1: Să se det. sol. gen. a ec. dif $y'' + 2y' + y = xe^x + \frac{1}{xe^x}$, $x \neq 0$

Ec caract asociată: $\lambda^2 + 2\lambda + 1 = 0 \Leftrightarrow (\lambda+1)^2 = 0 \Leftrightarrow \lambda_{1,2} = -1$, $m_{\lambda=-1} = 2$

$f_1(x) = e^{-x}$; $f_2(x) = x e^{-x}$

$\Rightarrow \{f_1, f_2\}$ bază pt sp sol. ec. dif. omogene asociate $\Rightarrow$

$$y_n(x) = c_1 f_1(x) + c_2 f_2(x) \quad , c_1, c_2 \in \mathbb{R}$$
$$= e^{-x}(c_1 + c_2 x)$$

y_p Met obs. că $f(x) = \underbrace{xe^x}_{f_1(x)} + \underbrace{\frac{1}{xe^x}}_{f_2(x)}$ (f_1 este cvasipolinom)

At o sol. particulară $y_p = y_{p_1} + y_{p_2}$, unde

y_{p_1} este o sol. particulară pt ec. dif $y'' + 2y' + y = xe^x (= f_1(x))$

y_{p_2} este o sol. part. pt ec. dif $y'' + 2y' + y = \frac{1}{xe^x} (= f_2(x))$

y_{p_1}: $xe^x = e^{\alpha x}[P_n(x)\cos\beta x + P_m(x)\sin\beta x]$

$\Rightarrow \alpha = 1$

$\beta = 0$, $\alpha + i\beta$ nu este sol $\Rightarrow \boxed{S = 0}$

$P_n(x) = x \Rightarrow n = 1$

$Q_m(x) = 0 \Rightarrow m = -\infty$ $\Big| \Rightarrow N = 1$.

$\Rightarrow y_{p_1} = x^S e^{\alpha x}[\tilde{P}_N(x)\cos\beta x + \tilde{Q}_{pq}(x)\sin\beta x]$, $\tilde{P}_N \in \mathbb{R}_N[x]$

$= e^x(ax + b)$

$\Rightarrow (y_{p_1}(x))'' + 2(y_{p_1}(x))' + y_{p_1}(x) = xe^x$

$(y_{p_1}(x))' = e^x(ax + b + a) \Rightarrow (y_{p_1}(x))'' = e^x(ax + 2a + b)$

$\Leftrightarrow e^x(ax + 2a + b + 2ax + 2b + 2a + ax + b) = xe^x$

$\Leftrightarrow 4ax + 4a + 4b = x$

$\Leftrightarrow \begin{cases} 4a = 1 \Rightarrow a = \frac{1}{4} \\ a + b = 0 \end{cases} \Big| \Rightarrow b = -\frac{1}{4}$

$\Rightarrow y_{p_1}(x) = e^x\left(\frac{x}{4} - \frac{1}{4}\right) = \frac{e^x}{4}(x - 1)$

y_{p_2}: Met. v. c.

$y_{p_2}(x) = c_1(x) f_1(x) + c_2(x) f_2(x)$

$= c_1(x) e^x + c_2(x) x e^{-x}$, unde

$\begin{cases} c_1' e^{-x} + c_2' x e^{-x} = 0 \\ c_1'(-e^{-x}) + c_2'(e^{-x} - x e^{-x}) = \frac{1}{x e^x} \end{cases}$ ($f_2(x) = \frac{1}{xe^x}$)

$(-) \begin{cases} c_1' + c_2' x = 0 \\ -c_1' + c_2'(1-x) = \frac{1}{x} \end{cases} \Leftrightarrow \begin{cases} c_1' + c_2' x = 0 \\ c_1' + c_2'(x-1) = -\frac{1}{x} \end{cases}$

$"-" \Rightarrow -c_2' = -\frac{1}{x}$

$\Rightarrow c_2(x) = \int \frac{1}{x} dx = \ln|x| + A_2$, $A_2 \in \mathbb{R}$

$c_1' = -x c_2'$

$c_1' = -1 \Leftrightarrow c_1(x) = -x + A_1$, $A_1 \in \mathbb{R}$

Pentru $A_1 = A_2 = 0$, obținem $\boxed{Y_{p_2}(x) = -xe^{-x} + \ln|x| x e^{-x}}$

$\Rightarrow \underline{Y_{p_2}(x) = xe^{-x}(-1 + \ln|x|)}$

Sol. gen. a ec. dif:

$y'' + 2y' + y = xe^x + \frac{1}{xe^x}$ este

$y(x) = c_1 e^{-x} + c_2 x e^{-x} + \frac{e^x}{4}(x-1) + x e^{-x}(-1 + \ln|x|)$.

$\Downarrow c_1, c_2 \in \mathbb{R}$.

$(\Leftrightarrow y(x) = k_1 e^{-x} + k_2 x e^{-x} + \frac{e^x}{4}(x-1) + x e^{-x} \ln|x|)$

Ec. dif. Euler

Forma generală:

(3) $a_n x^n y^{(n)} + a_{n-1} x^{n-1} y^{(n-1)} + \dots + a_1 x y' + a_0 y = f(x)$

$f: I \left(\subseteq (0, +\infty)\right)$ cont.

$a_0, a_1, \dots, a_n \in \mathbb{R}$; $a_n \neq 0$.

Se rezolvă fol. sch. de var $x = e^t$ și ch. de funcție

$z(t) = y(e^t) = y(x)$

$\dot{z}(t) = y'(e^t) \cdot e^t = y'(x) \cdot x$

$\ddot{z}(t) = \frac{d}{dt}[y'(e^t) \cdot e^t] = y''(e^t) \cdot e^t + y'(e^t) \cdot e^t$

$= y''(x) \cdot x^2 + y'(x) \cdot x = y''(x) \cdot x^2 + \dot{z}(t)$

$\Rightarrow y''(x) \cdot x^2 = \ddot{z}(t) - \dot{z}(t)$

Ex 2: Să se det. sol generală a ec. dif. Euler $x^2y'' - 3xy' + 5y = 3x^2$, $x > 0$.

$x = e^t$, $z(t) \overset{not}{=} y(e^t) = y(x)$.

$\ddot{z} - \dot{z} - 3\dot{z} + 5z = 3e^{2t}$

$\ddot{z} - 4\dot{z} + 5z = 3e^{2t}$

Ec. caract. asociată: $\lambda^2 - 4\lambda + 5 = 0$

$\Delta = -4$

$\lambda_{1,2} = \frac{4 \pm 2i}{2} = 2 \pm i$ ~~$\lambda_1 = \lambda_2$~~

$f_1(t) = e^{2t}\cos t$

$f_2(t) = e^{2t}\sin t$

$\Rightarrow z_h = c_1 e^{2t}\cos t + c_2 e^{2t}\sin t = e^{2t}(c_1\cos t + c_2 \sin t)$; $c_1, c_2 \in \mathbb{R}$

z_p: M.N.C.

$f(t) = 3e^{2t} = e^{\alpha t}[P_n \cos \beta t + Q_m \sin t]$

$\alpha = 2$, $\beta = 0$ $\Rightarrow \alpha + i\beta$ nu este sol a ec. caract $\rightarrow$ $\underline{S = 0}$.

$P_n(t) = 3$; $Q_m(t) = 0 \Rightarrow m = -\infty$

$n = 0$

$\Rightarrow \underline{N = 0}$

$\Rightarrow z_p = e^{2t}[a] = ae^{2t} \Rightarrow z_p'' = 4ae^{2t}$

$\Rightarrow z_p' = 2ae^{2t}$

$\Rightarrow 4ae^{2t} - 8ae^{2t} + 5ae^{2t} = 3e^{2t}$

$ae^{2t} = 3e^{2t} \Rightarrow \underline{a = 3} \Rightarrow z_p(t) = 3e^{2t}$

$z(t) = ~~e^{2t}(c_1\cos t + c_2 \sin t + 3)~~$

$= e^{2t}(c_1\cos t + c_2 \sin t) + 3e^{2t}$

$\Downarrow$

$y(x) = x^2[3 + c_1\cos(\ln x) + c_2 \sin(\ln x)]$

$= 3x + 2$
$x > 0.$

Sisteme diferențiale liniare omogene

Teoremă: Fie sist. dif. liniar (4) $x'(t) = Ax(t)$, unde $A \in \mathcal{M}_n(\mathbb{R})$ diagonalizabilă peste $\mathbb{R}$. Și $x(t) = \begin{pmatrix} x_1(t) \\ x_2(t) \\ \vdots \\ x_n(t) \end{pmatrix}$; $t \in \mathbb{R}$

1) Dacă $\lambda \in \mathbb{R} \in \sigma(A)$ cu $m_\lambda = m \geq 1 \Rightarrow \dim_{\mathbb{R}} V_\lambda = m$. Fie $\{v_1, \dots, v_m\}$ o bază pt V_λ. Atunci $\{e^{\lambda t} v_1, e^{\lambda t} v_2, \dots, e^{\lambda t} v_m\}$ sunt m sol. liniar indep. pentru sist dif. liniar (4)

2) Dacă $\alpha \pm i\beta \in \sigma(A)$ cu mult. alg. ≥ 1, at $\dim_{\mathbb{C}} V_{\alpha+i\beta} = s$.
$\mathbb{C} \setminus \mathbb{R}$

Fie $\{v_1, \dots, v_s\}$ o bază pt $V_{\alpha+i\beta}$ (peste $\mathbb{C}$). Atunci $\{\mathrm{Re}(e^{(\alpha+i\beta)t} v_1), \dots, \mathrm{Re}(e^{(\alpha+i\beta)t} v_s), \mathrm{Im}(e^{(\alpha+i\beta)t} v_1), \dots, \mathrm{Im}(e^{(\alpha+i\beta)t} v_s)\}$ sunt $2s$ soluții liniar independente pt. sistemul (4).

Reunind toate sol. liniar indep. pentru (4) obținute la 1) și 2) se obț. un sist. de funcții $\{\varphi_1, \varphi_2, \dots, \varphi_n\}$ care constt. o bază pt sp. soluțiilor sist (4). Atunci $x(t) = C_1\varphi_1(t) + \dots + C_n\varphi_n(t)$.
$(C_1, \dots, C_n \in \mathbb{R})$

Ex 3. Să se det. sol gen. a sist. dif. lin. omo.

$$\begin{cases} x' = y - 2x - 2z \\ y' = x - 2y + 2z \\ z' = 3x - 3y + 5z \end{cases} \Leftrightarrow \begin{pmatrix} x' \\ y' \\ z' \end{pmatrix} = \underbrace{\begin{pmatrix} -2 & 1 & -2 \\ 1 & -2 & 2 \\ 3 & -3 & 5 \end{pmatrix}}_{A} \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

$(x = x(t);\ y = y(t);\ z = z(t))$

$$P_A(\lambda) \overset{not}{=} \det(A - \lambda I_3) = \begin{vmatrix} -2-\lambda & 1 & -2 \\ 1 & -2-\lambda & 2 \\ 3 & -3 & 5-\lambda \end{vmatrix}$$

$$= \begin{vmatrix} -2-\lambda & 1 & -2 \\ 1 & -2-\lambda & 2 \\ 3 & -3 & 5-\lambda \end{vmatrix} = \begin{vmatrix} -1-\lambda & 1 & -2 \\ -1-\lambda & -2-\lambda & 2 \\ 0 & -3 & 5-\lambda \end{vmatrix}$$

$$= -(1+\lambda) \begin{vmatrix} 1 & 1 & -2 \\ 1 & -2-\lambda & 2 \\ 0 & -3 & 5-\lambda \end{vmatrix} =$$

$$= -(1+\lambda) \begin{vmatrix} 1 & 1 & -2 \\ 0 & -3-\lambda & 4 \\ 0 & -3 & 5-\lambda \end{vmatrix} = -(1+\lambda)[\underbrace{(\lambda-5)(\lambda+3)}_{\lambda^2-2\lambda-3} + 12]$$

$$= -(1+\lambda)(\lambda-3)(\lambda+1)$$

$\lambda = -1$; $m_{\lambda=-1} = 2$

$$A - \lambda I_3 = A + I_3 = \begin{pmatrix} -1 & 1 & -2 \\ 1 & -1 & 2 \\ 3 & -3 & 6 \end{pmatrix} = \begin{pmatrix} -1 & 1 & -2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$(A + I_3)\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \Leftrightarrow x_2 = x_1 + 2x_3 \Leftrightarrow \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = x_1 \underbrace{\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}}_{v_1} + x_3 \underbrace{\begin{pmatrix} 0 \\ 2 \\ 1 \end{pmatrix}}_{v_2}$$

$\Rightarrow \{v_1, v_2\}$ baza pt $V_{\lambda=-1}$ $\Rightarrow \dim_{\mathbb{R}} V_{\lambda=-1} = m_{\lambda=-1} = 2$

$\Rightarrow$ dar $\dim_{\mathbb{R}} V_{\lambda=3} = m_{\lambda=3} = 1$ $\Rightarrow$

A diagonalizeaza.

$\boxed{\lambda = 3}$: $A - 3I_3 = \begin{pmatrix} -5 & 1 & -2 \\ 1 & -5 & 2 \\ 3 & -3 & 2 \end{pmatrix} = \begin{pmatrix} 1 & -5 & 2 \\ -5 & 1 & -2 \\ 3 & -3 & 2 \end{pmatrix}$

$$= \begin{pmatrix} 1 & -5 & 2 \\ 0 & -24 & 12 \\ 0 & 12 & -4 \end{pmatrix} \to \begin{pmatrix} 1 & -5 & 2 \\ 0 & -3 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

$$(2) \begin{cases} x_1 - 5x_2 + 2x_3 = 0 \\ -3x_2 + x_3 = 0 \end{cases} \Leftrightarrow \begin{cases} x_1 = -x_2 \\ x_3 = 3x_2 \end{cases}$$

$$\Rightarrow \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} -x_2 \\ x_2 \\ 3x_2 \end{pmatrix} = x_2 \underbrace{\begin{pmatrix} -1 \\ 1 \\ 3 \end{pmatrix}}_{v_3}$$

$\Rightarrow \{v_3\}$ baza pt $V_{\lambda=3}$

O baza pt sp. sol. sist (S) este:

$$\{f_1(t) = e^{-t}v_1, f_2(t) = e^{-t}v_2, f_3(t) = e^{3t}v_3\}$$

$$\Rightarrow \begin{pmatrix} x(t) \\ y(t) \\ z(t) \end{pmatrix} = c_1 e^{t}\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} + c_2 e^{t}\begin{pmatrix} 0 \\ 2 \\ 1 \end{pmatrix} + c_3 e^{3t}\begin{pmatrix} -1 \\ 1 \\ 3 \end{pmatrix} \Leftrightarrow$$

$$\begin{cases} x(t) = c_1 e^{-t} - c_3 e^{3t} \\ y(t) = c_1 e^{-t} + 2c_2 e^{-t} + c_3 e^{3t} \\ z(t) = c_2 e^{-t} + 3c_3 e^{3t} \end{cases}$$

V Ec. dif. liniare de ord I

$\boxed{y' + f(x)y = g(x)}$, unde $f, g : \underset{\text{int.}}{I} (\subseteq \mathbb{R}) \to \mathbb{R}$ cont.

M I: $\boxed{y' + f(x)\cdot y = g(x) \;/\cdot e^{\int f(x)\,dx}}$

$\Rightarrow \underbrace{y' \cdot e^{\int f(x)dx} + f(x)\cdot y \cdot e^{\int f(x)dx}}_{(y \cdot e^{\int f(x)dx})'} = e^{\int f(x)dx} g(x)$.

$\Leftrightarrow y \cdot e^{\int f(x)dx} \overset{\text{R}}{=} C + \int g(x) \cdot e^{\int f(x)dx}\,dx$

$y = e^{-\int f(x)dx} \left(C + \int g(x) \cdot e^{\int f(x)dx}\,dx \right)$

M II: MVC

$y(x) = y_h(x) + y_p(x)$, unde $y_h(x)$ este sol. gen. a ec. dif. $y' + f(x)\cdot y = 0$, iar $y_p(x)$ este a sol. part a ec. dif $y' + f(x)y = g(x)$

Ex:

a) $y' = \frac{y}{2x} + e^{3x}\cdot\sqrt{x}$, $x > 0$.

M I: $y' - \frac{1}{2x}y = e^{3x}\sqrt{x} \;/\cdot e^{\int -\frac{1}{2x}dx} \left(= e^{\ln \frac{1}{\sqrt{x}}} = \frac{1}{\sqrt{x}} \right) \Leftrightarrow$

$y' \cdot \frac{1}{\sqrt{x}} - \frac{1}{2x\sqrt{x}} y = e^{3x}$

$\Leftrightarrow \left(y \cdot \frac{1}{\sqrt{x}} \right)' = e^{3x} \Leftrightarrow$

$y \cdot \frac{1}{\sqrt{x}} = \int e^{3x}dx \Leftrightarrow \frac{y}{\sqrt{x}} = \frac{e^{3x}}{3} + C \Leftrightarrow$

$y = \sqrt{x}\left(\frac{e^{3x}}{3} + C \right)$, $C \in \mathbb{R}$

M III: $y' - \frac{y}{2x} = 0$; $y' = \frac{y}{2x}$, $x > 0$; $\to y = 0$, sol. cst

$\downarrow y \neq 0$

ec cu vs

$\frac{dy}{dx}$ // $\frac{y'}{y} = \frac{1}{2x} \Leftrightarrow \frac{1}{y}dy = \frac{1}{2x}dx \;/\int$

$\Leftrightarrow \ln|y| = \ln\sqrt{x} + C$, $C \in \mathbb{R}$

$\Leftrightarrow \ln\left(|y|\cdot\frac{1}{\sqrt{x}}\right) = C \Leftrightarrow |y|\cdot\frac{1}{\sqrt{x}} = K$, $K \in \mathbb{R}_+^*$

$\Leftrightarrow y = K\sqrt{x}$

$\Rightarrow y = \frac{k}{\sqrt{x}}, k \in \mathbb{R}^*$ / $\Rightarrow y = \frac{k\sqrt{x}}{x}, k \in \mathbb{R}$.

$y = 0$

$\Rightarrow y_h = \frac{k}{\sqrt{x}}$; $k \in \mathbb{R}$; $y_p = k(x) \cdot x^{1/2}$

Căutăm y_p de forma $y_p = \frac{k(x)}{\sqrt{x}}$ / $\Rightarrow$

$y_p' = \frac{y_p}{2x} + e^{3x}\sqrt{x}$

$k'(x) \cdot x^{1/2} + \frac{1}{2} x^{-\frac{1}{2}} \cdot k(x) = \frac{1}{2} k(x) \cdot x^{-\frac{1}{2}} + e^{3x}\sqrt{x}$

! Aici trebuie să simplifici.

$k'(x) = e^{3x} \Rightarrow k(x) = \frac{e^{3x}}{3} + (k = 0)$

$\Rightarrow y_p(x) = \frac{e^{3x}}{3} \cdot \sqrt{x}$

$\Rightarrow y(x) = y_h(x) + y_p(x) = \sqrt{x} \cdot k + \frac{e^{3x}}{3}\sqrt{x}$

$= \sqrt{x}\left(\frac{e^{3x}}{3} + k\right)$, $k \in \mathbb{R}$.

b) $y' = 2xy + \frac{e^{x^2}}{x^2}$; $x \neq 0$

$y' - 2xy = \frac{e^{x^2}}{x^2}$ / $\cdot e^{-\int 2x\,dx}$ $\left(= e^{-x^2}\right)$

$\Rightarrow$ $y' \cdot e^{-x^2} + 2x \cdot y \cdot e^{-x^2} = \frac{1}{x^2}$

$(y \cdot e^{-x^2})' = \frac{1}{x^2}$ / $\int dx$

$\Rightarrow y \cdot e^{-x^2} = -\frac{1}{x} + b$

$\Rightarrow y = e^{x^2}\left(-\frac{1}{x} + b\right)$, $b \in \mathbb{R}$

VI. Ec. dif. BERNOULLI

$y' = f(x) \cdot y + g(x) \cdot y^\alpha$, $\alpha \in \mathbb{R} \setminus \{0, 1\}$

P1: stabilim dacă ec. y^α are soluții. Pt $\alpha > 0$: $y = 0$ sol. const.

P2: pt $y^\alpha \neq 0 \Rightarrow y' \cdot y^{-\alpha} = f(x) \cdot \boxed{y^{1-\alpha}} + g(x)$

liniară în z.

P3: Facem sch de fcț. $y^{1-\alpha} = z$, unde $y = y(x)$, $z = z(x)$.

$= (1-\alpha)\cdot y^{-\alpha}\cdot y'$

Înmulț. $\boxed{P_2}$ cu $(1-\alpha)$

$z)\ (1-\alpha)y'y^{-\alpha} = (1-\alpha)f(x)y^{1-\alpha} + (1-\alpha)g(x)$

$(\Rightarrow \underline{z'(x) = (1-\alpha)f(x)\cdot z(x) + (1-\alpha)g(x)}$

ec. dif. lin. de ord $\overline{I}$ în z.

c) $y' = y^2e^x - 2y$. Este ecuație Bernoulli.

$y' = -2y + e^x\cdot y^2 \;/\cdot \frac{1}{y^2};\ \alpha = 2.$

$y^2 = 0 \Rightarrow y = 0$

$y^2 \neq 0 \Rightarrow y'\cdot y^{-2} = -2y^{-1} + e^x \;/\cdot(-1)$

$-y'\cdot y^{-2} = 2y^{-1} - e^x$. Facem sch. de fct. $z = y^{-1}$.

$z' = -1\,y^{-2}y'$

$\Rightarrow z' = -y'\cdot y^{-2}$.

$\Rightarrow z' = 2z - e^x$.

$z' - 2z = -e^x \;/\cdot \int_e -2\,dx \;(= e^{-2x})$

$\underbrace{z'e^{-2x} - 2ze^{-2x}}_{(z\cdot e^{-2x})'} = -e^{-x}\ (= (e^{-x})'$

$\Rightarrow z\cdot e^{-2x} = e^{-x} + C$

$z = e^{2x}(e^{-x} + C)_{C\in\mathbb{R}}$

$y^{-1} = e^{2x}(e^{-x} + C)$

$y = \frac{1}{e^{2x}(e^{-x}+C)} = e^{-2x}\cdot\frac{1}{e^{-x}+C}$

d) $y' = \frac{y}{x} + \frac{x^3}{3y^2}$, $x\neq 0, y\neq 0$. Pres $x>0$.

$y' = \frac{1}{x}y + \frac{x^3}{3}y^{-2} \;/\cdot \frac{1}{y^{-2}};\ \alpha = -2.$

$y'\cdot y^{-2} = \frac{1}{x}y^3 + \frac{x^3}{3} \;/\cdot 3 \Leftrightarrow 3y'\cdot y^{-2} = \frac{3}{x}y^3 + x^3$. Facem sch. de fct $z = y^3$

$\Leftrightarrow z' = \frac{3}{x}z + x^3;\ z' - \frac{3}{x}z = x^3 \;/\cdot \int_e -\frac{3}{x}dx \left(= e^{-3\ln|x|} = \frac{1}{|x|^3}\right) \Rightarrow z' = 3y^2\cdot y'$

c) $\underbrace{z' \cdot x^3 - 3x^2 z}_{(z \cdot x^{-3})'} = x^4 + 1$

$(z \cdot x^{-3})' = x' \Rightarrow z \cdot x^{-3} = x + C$

$z = x^3(x + C)$

$y^3 = x^3(x + C)$

$\Rightarrow y = \sqrt[3]{x^3(x+C)}$

$\Rightarrow y = x \sqrt[3]{x + C}$, $C \in \mathbb{R}$

Ex2: Să se rezolve:

a) $y^{(4)} - y = e^x \sin x$

$y = y_h + y_p$

ec. am. asoc: $\lambda^4 - 1 = 0$

$(\lambda^2+1)(\lambda-1)(\lambda+1) = 0$ $\begin{cases} \lambda_{3,4} = \pm i \\ \lambda_2 = 1 \\ \lambda_1 = -1 \end{cases}$ cu $m_{\lambda_i} = 1$, $\forall i \in \{1,\dots,4\}$.

$(\alpha = 0, \beta = 1)$

y_h: $y_{h_3}(x) = \cos x$, $y_{h_4}(x) = \sin x$, $y_{h_2}(x) = e^x$, $y_{h_1}(x) = e^{-x}$

; $\{y_1, y_2, y_3, y_4\}$ baza pt sp. sol. ec. dif $y^{(4)} - y = 0$.

$\Rightarrow y_h(x) = C_1 e^{-x} + C_2 e^x + C_3 \cos x + C_4 \sin x$

y_p: $e^x \sin x = e^{\alpha x}[P_n(x)\cos\beta x + Q_m(x)\sin\beta x]$

$\Rightarrow \alpha = 1, \beta = 1, P_n(x) = 0 \Rightarrow n = -\infty$

$\alpha + i\beta$ nu este sol $\Rightarrow$ dacă $Q_m(x) = 1 \Rightarrow m = 0$ $\Big| \Rightarrow N = 0$.

$\Rightarrow y_p = e^{\alpha x} x^s [\tilde{P}_N(x)\cos\beta x + \tilde{Q}_N(x)\sin\beta x]$

$\Rightarrow y_p(x) = e^x (a\cos x + b\sin x)$

$(f \cdot g)^{(n)} = \sum_{k=0}^{n} C_n^k \cdot g^{(k)} \cdot f^{(n-k)}(x)$

$(y_p(x))^{(4)} = \sum_{k=0}^{4} C_4^k e^x \cdot (a\cos x + b\sin x)^{(k)}$; $C_4^0 = C_4^4 = 1$; $C_4^1 = C_4^3 = 4$

$= e^x [(a\cos x + b\sin x)] + 4(-a\sin x + b\cos x) + 6(-a\cos x - b\sin x)$

$+ 4(a\sin x - b\cos x) + (a\cos x + b\sin x)]$

$y_p(x)^{(4)} = e^x \left(a\cos x + b\sin x - 4a\sin x + 4b\cos x - 6a\cos x - 6b\sin x + 4a\sin x - 4b\cos x + a\cos x + b\sin x \right)$

$= e^x (-4a\cos x - 4b\sin x)$

$\Rightarrow (y_p(x))^{(4)} = -4e^x (a\cos x + b\sin x)$

$y_p(x) = e^x (a\cos x + b\sin x)$

$\Rightarrow -4e^x (a\cos x + b\sin x) + e^x (a\cos x + b\sin x) = e^x \sin x$

$-5e^x (a\cos x + b\sin x) = e^x \sin x$

$\Rightarrow -5a\cos x - 5b\sin x = \sin x \;/$
$\{\sin x, \cos x\}$ SLi

$\Rightarrow \begin{cases} -5a = 0 \\ -5b = 1 \end{cases} \Rightarrow \begin{cases} b = -\frac{1}{5} \\ a = 0 \end{cases}$

$\Rightarrow y_p(x) = -\frac{1}{5} e^x \sin x$

$\Rightarrow y = c_1 e^{-x} + c_2 e^x + c_3 \cos x + c_4 \sin x - \frac{1}{5} e^x \sin x.$

h) $y'' + y = \frac{2}{\cos^3 x}$; $x \in [0, \pi/2)$

Ecl. M.v.c.

$\lambda^2 + 1 = 0 \Rightarrow \lambda_{1,2} = \pm i$; $y_h(x) = c_1 \cos x + c_2 \sin x$

Căutăm $y_p(x)$ de forma

$y_p(x) = c_1(x)\cos x + c_2(x)\sin x$

unde $\begin{cases} c_1'(x)\cos x + c_2'(x)\sin x = 0 \\ -c_1'(x)\sin x + c_2'(x)\cos x = \frac{2}{\cos^3 x} \end{cases}$

$c_2'(x) = \frac{2}{\cos^2 x} \cdot \frac{1}{\cos x}$

$\Rightarrow c_2(x) = 2\operatorname{tg} x + (A_2 = 0)$

$(\cos)' = -\sin$

$c_1' = -c_2 \operatorname{tg} x = -2\frac{\sin x}{\cos^3 x} \;/\int$

$c_1(x) = -\frac{1}{\cos^2 x} + (A_1 = 0)$

Pt $A_1 = A_2 = 0 \Rightarrow y_p = -\frac{1}{\cos^2 x}\cos x + 2\operatorname{tg} x \cdot \sin x$

$y_p = -\frac{1}{\cos x} + 2\operatorname{tg} x \cdot \sin x = \frac{-1 + 2\sin^2 x}{\cos x} = -\frac{\cos 2x}{\cos x}$

$\Rightarrow y(x) = c_1\cos x + c_2 \sin x - \frac{\cos 2x}{\cos x}$; $c_1, c_2 \in \mathbb{R}$

Ex1: Să se rezolve SDLO lin. dif. omo:

$$\begin{cases} x' = 2x + y \\ y' = x + 3y - z \\ z' = 2y + 3z - x \end{cases}, \text{ unde } x = x(t);\ y = y(t);\ z = z(t)$$

$$\begin{pmatrix} x' \\ y' \\ z' \end{pmatrix} = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 3 & -1 \\ -1 & 2 & 3 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

A = matr. sist.

$$P_A(\lambda) = \det(A - \lambda I_3) = \begin{vmatrix} 2-\lambda & 1 & 0 \\ 1 & 3-\lambda & -1 \\ -1 & 2 & 3-\lambda \end{vmatrix} = \begin{vmatrix} 2-\lambda & 1 & 0 \\ 0 & 3-\lambda & -1 \\ 2-\lambda & 2 & 3-\lambda \end{vmatrix}$$

("+1")

$$= (2-\lambda) \begin{vmatrix} 1 & 1 & 0 \\ 0 & 3-\lambda & -1 \\ 1 & 2 & 3-\lambda \end{vmatrix} = (2-\lambda) \begin{vmatrix} 1 & 1 & 0 \\ 0 & 3-\lambda & -1 \\ 0 & 1 & 3-\lambda \end{vmatrix}$$

$$= (2-\lambda)\left[(\lambda-3)^2 + 1\right]$$

$\Rightarrow \sigma(A) = \{\underbrace{3 \pm i}, 2\}$ se alege una

$m_{\lambda = 3 \pm i} = m_{\lambda = 2} = 1 \Rightarrow A$ diag. peste $\mathbb{C}$

$$V_{\lambda=2} \overset{\text{def}}{=} \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 \;\middle|\; A \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 2 \begin{pmatrix} x \\ y \\ z \end{pmatrix} \right\}$$

$\Leftrightarrow$

$$\begin{cases} 2x + y = 2x \\ x + 3y - z = 2y \\ -x + 2y + 3z = 2z \end{cases} \to \begin{cases} y = 0 \\ x = z \\ x = z \end{cases}$$

$$\Rightarrow \begin{pmatrix} x \\ y \\ z \end{pmatrix} = z \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$$

v_1

$\Rightarrow \{v_1\}$ bază pt $V_{\lambda=2}$. ($\exists$ $\dim_{\mathbb{R}} V_{\lambda=2} = 1$)

$$V_{\lambda=3+i} \overset{\text{def}}{=} \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{C}^3 \;\middle|\; A \begin{pmatrix} x \\ y \\ z \end{pmatrix} = (3+i) \begin{pmatrix} x \\ y \\ z \end{pmatrix} \right\}$$

$\Leftrightarrow$

$$\begin{cases} 2x + y = (3+i)x \\ x + 3y - z = (3+i)y \\ -x + 2y + 3z = (3+i)z \end{cases} \Leftrightarrow \begin{cases} (1+i)x - y = 0 \\ x - iy - z = 0 \\ x - 2y + iz = 0 \end{cases}$$

$\Rightarrow \boxed{y = (1+i)x}$

$$\Rightarrow \begin{cases} y = (1+i)x \\ x - (i-1)x - z = 0 \\ x - 2(1+i)x + iz = 0 \end{cases} \Leftrightarrow \begin{cases} \boxed{(2-i)x = z} \\ (-1-2i)x + iz = 0 \;/\cdot i \end{cases} \Leftrightarrow (-i+2)x = z$$

$\Rightarrow \begin{pmatrix} x \\ y \\ t \end{pmatrix} = \begin{pmatrix} (1+i)x \\ (2-i)x \end{pmatrix} = x \underbrace{\begin{pmatrix} 1 \\ 1+i \\ 2-i \end{pmatrix}}_{v_2} \Rightarrow \{v_2\}$ o bază pt $V_{\lambda=3+i}$

$\Rightarrow \dim_{\mathbb{C}} V_{\lambda=3+i} = 1$

$\begin{pmatrix} 1 \\ 1+i \\ 2-i \end{pmatrix}$

O bază pt sp. sol. SLDO pe este $\{f_1 = e^{2t}v_1,\ f_2 = Re(e^{(3+i)t}v_2),\ f_3 = Im(e^{(3+i)t}v_2)\}$

v_2

Obs: $\forall t \in \mathbb{R}$: $\boxed{e^{it} = \cos t + i \sin t}$ (EULER)

$$e^{(3+i)t}v_2 = e^{3t}(\cos t + i\sin t)\begin{pmatrix} 1 \\ 1+i \\ 2-i \end{pmatrix}$$

$$= e^{3t}\begin{pmatrix} \cos t + i\sin t \\ \cos t - \sin t + i(\cos t + \sin t) \\ 2\cos t + \sin t + i(2\sin t - \cos t) \end{pmatrix}$$

$$\Rightarrow Re(e^{(3+i)t}v_2) = e^{3t}\begin{pmatrix} \cos t \\ \cos t - \sin t \\ 2\cos t + \sin t \end{pmatrix}$$

$$Im(e^{(3+i)t}v_2) = e^{3t}\begin{pmatrix} \sin t \\ \cos t + \sin t \\ 2\sin t - \cos t \end{pmatrix}$$

Sol gen: $\begin{pmatrix} x(t) \\ y(t) \\ z(t) \end{pmatrix} = c_1 e^{2t}\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} + c_2 e^{3t}\begin{pmatrix} \cos t \\ \cos t - \sin t \\ 2\cos t + \sin t \end{pmatrix} + c_3 e^{3t}\begin{pmatrix} \sin t \\ \cos t - \sin t \\ 2\cos t + \sin t \end{pmatrix}$

; $c_1, c_2, c_3 \in \mathbb{R}$.

Rezolvarea SLDN $\rightarrow$ neomogene. ! (unde diagonalizează A!) !

Fie ① $\boxed{X' = AX + b(t)}$, unde $A \in \mathcal{M}_n(\mathbb{R})$, $X(t) \overset{not}{=} \begin{pmatrix} x_1(t) \\ x_2(t) \\ \vdots \\ x_n(t) \end{pmatrix}$,

coloana funcțiilor necunoscute

$(X'(t) = A \cdot X(t) + b(t))$, iar $b(t) \overset{not}{=} \begin{pmatrix} b_1(t) \\ b_2(t) \\ \vdots \\ b_n(t) \end{pmatrix}$, cont $b_1, \dots, b_n : I (\subseteq \mathbb{R}) \to \mathbb{R}$

și ② $\boxed{X' = AX}$

Teoremă: a) Soluția generală a SLDN ① este:

$X(t) = X_g(t) + X_p(t)$, unde

$X_g(t)$ = sol. generală a SLDO ②

$X_p(t)$ = sol. particulară a SLDN ①

$\forall$ vc. Dacă $\{f_1, f_2, \dots, f_n\}$ este o bază pentru sp. vect real al sol. omog. SLO ②, atunci ($\exists$) o sol. part. $x_p(t)$ pt ① de forma

$$\boxed{x_p(t) = c_1(t) f_1(t) + c_2(t) f_2(t) + \dots + c_n(t) f_n(t).}$$

unde $c_1(\cdot), c_2(\cdot), \dots, c_n(\cdot) : I \to \mathbb{R}$ derivabile

$$\boxed{c_1' f_1 + c_2' f_2 + \dots + c_n' f_n = b}$$

Ex 2: Să se rezolve SDLN $\begin{cases} x' = 2x - 4y + 4e^{-2t} \\ y' = 2x - 2y \end{cases}$; $x = x(t), y = y(t)$

(Derivăm ec.)

MI: $y'' = 2x' - 2y'$

$y'' + 2y' = 2(2x - 4y + 4e^{-2t}) \iff y'' + 2y' + 8y = 4x + 8e^{-2t} \iff$

$4x = 2y' + 4y$

$\Rightarrow y'' + 4y = 8e^{-2t}$

Ec. caract: $\lambda^2 + 4 = 0 \iff \lambda_{1,2} = \pm 2i \Rightarrow$

$$y_h(t) = c_1 \cos 2t + c_2 \sin 2t \quad ; c_1, c_2 \in \mathbb{R}.$$

$f(t) = 8e^{-2t} = e^{\alpha t}[P_n(t)\cos\beta t + Q_m(t)\sin\beta t]$

$\Rightarrow \alpha = -2$, $\beta = 0$ $\Rightarrow \alpha + i\beta = -2$ nu este rădăcină a ec. caract $\Rightarrow \boxed{\beta = 0}$

$P_n(t) = 8 \Rightarrow n = 0$
$Q_m(t) = 0 \Rightarrow m = -\infty$ $\Big| \Rightarrow N = 0.$

$\Rightarrow$ Căutăm $\Rightarrow y_p(t) = e^{-2t}[\tilde{P}_N(t)] = ae^{-2t}$

$y_p'' + 4y_p = 8e^{-2t}$
$y_p'' = 4ae^{2t}$ $\Big| \Rightarrow 4a + 4a = 8 \Rightarrow \underline{a = 1}$

$\Rightarrow y_p(t) = e^{-2t}$

$$\Rightarrow \boxed{y(t) = y_h(t) + y_p(t) = c_1 \cos 2t + c_2 \sin 2t + e^{-2t}}$$

$x(t) = \frac{1}{2}(y'(t) + 2y(t))$

$y'(t) = -2c_1 \sin 2t + 2c_2 \cos 2t - 2e^{-2t}$

$$\Rightarrow \boxed{x(t) = \frac{1}{2}\left((2c_2 + 2c_1)\cos 2t + (2c_2 - 2c_1)\sin 2t\right)}$$

M II: Mvc $A = \begin{pmatrix} 2 & -4 \\ 2 & -2 \end{pmatrix}$; $P_A(\lambda) = \begin{vmatrix} 2-\lambda & -4 \\ 2 & -2-\lambda \end{vmatrix} = \lambda^2 - 4 + 8 = \lambda^2 + 4$

$\Rightarrow \sigma(A) \in \{\pm 2i\}$ cu $m_{\lambda=2i} = 1$

$\lambda = 2i$: $V_{\lambda=2i}$: $A\begin{pmatrix} x \\ y \end{pmatrix} = 2i\begin{pmatrix} x \\ y \end{pmatrix} \Leftrightarrow$

$$\begin{cases} 2x - 4y = 2ix \\ 2x - 2y = 2iy \end{cases} \Leftrightarrow \begin{cases} (1-i)x - 2y = 0 \\ x - (1+i)y = 0 \end{cases} \quad |\cdot(2)$$

$\frac{1-i}{1+i} \cdot \frac{2}{2} = \frac{2(1-i)}{2} = 1-i$

$x = (1+i)y$

$\Rightarrow \begin{pmatrix} x \\ y \end{pmatrix} = y\underbrace{\begin{pmatrix} 1+i \\ 1 \end{pmatrix}}_{v}$

; $\{v\}$ o bază pt ~~xxxx~~ $V_{\lambda=2i}$ (Φ)

o bază pt sp. sol. sist. omo asoc. $\begin{cases} x' = 2x - 4y \\ y' = 2x - 2y \end{cases}$

este $\{f_1 = Re(e^{2it}v),\ f_2 = Im(e^{2it}v)\}$

$$e^{2it}v = (\cos 2t + i\sin 2t)\begin{pmatrix} 1+i \\ 1 \end{pmatrix} = \begin{pmatrix} \cos 2t - \sin 2t + i(\cos 2t + \sin 2t) \\ \cos 2t + i\sin 2t \end{pmatrix}$$

$\Rightarrow Re\{e^{2it}v\}$

$$\begin{pmatrix} x \\ y \end{pmatrix}_0 = c_1\begin{pmatrix} \cos 2t - \sin 2t \\ \cos 2t \end{pmatrix} + c_2\begin{pmatrix} \cos 2t + \sin 2t \\ \sin 2t \end{pmatrix} : c_1, c_2 \in \mathbb{R}$$

Căutăm o sol. part. a sist. DLN de mai sus sub forma

$\begin{pmatrix} x \\ y \end{pmatrix}_p = c_1(t) f_1(t) + c_2(t) f_2(t)$ pt care

$$c_1'(t) f_1(t) + c_2'(t) f_2(t) = \begin{pmatrix} 4e^{-2t} \\ 0 \end{pmatrix} \Leftrightarrow$$

$$\begin{cases} c_1'(\cos 2t - \sin 2t) + c_2'(\cos 2t + \sin 2t) = 4e^{-2t} \\ c_1'\cos 2t + c_2'\sin 2t = 0 \end{cases} \quad \begin{matrix} f_1 \\ f_2 \end{matrix}$$

$$\begin{cases} -c_1'\sin 2t + c_2'\cos 2t = 4e^{-2t} \,/\cdot \cos 2t \\ c_1'\cos 2t + c_2'\sin 2t = 0 \,/\cdot \sin 2t \end{cases} \quad "+" \Rightarrow$$

$c_2'(t) = 4e^{-2t}\cos 2t$

$c_1'(t) = -tg2t \cdot 4e^{-2t}\cdot\cos 2t = 4e^{-2t}\sin 2t$

$$(2)\begin{cases} c_1(t) = e^{-2t}(\sin 2t + \cos 2t) \\ c_2(t) = e^{-2t}(\sin 2t - \cos 2t) \end{cases}$$

!!!

$$\boxed{\begin{matrix} e^{it} = \cos t + i\sin t \\ e^{-it} = \cos t - i\sin t \end{matrix}} \Rightarrow \underline{\cos t} = \frac{e^{it} + e^{-it}}{2}$$

obs: $\int e^{-2t}\cos 2t\,dt =$

$\int e^{-2t}(e^{2it}+e^{-2it})\,dt = \frac{1}{2}\int e^{2(i-1)t} + e^{-2(1+i)t}\,dt$

$\frac{1}{2}\left[\frac{e^{2(i-1)t}}{2(i-1)} - \frac{e^{-2(i+1)t}}{2(i+1)}\right] + C.$

$\Rightarrow \begin{pmatrix} x \\ y \end{pmatrix}_p = e^{-2t}\left[(\sin 2t+\cos 2t)\begin{pmatrix}\cos 2t - \sin 2t \\ \cos 2t\end{pmatrix} + (\sin 2t - \cos 2t)\begin{pmatrix}\cos 2t + \sin 2t \\ \sin 2t\end{pmatrix}\right].$

$\Rightarrow \begin{pmatrix} x \\ y \end{pmatrix}_p = e^{-2t}\begin{pmatrix}\cos^2 2t - \sin^2 2t + \sin^2 2t - \cos^2 2t \\ 1\end{pmatrix} = \begin{pmatrix}0 \\ e^{-2t}\end{pmatrix}$

$\Rightarrow \begin{cases} x(t) = C_1(\cos 2t - \sin 2t) + C_2(\cos 2t + \sin 2t) \\ y(t) = C_1\cos 2t + C_2 \sin 2t + e^{-2t}\end{cases}; C_1, C_2 \in \mathbb{R}.$

Exponențiala unei matrice

Fie $A \in \mathcal{M}_n(\mathbb{C})$. Atunci $e^A \overset{\text{def}}{=} \sum_{n=0}^{\infty} \frac{A^n}{n!} = I_n + \frac{1}{1!}A + \frac{1}{2!}A^2 + \ldots + \frac{1}{n!}A^n + \ldots$

Prop: 1) $(\forall) A, B \in \mathcal{M}_n(\mathbb{C})$ a.î. $A \cdot B$ comutativ $\Rightarrow e^{A+B} = e^A \cdot e^B$

2) $e^{O_n} = I_n$

3) $(\forall) A \in \mathcal{M}_n(\mathbb{C})$, e^A este inversabilă și $(e^A)^{-1} = e^{-A}$.

Teoremă (Metodă de calcul al lui e^{At} pt $A \in \mathcal{M}_n(\mathbb{R})$ și $t \in \mathbb{R}$)

Fie $A \in \mathcal{M}_n(\mathbb{R})$ și $t \in \mathbb{R}$. At $(\exists)\ \alpha_0, \alpha_1, \ldots, \alpha_{n-1} : \mathbb{R} \to \mathbb{R}$ a.î.

$$\boxed{e^{At} = \alpha_{n-1}t^{n-1}A^{n-1} + \alpha_{n-2}t^{n-2}A^{n-2} + \ldots + \alpha_1 tA + \alpha_0 I_n}$$

În plus, dacă $r(\lambda) \overset{\text{not}}{=} \alpha_{n-1}\lambda^{n-1} + \alpha_{n-2}\lambda^{n-2} + \ldots + \alpha_1\lambda + \alpha_0$

pt $(\forall)\mu \in \sigma(At)$ cu $m_\mu = m \geq 1$ au loc relațiile: $r(\mu) = e^\mu$; $r'(\lambda)\big|_{\lambda=\mu} = e^\mu$; $\ldots$; $r^{(m-1)}(\lambda)\big|_{\lambda=\mu} = e^\mu$.

Ex 3: Să se calc. e^{At} pt:

a) $A = \begin{pmatrix}2 & -4 \\ 2 & -2\end{pmatrix} \Rightarrow \sigma(A) = \{\pm 2i\}$ cu $m_{\lambda=\pm 2i} = 1 \Rightarrow \sigma(At) = \{\pm 2it\}$ cu $m_{\mu=\pm 2it} = 1$

$n = 2 \Rightarrow (\exists)\alpha_0, \alpha_1 : \mathbb{R} \to \mathbb{R}$ a.î.

$$e^{At} = \alpha_1 tA + \alpha_0 I_2$$

Fie $r(\lambda) = \alpha_1\lambda + \alpha_0$. At pt $\mu_1 = 2it$, $\mu_2 = -2it$ cu loc

rel : $r(\mu_1)=e^{\mu_1}$ si $r(\mu_2)=e^{\mu_2}$ $\Rightarrow$ $\begin{cases} \alpha_1 \cdot 2it + \alpha_0 = e^{2it} (= \cos 2t + i \sin 2t) \\ -\alpha_1 \cdot 2it + \alpha_0 = e^{-2it} (= \cos 2t - i \sin 2t) \end{cases}$

$\Rightarrow \begin{cases} \alpha_0 = \cos 2t \\ \alpha_1 = \dfrac{\sin 2t}{2t} \end{cases}$, $t \neq 0$.

$$\Rightarrow \boxed{e^{At} = \frac{\sin 2t}{2} \begin{pmatrix} 2 & -4 \\ 2 & -2 \end{pmatrix} + \cos 2t \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}}$$

$$\Rightarrow \boxed{e^{At} = \begin{pmatrix} \sin 2t + \cos 2t & -2\sin 2t \\ \sin 2t & -\sin 2t + \cos 2t \end{pmatrix}}$$

b) $A = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & -3 & 3 \end{pmatrix}$ $\Rightarrow P_A(\lambda) = \begin{vmatrix} -\lambda & 1 & 0 \\ 0 & -\lambda & 1 \\ 1 & -3 & 3-\lambda \end{vmatrix} = \begin{vmatrix} 1-\lambda & 1 & 0 \\ 1-\lambda & -\lambda & 1 \\ 1-\lambda & -3 & 3-\lambda \end{vmatrix}$

$= (1-\lambda) \begin{vmatrix} 1 & 1 & 0 \\ 1 & -\lambda & 1 \\ 1 & -3 & 3-\lambda \end{vmatrix} = (1-\lambda) \begin{vmatrix} 1 & 1 & 0 \\ 0 & -1-\lambda & 1 \\ 0 & -4 & 3-\lambda \end{vmatrix}$

$= (1-\lambda)[(\lambda-3)(\lambda+1)+4] = (1-\lambda)(\lambda^2-2\lambda+1) = -(\lambda+1)^3$

$\Rightarrow \sigma(A) = \{1\}$ cu $m_1 = 3$

$\Rightarrow \sigma(At) = \{t\}$ cu $\mu = t^3$

$\Rightarrow e^{At} = \alpha_2 t^2 A^2 + \alpha_1 t A + \alpha_0 I_3$

Not $r(\lambda) = \alpha_2 \lambda^2 + \alpha_1 \lambda + \alpha_0$ $\Rightarrow r'(\lambda) = 2\lambda \alpha_2 + \alpha_1$

$\Rightarrow r''(\lambda) = 2\alpha_2$

$\alpha_0, \alpha_1, \alpha_2$ le det din $\begin{cases} r(t) = e^t \\ r'(\lambda) = e^t \\ r''(\lambda) = e^t \end{cases}$ $\Rightarrow \begin{cases} \alpha_2 t^2 + \alpha_1 t + \alpha_0 = e^t \\ 2t\alpha_2 + \alpha_1 = e^t \\ 2\alpha_2 = e^t \end{cases}$

$\Rightarrow \begin{cases} \alpha_2 = \dfrac{e^t}{2} \\ \alpha_1 = e^t - te^t = (1-t)e^t \\ \alpha_0 = e^t - t(1-t)e^t - \dfrac{t^2}{2}e^t \end{cases}$

$= e^t \left(1 - t + \dfrac{t^2}{2}\right)$

$$\Rightarrow e^{At} = \frac{e^t \cdot t^2}{2} A^2 + t(1-t)e^t A + e^t\left(1 - t + \frac{t^2}{2}\right) I_3$$

a) $A=\begin{pmatrix}0&1&0\\0&0&1\\1&-3&3\end{pmatrix} \Rightarrow P_A(\lambda)=\begin{vmatrix}-\lambda&1&0\\0&-\lambda&1\\1&-3&3-\lambda\end{vmatrix}=\begin{vmatrix}1-\lambda&1&0\\1-\lambda&-\lambda&1\\1-\lambda&-3&3-\lambda\end{vmatrix}$

$=(1-\lambda)\begin{vmatrix}1&1&0\\1&-\lambda&1\\1&-3&3-\lambda\end{vmatrix}=(1-\lambda)\begin{vmatrix}1&1&0\\0&-1-\lambda&1\\0&-4&3-\lambda\end{vmatrix}$

REPACI PE EXERCIȚII

$=(1-\lambda)\begin{vmatrix}-1-\lambda&1\\-4&3-\lambda\end{vmatrix}=(1-\lambda)\left[-(\lambda+1)(3-\lambda)+4\right]$

$=(1-\lambda)(\lambda^2-2\lambda+1)$

$=(1-\lambda)(\lambda-1)^2=(\lambda-1)^3 \Rightarrow \lambda_{1,2,3}=1$, $m_{\lambda=1}=3$.

$\Rightarrow \sigma(A)\in\{1\}\Rightarrow \sigma(At)\in\{t\}$ și $m_\mu=3$

$n=3 \Rightarrow (-1)$ $\alpha_2,\alpha_1,\alpha_0\in\mathbb{R}$ a.î.

$e^{At}=\alpha_2 t^2A^2+\alpha_1 tA+\alpha_0 I_3$; $r(\lambda)=\alpha_2\lambda^2+\alpha_1\lambda+\alpha_0$

$\mu\in\sigma(At)\Rightarrow\mu=t$, $m_\mu=3$ $\Rightarrow$ $\begin{cases} r(\mu)=e^\mu \\ r'(\lambda)|_{\lambda=\mu}=e^\lambda \\ r''(\lambda)|_{\lambda=\mu}=e^\lambda \end{cases} \Leftrightarrow \begin{cases} \alpha_2\mu^2+\alpha_1\mu+\alpha_0=e^\mu \\ 2\alpha_2\mu+\alpha_1=e^\mu \\ 2\alpha_2=e^\mu \end{cases}$; $\mu=t$.

$\Leftrightarrow \begin{cases} \alpha_2 t^2+\alpha_1 t+\alpha_0=e^t \\ 2\alpha_2 t+\alpha_1=e^t \\ 2\alpha_2=e^t \end{cases} \Rightarrow \alpha_2=\frac{1}{2}e^t \Rightarrow \alpha_1=e^t-te^t=e^t(1-t)$

$\Rightarrow \alpha_0=e^t-\frac{1}{2}e^t t^2-te^t(1-t)$

$\Rightarrow \alpha_0=e^t\left(1-\frac{t^2}{2}-t+t^2\right)=e^t\left(1-t+\frac{t^2}{2}\right)$

$\Rightarrow \boxed{e^{At}=\frac{1}{2}e^t t^2A^2+e^t(1-t)tA+e^t\left(1-t+\frac{t^2}{2}\right)}$

$\begin{cases} x'=2x-4y+4e^{-2t} \\ y'=2x-2y \end{cases}$; $2x=y'+2y$

$y'=2x-2y \mid (\cdot)' \Leftrightarrow y''=2x'-2y' \Leftrightarrow y''=4x-8y+8e^{-2t}-2y'$

$\Leftrightarrow y''=2y'+4y-8y-2y'+8e^{-2t} \Rightarrow y''=-4y+8e^{-2t} \Leftrightarrow y''+4y=8e^{-2t}$

O sol $y=y_h+y_p$; y_h sol. ec. caract om. asoc ec. obț. ale rezolvat;

$\lambda^2+4=0 \Leftrightarrow \lambda_{1,2}=\pm 2i$ cu $m_{\lambda=\pm 2i}=1$.

$(\alpha=0,\beta=2)$

$\Rightarrow$ o transpusă sp. sol. ec. (...) este $\{y_1 = \cos 2t, y_2 = \sin 2t\}$

$\Rightarrow y_{sh}(t) = c_1 \cos 2t + c_2 \sin 2t$; $c_1, c_2 \in \mathbb{R}$.

y_p: $f(t) = 8e^{-2t} = e^{\alpha x}[P_n(t)\cos\beta t + Q_m(t)\sin\beta t]$

$\Rightarrow \alpha = -2$, $\beta = 0$ $\Rightarrow \alpha + i\beta = -2$ nu este sol. a ec. caract. atunci $\Rightarrow s = 0$

$P_n(t) = 8 \Rightarrow n = 0$

$Q_m(t) = 0 \Rightarrow m = -\infty$ $\Rightarrow N = 0$.

$\Rightarrow y_p(t) = e^{-2t} \cdot a = ae^{-2t}$

$y_p''(t) = 4ae^{-2t}$

$\Rightarrow y_p'' + 4y_p = 8e^{-2t} \Leftrightarrow 8ae^{-2t} = 8e^{-2t} \Rightarrow \boxed{a = 1} \Rightarrow$

$y_p(t) = e^{-2t}$

$\Rightarrow \boxed{y(t) = c_1 \cos 2t + c_2 \sin 2t + e^{-2t}}$

$x(t) = \frac{1}{2}(y'(t) + 2y(t))$

$x(t) = \frac{1}{2}[-2c_1 \sin 2t + 2c_2 \cos 2t - 2e^{-2t} + 2c_1 \cos 2t + 2c_2 \sin 2t + 2e^{-2t}]$

$\boxed{x(t) = \sin 2t(-c_1 + c_2) + \cos 2t(c_2 + c_1)}$

Fie SDLO:

(1): $x'(t) = A \cdot x(t)$, unde $t \in \mathbb{R}$

$x(t) \overset{not}{=} \left\{ \begin{pmatrix} x_1(t) \\ \vdots \\ x_n(t) \end{pmatrix} \mid \text{col. fct. nec.} \right\}$

, iar $A \in \mathcal{M}_n(\mathbb{R})$.

Def: S.n. MATRICE FUNDAMENTALĂ pt. SDLO (1) o matrice $M(t) = (f_1(t) | f_2(t) | \dots | f_n(t))$, unde $\{f_1, \dots, f_n\}$ const. o bază pt sp. vect real al sol. sist. (1)

Obs: ! $M(t)$ nu este unică !

Prop: Dacă $M(t)$ este o M.F. pentru SDLO (1) atunci: sol. generală a lui (1) este:

$$\boxed{x(t) = M(t) \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{pmatrix}}, \{c_1, \dots, c_n\} \in \mathbb{R}.$$

Fie SDLNO:

(2): $x'(t) = A \cdot x(t) + b(t)$; $t \in \underset{\text{int}}{I} \subseteq \mathbb{R}$

, iar $b(t) \overset{not}{=} \begin{pmatrix} b_1(t) \\ b_2(t) \\ \vdots \\ b_n(t) \end{pmatrix}$; $b_j : I \to \mathbb{R}$, $j = \overline{1,n}$, cont.

(coloana fct. cunoscute)

Teoremă:

1) Sol. generală a SDLN (2) este $x(t) = x_h(t) + x_p(t)$, unde $x_h(t)$ = sol. gen. a lui (1), iar $x_p(t)$ = sol. part. a lui (2).

2) Dacă $M(t)$ este o M.F. pt (1), atunci

$x_h(t) = M(t) \begin{pmatrix} c_1 \\ \vdots \\ c_n \end{pmatrix}$; $c_1, \dots, c_n \in \mathbb{R}$ at $x_p(t)$ este def.

$x_p(t) = M(t) \begin{pmatrix} c_1(t) \\ \vdots \\ c_n(t) \end{pmatrix}$; $c_1(\cdot), \dots, c_n(\cdot) : I \to \mathbb{R}$

$c_j(\cdot) \in \mathcal{C}^1$; $j = \overline{1,n}$

$\Leftrightarrow M(t) \begin{pmatrix} c_1' \\ \vdots \\ c_n' \end{pmatrix} = b(t)$; $\forall t \in I$. $\begin{pmatrix} c_1' \\ \vdots \\ c_n' \end{pmatrix} = \underset{\exists}{M^{-1}(t)} \cdot b(t)$

At $x_p(t) = M(t) \left[\begin{pmatrix} c_1 \\ \vdots \\ c_n \end{pmatrix} + \int M^{-1}(t) \cdot b(t)\, dt \right] \begin{pmatrix} c_1(t) \\ c_n(t) \end{pmatrix} = \int [M(t)]^{-1} b(t)\, dt$

part. eventual $c_1 = \dots = c_n = 0$

Ca să obț. sol. part pe (2).

Așadar sol gen. pt (2): $M(t)\begin{pmatrix} c_1 \\ \vdots \\ c_n \end{pmatrix} + M(t)\cdot\int M(t)^{-1}\cdot b(t)\,dt$

$\Rightarrow X(t) = X_h(t) + X_p(t)$

$= M(t)\left[\begin{pmatrix} c_1 \\ \vdots \\ c_n \end{pmatrix} + \int M^{-1}(t)\cdot b(t)\,dt\right]$. ; $c_1,\dots,c_n \in \mathbb{R}$

Prop: 1) e^{At}, $t\in\mathbb{R}$ este o M.F. pentru SDLO (1).

2) Sol. gen. a SDLN (2) este:

$$x(t) = e^{At}\left[\begin{pmatrix} c_1 \\ \vdots \\ c_n \end{pmatrix} + \int e^{-At}\cdot b(t)\,dt\right]$$

3) Dacă $t_0 \in J$ și

$$x(t_0) = \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_n \end{pmatrix} ; \alpha_1,\dots,\alpha_n \in \mathbb{R} \text{ fixate.}$$

At. sol. SDLN (2) cu cond. Cauchy $x(t_0)$ este:

$$x(t) = e^{A(t-t_0)}\begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_n \end{pmatrix} + \int_{t_0}^{t} e^{A(t-s)}\,b(s)\,ds$$

4) Dacă $0\in J$, at. sol. generală pt (2) mai poate fi scrisă:

$$\Rightarrow x(t) = e^{At}\begin{pmatrix} c_1 \\ \vdots \\ c_n \end{pmatrix} + \int_0^t e^{A(t-s)}\,b(s)\,ds.$$

Ex: Să se re a SDLN:

$$\begin{cases} x' = 2y - x + 1 \\ y' = 3y - 2x \end{cases}$$

MI: $x'' = 2y' - x' = 2(3y - 2x) - x'$ ($y = \frac{x'+x-1}{2}$)

$\Rightarrow x'' = 3x' + 3x - 3 - 4x - x'$

$\Rightarrow x'' = 2x' - x - 3$ (:) $x'' - 2x' + x + 3 = 0$

(:) $x'' - 2x' + x = -3$

X_h: $\lambda^2 - 2\lambda + 1 = 0$ (:) $(\lambda-1)^2 = 0$ $\Rightarrow \lambda = 1$; $m_{\lambda=1} = 2$ ($\lambda \in \mathbb{R}$)

$\Rightarrow \varphi_1 = e^{2t}$, $\varphi_2 = te^{2t}$ $\Rightarrow$ $\underline{x_h(t) = c_1 e^{t} + c_2 \cdot te^{t}}$, $c_1, c_2 \in \mathbb{R}$

X_p: $-3 = e^{\alpha t}(P_n(t)\cos\beta t + Q_m(t)\sin\beta t)$

$\alpha, \beta = 0 \Rightarrow \beta = 0$

$P_n(t) = -3 \Rightarrow \begin{matrix} n = 0 \\ m = -\infty \end{matrix} \Rightarrow N = 0$

2) $x_p(t) = a$

$\Rightarrow x_p'' - 2x_p' + x_p = -3 \Rightarrow \boxed{a = -3} \Rightarrow x_p(t) = -3$

$$\Rightarrow \begin{cases} x(t) = C_1 e^t + C_2 t e^t - 3 \\ y(t) = \frac{1}{2}(x' + x - 1) = \frac{1}{2}\left[C_1 e^t + C_2(e^t + te^t) - 4\right] \end{cases}$$

MII: Fol. e^{At}, $t \in \mathbb{R}$.

$$\begin{pmatrix} x(t) \\ y(t) \end{pmatrix} = e^{At}\begin{pmatrix} c_1 \\ c_2 \end{pmatrix} + \int_0^t e^{A(t-s)} \cdot b(s)\,ds$$

$$A = \begin{pmatrix} -1 & 2 \\ -2 & 3 \end{pmatrix} \Rightarrow \lambda = 1, \; m_\lambda = 2$$

$(\dots) \Rightarrow f(At) = \lambda + t$ cu $m_{\mu = t} = 2$

$n = 2 \Rightarrow \exists\, \alpha_1, \alpha_0 \in \mathbb{R}$ a.i. $e^{At} = \alpha_1 t A + \alpha_0 I_2$

funcții reale

În plus, pt $r(\lambda) = \alpha_1 \lambda + \alpha_0$ au loc rel:

$$\begin{cases} r(\mu) = e^\mu \\ r'(\lambda)\big|_{\lambda = \mu} = e^\lambda \end{cases}$$

$\mu = t$; $m = 2$.

$r(\lambda) \overset{not}{=} \alpha_1 \lambda + \alpha_0 \Rightarrow r'(\lambda) = \alpha_1$ $\Big| \Rightarrow$

$r(\mu) = e^\mu$; $r'(\lambda)\big|_{\lambda = \mu} = e^\mu$

$$\begin{cases} \alpha_1 \cdot t + \alpha_0 = e^t \\ \alpha_1 = e^t \end{cases} \Big| \Rightarrow te^t - e^t = -\alpha_0 \Rightarrow \alpha_0 = e^t(1 - t)$$

$$e^{At} = t \cdot e^t A + e^t(1-t) I_2$$

$$\Rightarrow e^{At} = e^t \begin{bmatrix} -t & 2t \\ -2t & 3t \end{bmatrix} + e^t \begin{pmatrix} 1-t & 0 \\ 0 & 1-t \end{pmatrix}$$

$$\Rightarrow \boxed{e^{At} = e^t \begin{pmatrix} 1-2t & 2t \\ -2t & 1+2t \end{pmatrix}}$$

Sol. gen:

$$\begin{pmatrix} x(t) \\ y(t) \end{pmatrix} = e^{At}\begin{pmatrix} c_1 \\ c_2 \end{pmatrix} + \int_0^t e^{A(t-s)} b(s)\,ds$$

$$\int_0^t e^{A(t-s)} b(s)\,ds = \int_0^t e^{t-s} \begin{pmatrix} 1 - 2(t-s) & 2(t-s) \\ -2(t-s) & 2(t-s) + 1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} ds =$$

$$= \int_0^t e^{t-s} \begin{pmatrix} 1-2t+2s \\ -2t+2s \end{pmatrix} ds = e^t \begin{pmatrix} \int_0^t e^{-s}[(1-2t)+2s]ds \\ \int_0^t e^{-s}[-2t+2s]ds \end{pmatrix}$$

$$\int_0^t e^{-s} \cdot s \, ds = -\int_0^t (e^{-s})' \cdot s \, ds = e^{-s} \cdot s \Big|_0^t + \int_0^t e^{-s} ds$$

$$= -te^{-t} - e^{-s}\Big|_0^t = -te^{-t} - e^{-t} + 1$$

$$\Rightarrow \begin{pmatrix} x(t) \\ y(t) \end{pmatrix} = e^t \begin{pmatrix} c_1(1-2t)+c_2(2t) \\ -2tc_1 + (2t+1)c_2 \end{pmatrix} +$$

$$e^t \begin{pmatrix} (1-2t)(-e^{-t}+1) + 2(-te^{-t} - e^{-t}+1) \\ -2t(-e^{-t}+1) + 2(-te^{-t} - e^{-t}+1) \end{pmatrix}$$

Obs: Poate era mai simplu

$$\begin{pmatrix} x(t) \\ y(t) \end{pmatrix} = e^{At} \left[\begin{pmatrix} c_1 \\ \vdots \\ c_2 \end{pmatrix} + \int e^{-At} \cdot b(t) dt \right]$$

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19.01.2023

1) Să se rez. urm. ec. dif:

a) $x^2y'' - 3xy' + 5y = 3x^2$, $x>0$.

b) $(2x+3)^3 y''' + 3(2x+3)y' - 6y = 0$, $x > -\frac{3}{2}$

ATENȚIE!!!

a) $\{x = e^t\}$; $z(t) = y(e^t) = y(x)$

$\dot{z}(t) = e^t \cdot y'(e^t) = x \cdot y'(x) = xy' = \dot{z}$

$\ddot{z}(t) = e^t \cdot y'(e^t) + e^{2t} \cdot y''(e^t) = \dot{z} + x^2 y''(x)$

$\Rightarrow x^2 y'' = \ddot{z} - \dot{z}$

$\Rightarrow \ddot{z} - \dot{z} - 3\dot{z} + 5z = 3e^{2t}$

$\ddot{z} - 4\dot{z} + 5z = 3e^{2t}$

z_h: $\lambda^2 - 4\lambda + 5 = 0$; $\Delta = 16 - 4 \cdot 5 = -4$

$\lambda_{1,2} = \frac{4 \pm 2i}{2} = 2 \pm i$; $\alpha = 2$, $\beta = 1$

$f_1 = e^{2t}\cos t$; $f_2 = e^{2t}\sin t$

$\Rightarrow z_h = c_1 e^{2t}\cos t + c_2 e^{2t}\sin t$; $c_1, c_2 \in \mathbb{R}$.

z_p: $3x^2 = e^{\alpha x}[P_n(x)\cos\beta x + Q_m(x)\sin\beta x]$

$\alpha = 0$, $\beta = 0$ $\Rightarrow s = 0$

$P_n(x) = 3x^2 \Rightarrow n = 2$

$Q_m(x) = 0 \Rightarrow m = -\infty$ $\Rightarrow N = 2$

$\Rightarrow z_p(t) = (at^2 + bt + c)$; $a, b, c \in \mathbb{R}$.

$\Rightarrow \ddot{z}_p = 2a$

$\dot{z}_p = 2at + b$

$\Rightarrow 2a - 8at + 4b + 5at^2 + 5bt + 5c = 3x^2$

$\Rightarrow 5at^2 + t(5b - 8a) + 7a + 4b + 5c = 3x^2$

$\Rightarrow \begin{cases} 5a = 3 \Rightarrow a = 3/5 \\ 5b - 8a = 0 \Rightarrow b = \frac{8}{5}a = \frac{24}{25} \\ 5c = -7a - 4b \end{cases}$

$3e^{2t} = e^{\alpha t}[P_n(t)\cos\beta t + Q_m(t)\sin\beta t]$

$\alpha = 2$, $\beta = 0$ $\Rightarrow s = 0$.

$P_n(t) = 3 \Rightarrow n = 0$

Aleg $Q_m(t) = 0 \Rightarrow m = -\infty$ $\Rightarrow N = 0$ $\Rightarrow z_p(t) = ae^{2t}$

$x_p = ae^{2t}$; $\dot{x}_p = 2ae^{2t}$; $\ddot{x}_p = 4ae^{2t}$.

$\Rightarrow 4ae^{2t} - 8ae^{2t} + 5ae^{2t} = 3e^{2t}$

$ae^{2t} = 3e^{2t} \Rightarrow a = 3 \Rightarrow x_p(t) = 3e^{2t}$

$\Rightarrow x(t) = c_1 \cdot e^{2t}\cos t + c_2 e^{2t}\sin t + 3e^{2t}$; $c_1, c_2 \in \mathbb{R}$

$\Rightarrow y(x) = c_1 \cdot e^{2\ln x}\cos(\ln x) + c_2 e^{2\ln x}\sin(\ln x) + 3e^{2\ln x}$

$\Rightarrow y(x) = x^2\left(c_1\cos(\ln x) + c_2\sin(\ln x) + 3\right)$

b) $2x+3 = e^t \Rightarrow 2x = e^t - 3 \Rightarrow x = \frac{e^t-3}{2}$

$z(t) = \underbrace{y(x)}_{(y(x))} = y\left(\frac{e^t-3}{2}\right)$

$\Rightarrow \dot{z}(t) = \frac{e^t}{2}y'\left(\frac{e^t-3}{2}\right) = \frac{2x+3}{2}y'\left(\frac{e^t-3}{2}\right) \Rightarrow \boxed{(2x+3)y' = 2\dot{z}}$

$\ddot{z}(t) = \frac{d}{dt}(\dot{z}(t)) = \frac{d}{dt}\left(\frac{e^t}{2}y'\left(\frac{e^t-3}{2}\right)\right)$

$= \underbrace{\frac{e^t}{2}y'\left(\frac{e^t-3}{2}\right)} + \frac{e^t}{2}\cdot\frac{e^t}{2}\cdot y''\left(\frac{e^t-3}{2}\right)$

$\ddot{z}(t) = \dot{z}(t) + \frac{e^{2t}}{4}\cdot y''\left(\frac{e^t-3}{2}\right)$

$\dddot{z}(t) = \ddot{z}(t) + y''(x) \cdot \frac{(2x+3)^2}{4}$

$\Rightarrow \boxed{y''(x)\cdot(2x+3)^2 = 4\ddot{z} - 4\dot{z}}$

$\dddot{z}(t) = \dot{z}(t) + \boxed{y'''\left(\frac{e^t-3}{2}\right)\cdot\frac{e^{3t}}{8}} + \underbrace{y''\left(\frac{e^t-3}{2}\right)\cdot\frac{e^{2t}}{2}}$

$\underbrace{\left\{(2x+3)^3 y''' = 8(\dddot{z} - \ddot{z} - 2(\ddot{z} - \dot{z}))\right\}}$

$\Rightarrow 8(\dddot{z} - \ddot{z} - 2(\ddot{z} - \dot{z})) + 3\cdot 2\dot{z} - 6z = 0$

$8\dddot{z} - 8\ddot{z} - 16\ddot{z} + 16\dot{z} + 6\dot{z} - 6z = 0$

$\Rightarrow 4\dddot{z} - 12\ddot{z} + 11\dot{z} - 3z = 0$;

$4\lambda^3 - 12\lambda^2 + 11\lambda - 3 = 0$

$4\lambda^2(\lambda-1) - 8\lambda(\lambda-1) + 3(\lambda-1) = 0 \Leftrightarrow (\lambda-1)(4\lambda^2 - 8\lambda + 3) = 0$

$\Delta = 64 - 48 = 16$

$\lambda_{2,3} = \frac{8 \pm 4}{8} \begin{cases} \lambda_2 = \frac{3}{2} \\ \lambda_3 = \frac{1}{2} \end{cases}$

$x(t) = c_1 e^t + c_2 e^{(3/2)t} + c_3 e^{(1/2)t}$; $c_1, c_2, c_3 \in \mathbb{R}$

$\Rightarrow y(x) = c_1(2x+3) + c_2(2x+3)\sqrt{2x+3} + c_3\sqrt{2x+3}$; $c_1, c_2, c_3 \in \mathbb{R}$.

2) $\begin{cases} X'(t) = \begin{pmatrix} x' \\ y' \\ z' \end{pmatrix} = A \begin{pmatrix} x \\ y \\ z \end{pmatrix} \end{cases}$; $\begin{matrix} x = x(t) \\ y = y(t) \\ z = z(t) \end{matrix}$

βi $A = \begin{pmatrix} 2 & 0 & -1 \\ 1 & -1 & 0 \\ 3 & -1 & -1 \end{pmatrix}$

Met I : Fo se rez. sist. reducându-l la o ec. dif. lin cu coef. const. de ord 3.

$$P_A(\lambda) = \begin{vmatrix} 2-\lambda & 0 & -1 \\ 1 & -1-\lambda & 0 \\ 3 & -1 & -1-\lambda \end{vmatrix} = (2-\lambda)(\lambda+1)^2 + 1 - 3(\lambda+1)$$

$$= (\lambda+1)[(2-\lambda)(\lambda+1) - 3] + 1$$

$$= (\lambda+1)(2\lambda + 2 - \lambda^2 - \lambda - 3) + 1$$

$$= (\lambda+1)(-\lambda^2 + \lambda - 1) + 1$$

$$= -\lambda^3 + \lambda^2 - \lambda - \lambda^2 + \lambda - 1 + 1 = -\lambda^3$$

$\Rightarrow \sigma(A) = \{0\}$ cu $m_{\lambda=0} = 3$

$\lambda = 0$: $V_{\lambda=0} = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 \mid A \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0 \begin{pmatrix} x \\ y \\ z \end{pmatrix} \right\}$

$$(*) \begin{cases} 2x - z = 0 \\ x - y = 0 \\ 3x - y + z = 0 \end{cases} \Rightarrow \begin{cases} y = x \\ z = 2x \end{cases} \Rightarrow \begin{pmatrix} x \\ y \\ z \end{pmatrix} = x \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}$$

SG — nenul — V

$\Rightarrow \{v\}$ este o bază pt $V_{\lambda=0}$ $\Rightarrow$ SLi

$\Rightarrow \dim_{\mathbb{R}} V_{\lambda=0} = 1 \neq m_{\lambda=0} \Rightarrow$ Ane diagonalizează.

$\Rightarrow \begin{pmatrix} x \\ y \\ z \end{pmatrix} = e^{At} \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix}$

$e^{At}_{n=3} = \alpha_2 t^2 A^2 + \alpha_1 t A + \alpha_0 \bar{I}_3$, unde pt $r(\lambda) = \alpha_2\lambda^2 + \alpha_1\lambda + \alpha_0$

βi $\sigma(At) = \{0\}$ cu $m_{\mu=0} = 3$ are loc rel

$$\begin{cases} r(\mu) = e^{\mu} \\ r'(\lambda)|_{\lambda=\mu} = e^{\mu} \\ r''(\lambda)|_{\lambda=\mu} = e^{\mu} \end{cases} \Leftrightarrow \begin{cases} \alpha_0 = 1 \\ \alpha_1 = 1 \\ 2\alpha_2 = 1 \end{cases} \Rightarrow \alpha_2 = \frac{1}{2}$$

$$A^2 = \begin{pmatrix} 2 & 0 & -1 \\ 1 & -1 & 0 \\ 3 & -1 & -1 \end{pmatrix}\begin{pmatrix} 2 & 0 & -1 \\ 1 & -1 & 0 \\ 3 & -1 & -1 \end{pmatrix} = \begin{pmatrix} 1 & 1 & -1 \\ 1 & 1 & -1 \\ 2 & 2 & -2 \end{pmatrix}$$

$$\Rightarrow e^{At} = \frac{t^2}{2}\begin{pmatrix} 1 & 1 & -1 \\ 1 & 1 & -1 \\ 2 & 2 & -2 \end{pmatrix} + t\begin{pmatrix} 2 & 0 & -1 \\ 1 & -1 & 0 \\ 3 & -1 & -1 \end{pmatrix} + \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} \frac{t^2}{2} + 2t + 1 & \frac{t^2}{2} & -\frac{t^2}{2} - t \\ \frac{t^2}{2} + t & \frac{t^2}{2} + t + 1 & -\frac{t^2}{2} \\ t^2 + 3t & t^2 - t & -t^2 - t + 1 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} t^2+4t+2 & t^2 & -t^2-2t \\ t^2+2t & t^2+2t+2 & -t^2 \\ 2t^2+6t & 2t^2-2t & -2t^2-t+2 \end{pmatrix}\begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix}$$

SDLN

$\Rightarrow \begin{cases} x_h(t) = e^{At}\begin{pmatrix} c_1 \\ \vdots \\ c_n \end{pmatrix}; c_1, \dots, c_n \in \mathbb{R} \\ x_p(t) = e^{At}\begin{pmatrix} c_1(t) \\ \vdots \\ c_n(t) \end{pmatrix}; \end{cases}$ (metoda)

$e^{At}\begin{pmatrix} c_1' \\ \vdots \\ c_n' \end{pmatrix} = \begin{pmatrix} b_1(t) \\ \vdots \\ b_n(t) \end{pmatrix}$

$\begin{pmatrix} c_1' \\ \vdots \\ c_n' \end{pmatrix} = e^{-At}\begin{pmatrix} b_1(t) \\ \vdots \\ b_n(t) \end{pmatrix}$ | $\int$

$b_n(t)$ liniar

$\Rightarrow \begin{pmatrix} c_1 \\ \vdots \\ c_n \end{pmatrix} = \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} + \int e^{-At} b(t)\, dt.$

$\in \mathcal{M}_{n,1}(\mathbb{R})$ alegem $0_{n,1}$

$\Rightarrow x(t) = e^{At}\left[\begin{pmatrix} c_1 \\ \vdots \\ c_n \end{pmatrix} + \int e^{-At} b(t)\, dt\right]$

$\begin{pmatrix} x(0) \\ y(0) \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, $0 \in \mathbb{R}$

$\Rightarrow \boxed{x(t) = e^{At}\begin{pmatrix} 1 \\ 0 \end{pmatrix} + \int_0^t e^{A(t-s)} b(s)\, ds}$

EXEMPLU EXAMEN 2024.

EX: $\begin{cases} x' = 3x + y + 1 \\ y' = 2x + 2y \end{cases}$; $a = 2$, $a = 1$

$A = \begin{pmatrix} 3 & 1 \\ 2 & 2 \end{pmatrix}$; $P_A(\lambda) = \begin{vmatrix} 3-\lambda & 1 \\ 2 & 2-\lambda \end{vmatrix} = (3-\lambda)(2-\lambda) - 2$

$= 6 - 5\lambda + \lambda^2 - 2$

$= \lambda^2 - 5\lambda + 4$

$= (\lambda - 1)(\lambda - 4) \Rightarrow \sigma(A) \in \{1, 4\}$

$\Rightarrow \lambda_1 = 1$; $m_{\lambda=1} = 1$

$\lambda_2 = 4$; $m_{\lambda=4} = 1$

$\Rightarrow \sigma(At) \in \{t, 4t\}$ cu $m_{\mu_1 = t} = 1$ și $m_{\mu_2 = 4t} = 1$.

$(\exists)\ \alpha_1, \alpha_0 : \underset{\text{int}}{I} \subseteq \mathbb{R} \to \mathbb{R}$ a.î. $e^{At} = \alpha_1 t A + \alpha_0 I_2$.

În plus, def. $r(\lambda) = \alpha_1 \lambda + \alpha_0$ a.î.

$\begin{cases} r(\mu_1) = e^{\mu_1} \\ r(\mu_2) = e^{\mu_2} \end{cases} \Leftrightarrow \begin{cases} \alpha_1 t + \alpha_0 = e^t \\ 4\alpha_1 t + \alpha_0 = e^{4t} \end{cases}$ (-)

$\Rightarrow 3\alpha_1 t = e^{4t} - e^t \Rightarrow \boxed{\alpha_1 = \frac{e^{4t} - e^t}{3t}}$ (:3t)

$\Rightarrow \alpha_0 = e^t - \frac{t(e^{4t} - e^t)}{3t} = \frac{2t + e^t - te^{4t}}{3t}$

$\Rightarrow \boxed{\alpha_0 = \frac{2 + e^t - te^{4t}}{3t}}$

$$\Rightarrow e^{At} = \frac{e^{4t}-e^{t}}{3t}\cdot t\begin{pmatrix}3 & 1\\ 2 & 2\end{pmatrix} + \frac{2+e^{t}-te^{4t}}{3t}\begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix}$$

$$\Rightarrow e^{At} = \begin{pmatrix} 3\cdot\frac{te^{4t}-te^{t}}{3t} + \frac{2+e^{t}-te^{4t}}{3t} & \frac{te^{4t}-te^{t}}{3t} \\ 2\cdot\frac{te^{4t}-te^{t}}{3t} & 2\cdot\frac{te^{4t}-te^{t}}{3t} + \frac{2+e^{t}-te^{4t}}{3t}\end{pmatrix}$$

$$\Rightarrow e^{At} = \begin{pmatrix} \frac{2te^{4t}-te^{t}}{3t} & \frac{te^{4t}-te^{t}}{3t} \\ \frac{2te^{4t}-2te^{t}}{3t} & \frac{te^{4t}}{3t}\end{pmatrix}$$

$$\Rightarrow e^{At} = \begin{pmatrix} \frac{2e^{4t}-e^{t}}{3} & \frac{e^{4t}-e^{t}}{3} \\ \frac{2e^{4t}-2e^{t}}{3} & \frac{e^{4t}}{3}\end{pmatrix}$$

$$\begin{pmatrix}x\\ y\end{pmatrix} = e^{At}\begin{pmatrix}1\\ 0\end{pmatrix} + \int_0^t e^{A(t-s)}b(s)\,ds$$

$$2)\ e^{A(t-s)}b(s) \overset{\frac{1}{3}}{=} \begin{pmatrix} 2e^{4(t-s)}-e^{(t-s)} & e^{4(t-s)}-e^{(t-s)} \\ 2e^{4(t-s)}-2e^{(t-s)} & e^{4(t-s)}\end{pmatrix}\begin{pmatrix}1\\ 0\end{pmatrix}$$

$$= \frac{1}{3}\begin{pmatrix} 2e^{4(t-s)}-e^{(t-s)} \\ 2e^{4(t-s)}-2e^{(t-s)}\end{pmatrix}$$

$$\Rightarrow \int_0^t e^{A(t-s)}b(s)\,ds = \frac{1}{3}\begin{pmatrix} \int_0^t (2e^{4(t-s)}-e^{(t-s)})\,ds \\ \int_0^t (2e^{4(t-s)}-2e^{(t-s)})\,ds\end{pmatrix}$$

$$= \frac{1}{3}\begin{pmatrix} 2e^{4t}\cdot(-4)\cdot e^{-4s} - e^{t}(-1)\cdot e^{-s} \\ 2e^{4t}\cdot(-4)\cdot e^{-4s} - 2e^{t}\cdot(-1)\cdot e^{-s}\end{pmatrix}\Big|_0^t$$

$$= \frac{1}{3}\begin{pmatrix} -8e^{4t}(e^{-4t}-1)+e^{t}(e^{-t}-1) \\ -8e^{4t}(e^{-4t}-1)+2e^{t}(e^{-t}-1)\end{pmatrix}$$

$$= \frac{1}{3}\begin{pmatrix} 8e^{4t}-8+1-e^{t} \\ 8e^{4t}-8+2-2e^{t}\end{pmatrix}$$

$$\Rightarrow \begin{pmatrix}x(t)\\ y(t)\end{pmatrix} = \frac{1}{3}\begin{pmatrix}2e^{4t}-e^{t}\\ 2e^{4t}-2e^{t}\end{pmatrix} + \frac{1}{3}\begin{pmatrix}8e^{4t}-8+1-e^{t}\\ 8e^{4t}-8+2-2e^{t}\end{pmatrix} = \dots$$